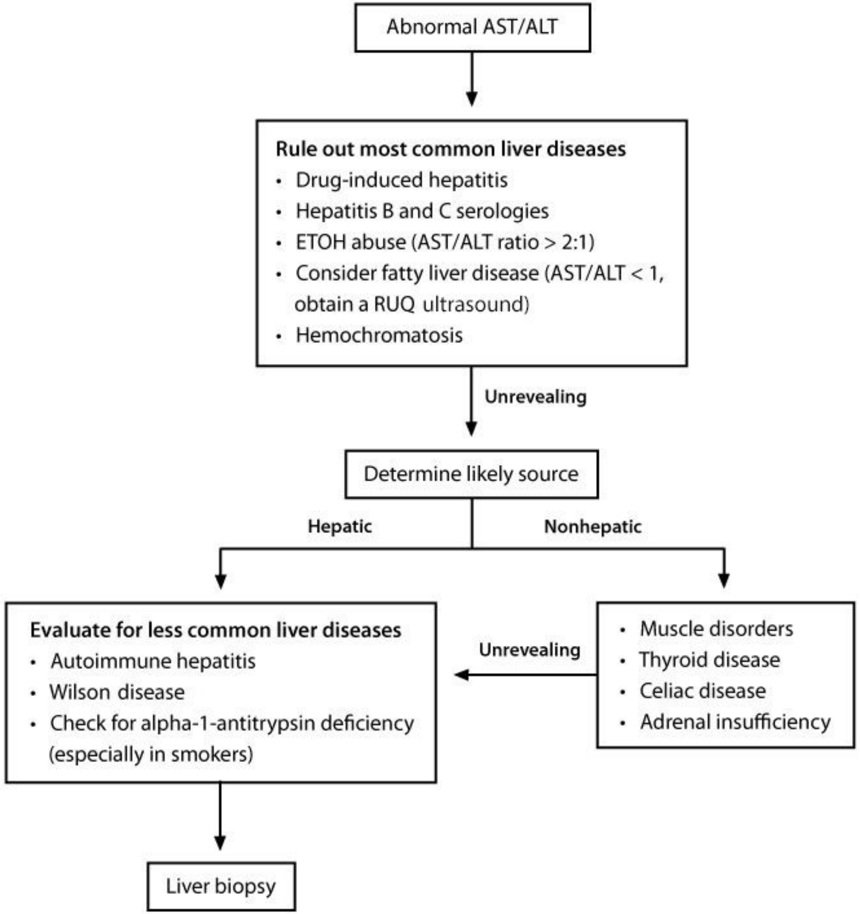
Boards Flow Sheets & Tables (UWORLD)

Approach to abnormal aminotransferases



ALT = alanine aminotransferase; **AST** = aspartate aminotransferase;
ETOH = ethyl alcohol; **RUQ** = right upper quadrant.

Clinical features of testicular cancer	

Clinical presentation	• Nodule or painless swelling of one testicle
	• Dull ache or heavy discomfort in lower abdomen, scrotum, or perianal area
	• Gynecomastia
	• Metastatic symptoms/signs
	  ○ Neck mass (lymph node)
	  ○ Cough, dyspnea (pulmonary)
	  ○ Bone pain (skeletal)
	  ○ CNS symptoms
	  ○ Unilateral or bilateral lower extremity swelling (iliac thrombosis)

Diagnostic evaluation	• Scrotal ultrasound - differentiate between cystic and hypoechoic lesions • CT abdomen/pelvis to evaluate for retroperitoneal lymph nodes • Serum tumor markers: alpha-fetoprotein, β-hCG, LDH • Radical inguinal orchiectomy for histological diagnosis

Abdominal aortic aneurysm	
Anatomy	Most commonly affects **infrarenal** aorta (≥3 cm)
Risks	• **Smoking** • Male sex • Older • White ethnicity • Family history of AAA • Atherosclerotic disease
Screening	Abdominal ultrasound in men age 65-75 who have ever smoked
Symptoms	• Mostly asymptomatic • May have abdominal, back, or flank pain • Lower limb ischemia &/or thromboembolism • Rupture often presents with abdominal distension & shock

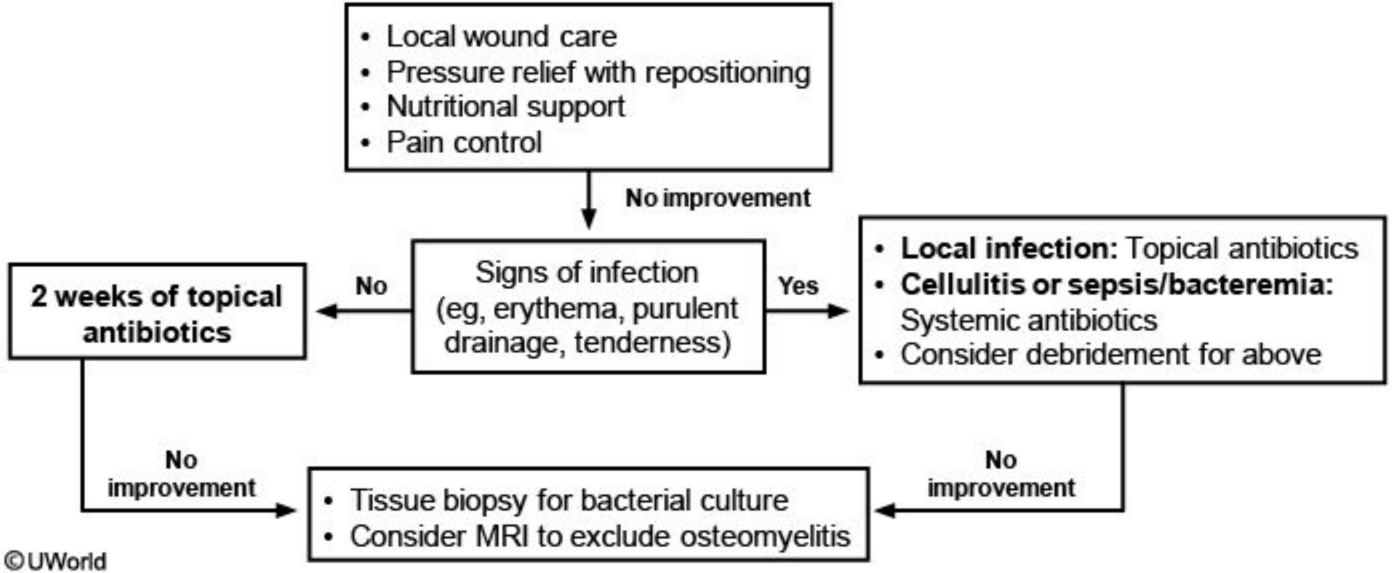
Management	• **Smoking cessation**
	• Aspirin & statin therapy
	• Elective repair recommended for:
	  ○ Large (≥5.5 cm) aneurysms
	  ○ Rapidly enlarging aneurysms (≥0.5 cm in 6 months)
	  ○ AAA associated with peripheral artery disease or aneurysm

Follow-up imaging	• Medium (4-5.4 cm): Ultrasound every 6-12 months • Smaller: Ultrasound every 2-3 years

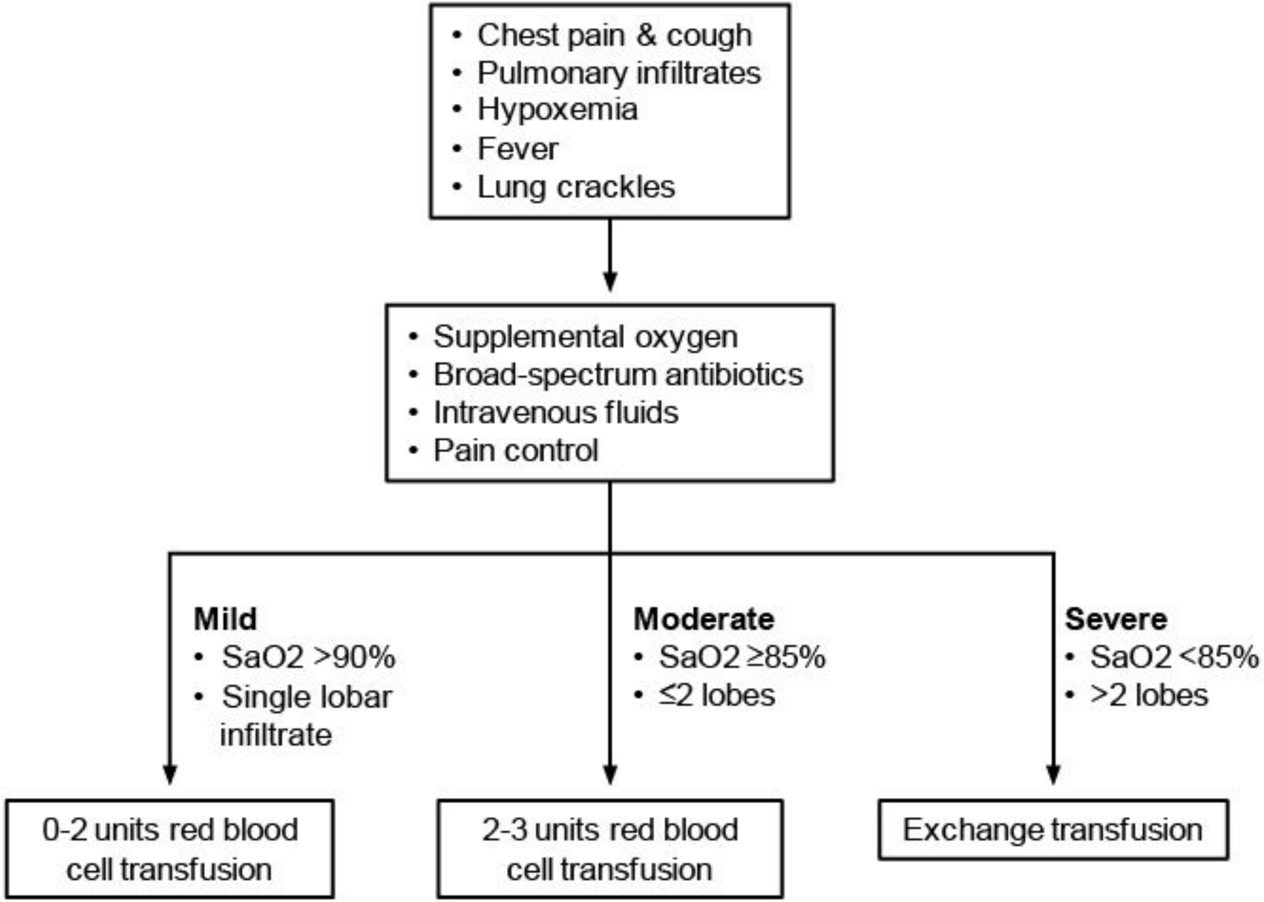
AAA = abdominal aortic aneurysm.

Chronic venous stasis	
Clinical features	**Pain** • Aching, cramping, or heaviness • Often improves with walking **Edema** • Worse with prolonged standing & improves with walking **Venous dilation** • Varicosities & telangiectasias **Dermatitis** • Erythema & serosanguineous seepage • Scaling & pruritus • Chronic brawny **discoloration** **Ulceration** • Most often near **medial malleoli** • No or mild pain at ulcer
Treatment	• Leg elevation, when possible • Leg exercises (ankle flexion, walking) to increase calf muscle strength • **Continuous graduated compression** stockings • **Aspirin** to accelerate ulcer healing

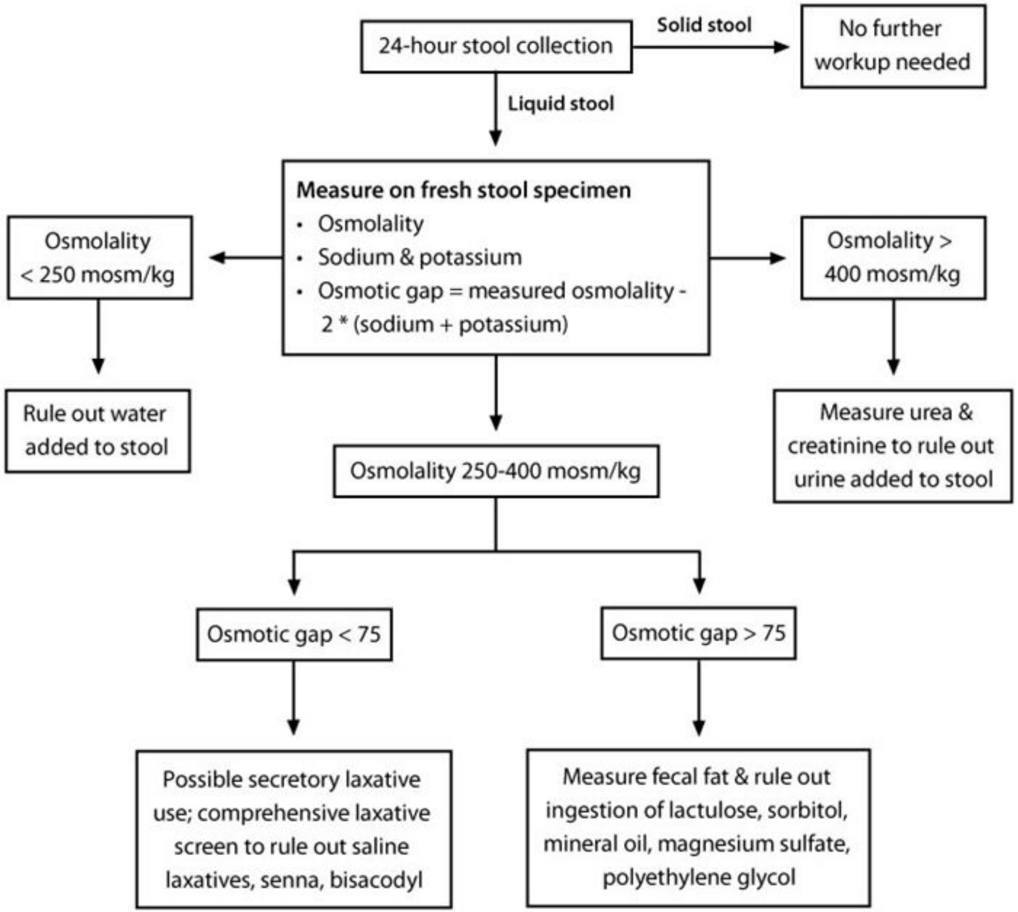
Treatment of nonhealing pressure ulcers



Acute sickle cell chest syndrome



Evaluation of suspected factitious diarrhea



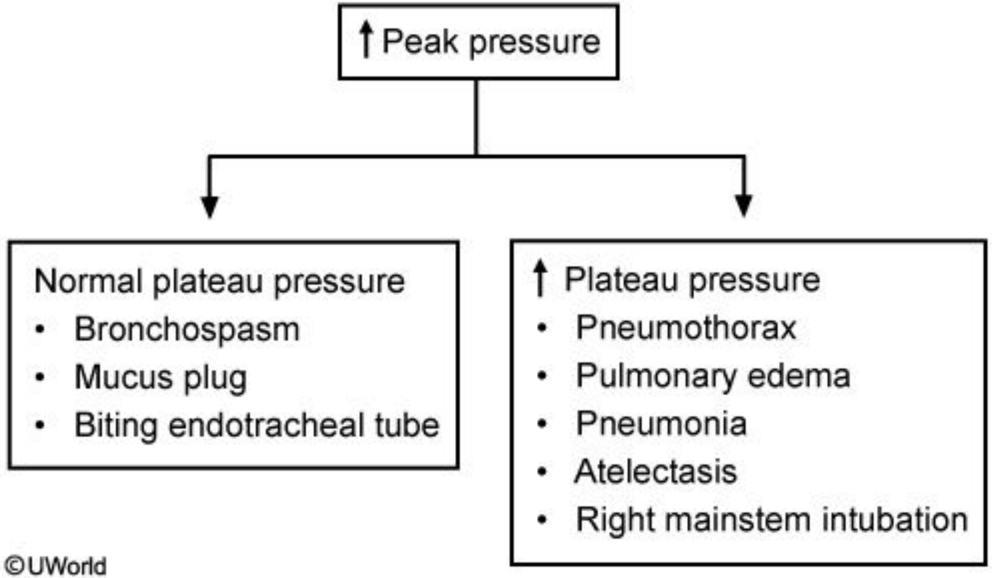
Polymyalgia rheumatica	
Clinical features	**Findings**

Symptoms	• Age >50
	• Bilateral pain & morning stiffness >1 month
	• Involvement of 2 of following:
	  ○ Neck or torso
	  ○ Shoulders or proximal arms
	  ○ Proximal thigh or hip
	  ○ Constitutional (fever, malaise, weight loss)

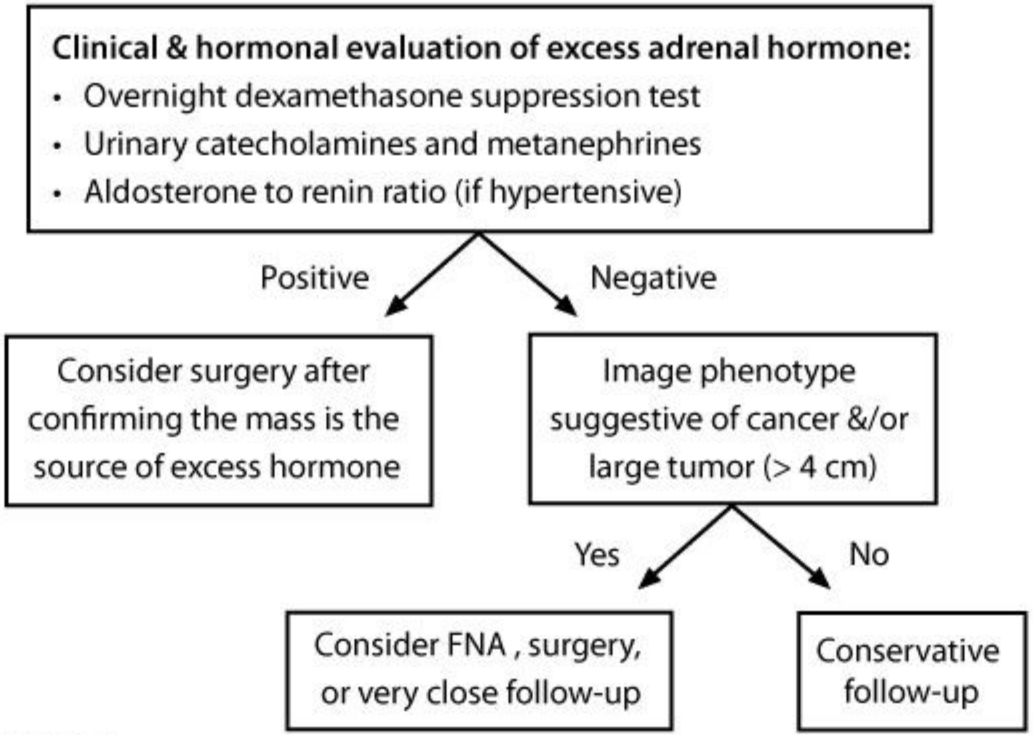
Physical examination	• Decreased active ROM in shoulders, neck & hips
Laboratory studies	• ESR >40 mm/h, sometimes >100 mm/h • Elevated CRP • Normocytic anemia possible • ~20% can have normal studies
Treatment	Response to glucocorticoids

CRP= C-reactive protein; **ESR** = erythrocyte sedimentation rate; **ROM** = range of motion.

Increased peak pressure



Adrenal incidentaloma



Magnesium deficiency	
Etiology	• Malnutrition (↓ intake) • Alcoholism (↓ intake) • Thiazide diuretics (↑ renal loss) • Malabsorption or diarrhea (↑ gastrointestinal loss) • Bartter & Gitelman syndrome
Clinical features	• **Neuromuscular excitability** (eg, tetany, seizures) • Generalized weakness • Atrial & ventricular arrhythmias
Laboratory studies	• **Hypokalemia** • **Hypocalcemia** (↓PTH release or resistance) • Low serum magnesium level • ECG abnormalities (widened QRS)

PTH = parathyroid hormone.

Imaging features suggestive of metastatic brain tumors

- Location at gray-white matter junction
- Multiple lesions
- Large vasogenic edema
- Circumscribed margins

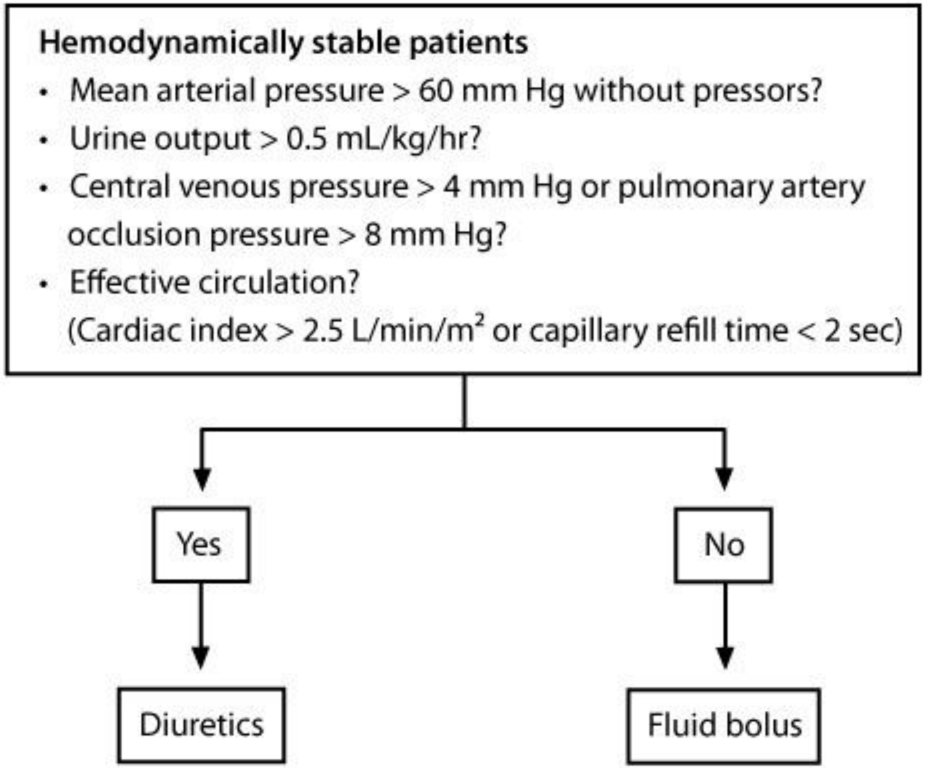
Clinical features of dermatomyositis	
Clinical signs/symptoms	• Muscle weakness (symmetric and more proximal) • Skin: Gottron's sign, heliotrope rash, shawl sign, erythroderma, mechanic's hands • Interstitial lung disease • Increased association with malignancy • Esophageal dysmotility • Myocarditis • Systemic symptoms (eg, fever, weight loss) • Raynaud's phenomenon
Diagnosis	• Elevated muscle enzymes (eg, CK, LDH, aldolase, AST) • Autoantibodies (eg, ANA, Anti-Jo-1) • EMG may be nonspecific or show signs of muscle irritability • Skin/muscle biopsy shows inflammatory changes

Common causes of persistent bowel symptoms in celiac sprue	
Condition	**Clinical clues**
Poor dietary compliance/ symptom recurrence	• Inadvertent gluten ingestion most common cause • Persistent histological abnormalities, high antibody titers & careful dietary history help diagnosis

Refractory sprue	• Type I: No initial response to gluten-free diet for 12 months
	• Type II: Initial response but symptoms return despite dietary adherence
	  ◦ Poor prognosis & frequent progression to EATL
	  ◦ Treatment involves glucocorticoids

EATL & ulcerative jejunoileitis	• Due to aberrant T cell populations • Presents with abdominal pain, B symptoms & GI bleeding • Often presents with intestinal obstruction or perforation • No response to glucocorticoids
Other coexisting conditions	• Lactose intolerance • Irritable bowel syndrome • Pancreatic insufficiency • Microscopic colitis • Small bowel bacterial bowel overgrowth

Fluid management in ARDS



Management of pressure ulcers		
Stage	**Clinical features**	**Treatment/Prevention**
1	• Intact skin • Localized, nonblanchable redness	• Transparent film dressing for protection
2	• Shallow, open ulcer • Red/pink wound with no sloughing • May have intact or ruptured blister	• Occlusive or semipermeable dressing to maintain moist wound environment
3	• **Full-thickness** skin loss • May have visible subcutaneous fat but no exposed bone, tendon, or muscle	• **Dry/light exudate**: Hydrocolloid or hydrogel dressing • **Moderate/heavy exudate**: Alginate, foam, Hydrofiber, or composite dressing
4	• Exposed bone, tendon, or muscle	• Same as stage 3 • Consider surgical wound closure

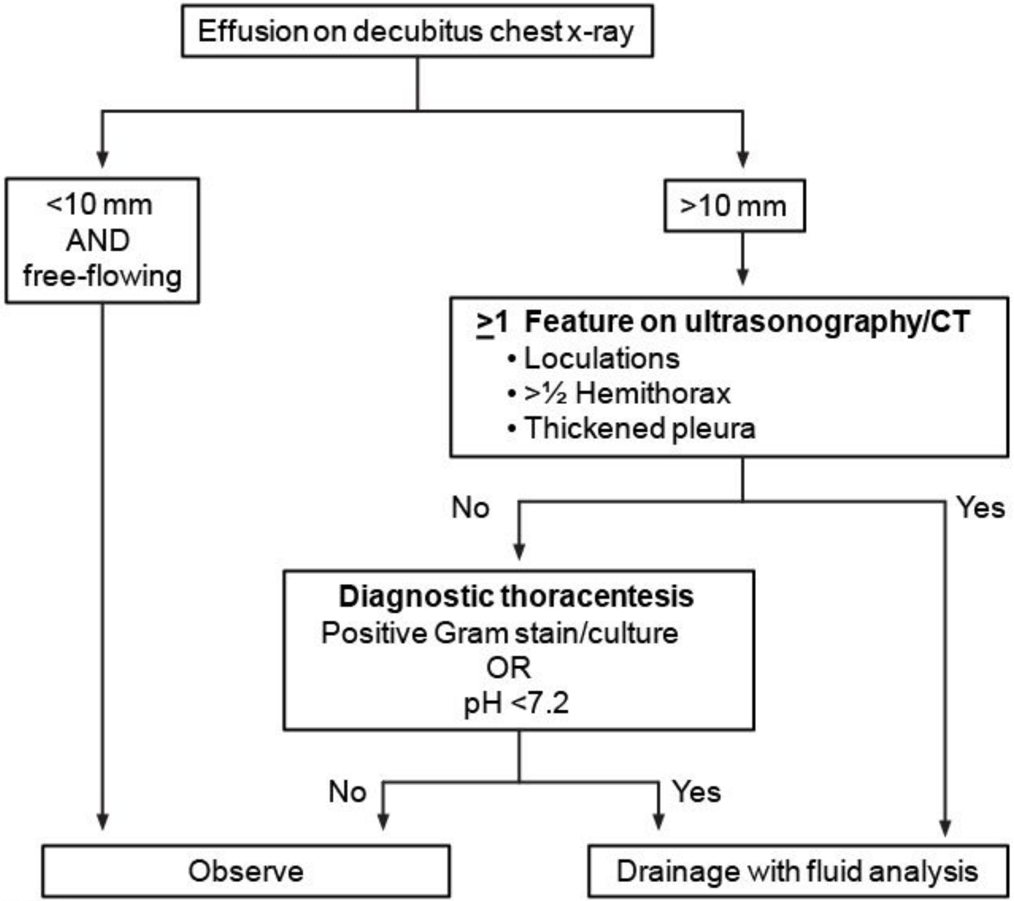
Hyperplastic polyposis syndrome
• 5 or more proximal (to sigmoid colon) hyperplastic polyps (at least 2 of which are > 1 cm) **or** • Any number of proximal hyperplastic polyps & family history of first-degree relative with HPS **or** • > 30 hyperplastic polyps throughout the colon

Intervals for follow-up colonoscopy after polypectomy	
Small rectal hyperplastic polyps	10 years
1 or 2 small (<1 cm) tubular adenomas	5 years
▪ 3-10 adenomas ▪ Any adenoma >1 cm ▪ Adenoma with high-grade dysplasia or **villous** features	3 years
More than 10 adenomas	<3 years Consider underlying familial syndrome
▪ Large (>2 cm) sessile polyp removed by piecemeal excision ▪ Polyp with adenocarcinoma (must have minimal invasion & ≥2 mm margin)	2-6 months 2-3 months

	Primary Raynaud's phenomenon	Secondary Raynaud's phenomenon
Etiology	• No underlying cause	• **Connective tissue diseases** • Occlusive vascular conditions • Sympathomimetic drugs • Vibrating tools • Hyperviscosity syndromes • Nicotine
Clinical presentation	• Usually women age <30 • No tissue injury • Negative ANA & ESR	• Usually men age >40 • Symptoms of underlying disease • **Tissue injury** or digital ulcers • Abnormal nail fold capillary examination
Management	• Avoid aggravating factors • **CCB** for persistent symptoms	• Evaluate & treat underlying disorder • **CCB** for persistent symptoms, **aspirin** for patients at risk for digital ulceration

ANA = antinuclear antibody; E**S**R = erythrocyte sedimentation rate; CCB = calcium channel blocker.

Approach to parapneumonic effusion



Clinical features of silicosis	
Associated occupations	• Coal mining (underground & surface) • Hard rock mining (eg, granite, slate, sandstone) • Sand blasting, quarrying, or stone cutting • Masonry, construction • Glass manufacturing, ceramic production
Clinical presentation	**Acute silicosis (less common)** • Symptoms within few weeks to few years after exposure • Rapid onset of cough, weight loss, fatigue & possible pleuritic chest pain • Chest x-ray shows basilar alveolar filling pattern **without** rounded opacities or lymph node calcifications • Poor prognosis with cyanosis, cor pulmonale & respiratory failure; <4 year survival rate **Chronic silicosis (most common)** • Asymptomatic or slowly developing symptoms 10-30 years after silica exposure • Chest x-ray shows <10 mm-diameter nodular opacities that are rounded, irregular & in mid to upper lung zones • Nodules can coalesce to form PMF with severe respiratory symptoms &/or cor pulmonale **Accelerated silicosis** • Asymptomatic or slowly developing symptoms within 10 years of exposure to high levels of silica • Chest x-ray similar to chronic silicosis • Increased risk for PMF
Associated conditions	• Increased risk of lung cancer • High risk for mycobacterial disease • Associated with scleroderma & rheumatoid arthritis • Airflow limitation & chronic bronchitis

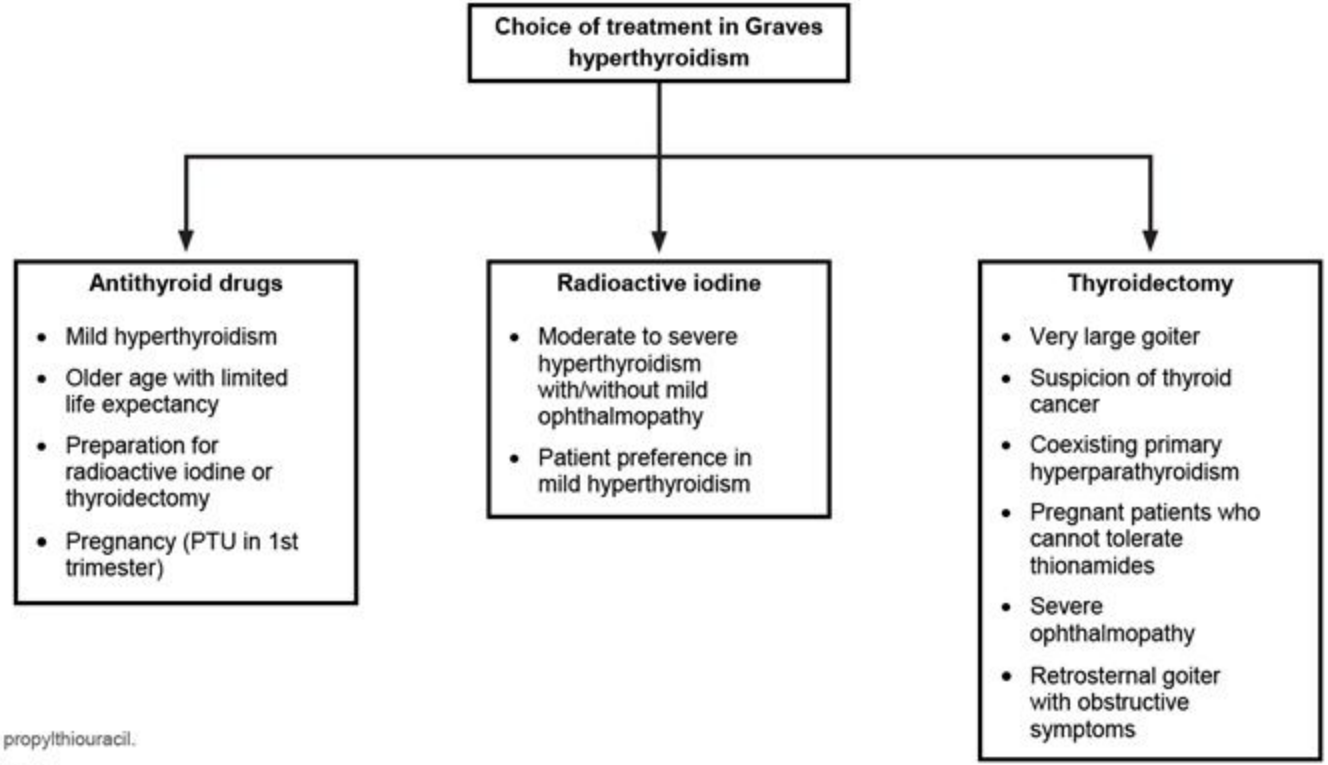
Babesiosis	
Epidemiology	• Transmission: **Tick** (*Ixodes scapularis*), blood transfusion • Geographic distribution: Northern United States • Risk factors for severe disease: Splenectomy, HIV, immunosuppression (cancer, organ transplantation)

Clinical manifestations	• Signs & symptoms:
	  ○ Mild: **Fevers**, fatigue, myalgias, headache
	  ○ **Severe**: ARDS, CHF, DIC, AKI, liver failure, splenic rupture
	• Laboratory studies: **Anemia & thrombocytopenia**

Diagnosis	• Thin blood smear (Wright/Giemsa stains): **Intraerythrocytic** pleomorphic ring forms (**"Maltese cross"**) with pale blue cytoplasm • PCR: Detection of *Babesia* DNA in the blood • Serology: May be negative in acute infection

AKI = acute kidney injury; **ARDS** = acute respiratory distress syndrome; CHF = congestive heart failure; DIC = disseminated intravascular coagulation; PCR = polymerase chain reaction.

Overview of subarachnoid hemorrhage	
Clinical features	• Most commonly due to ruptured arterial saccular ("berry") aneurysm • Severe headache at onset of neurologic symptoms • Meningeal irritation (eg, neck stiffness) • Focal deficits uncommon
Complications	• Rebleeding (first 24 hr) • Vasospasm (after 3 days) • Hydrocephalus/increased intracranial pressure • Seizures • Hyponatremia (usually from syndrome of inappropriate antidiuretic hormone secretion)
Diagnosis	• Noncontrast head CT scan is >90% sensitive • Lumbar puncture required to definitely rule out SAH • Xanthochromia in cerebrospinal fluid confirms diagnosis (usually seen 6 hr after onset) • Cerebral angiography to identify bleeding source
Treatment	• Angiographic procedure to stabilize aneurysm by coiling &/or stenting (endovascular therapy) • Nimodipine & hyperdynamic therapy to reduce vasospasm



PTU = propylthiouracil.

Clinical features of strongyloidiasis	
Epidemiology	• Southeastern United States • Immigrants of endemic areas such as Laos, Vietnam, and Cambodia

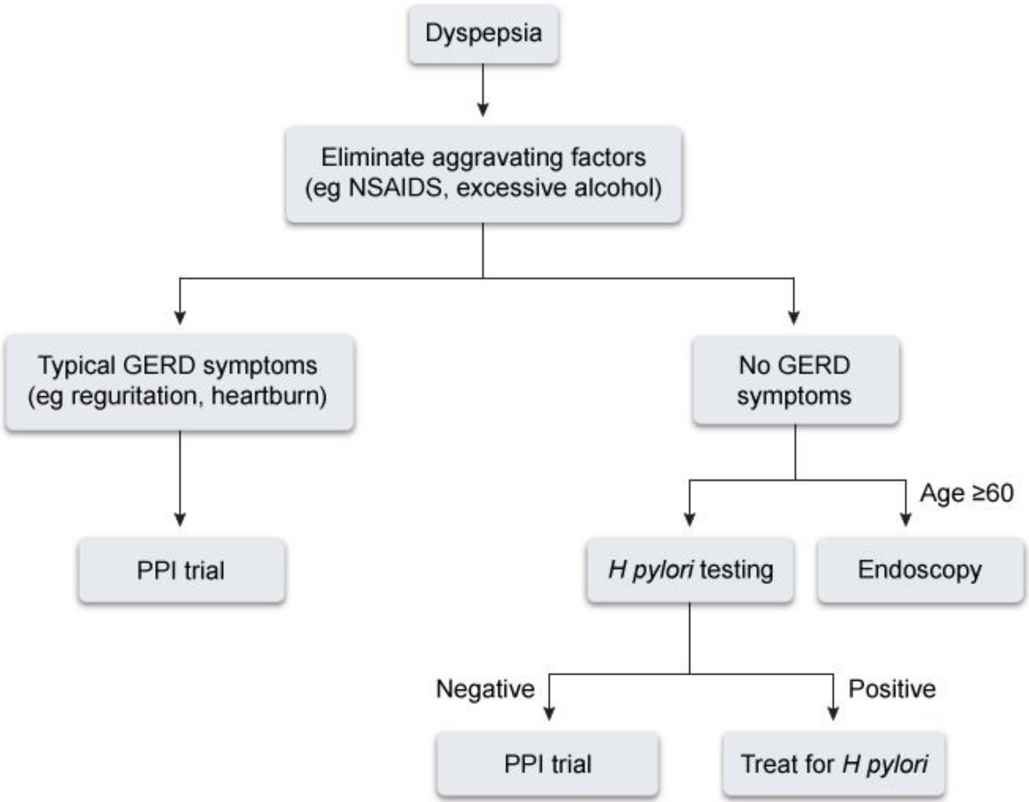
Clinical presentation	• Can be asymptomatic with peripheral eosinophilia
	• **Gastrointestinal**
	  ◦ Abdominal pain mimicking duodenal ulcer, but pain worse with eating
	  ◦ Chronic enterocolitis and malabsorption with high worm burden
	• **Pulmonary**
	  ◦ Migration or larvae can cause dry cough, hemoptysis, dyspnea, and wheezing (often misdiagnosed as asthma)
	  ◦ Fever with mild pneumonitis (can resemble pneumonia)
	• **Skin**
	  ◦ Serpiginous and raised erythematous track (larva currens or “running” larva)
	  ◦ Severe pruritus (ground itch)
	• Hyperinfection syndrome (due to autoinfection) leading to multiorgan dysfunction and possible septic shock

Diagnosis	• Stool samples can confirm but are frequently negative • Many patients require serology tests (ELISA IgG antibody), endoscopy for duodenal biopsy or aspirate, or string test (Entero-Test)

Features of symptomatic neurosyphilis		
Early neurosyphilis	Meningitis	• Headache, confusion, nausea/vomiting, stiff neck
	Ocular	• Posterior uveitis (most common) • Decreased visual acuity • Often with syphilitic meningitis
	Meningovascular	• Infectious CNS arteritis causing ischemia &/or infarction
Late neurosyphilis (tertiary)	General paresis	• Progressive dementia
	Tabes dorsalis (uncommon)	• Posterior column/dorsal root disease • Pupillary defect, diffuse neurologic signs • May have normal CSF

CNS = central nervous system; **CSF** = cerebrospinal fluid.

Initial management of dyspepsia



GERD = gastrointestinal reflux disease; *H pylori* = *Helicobacter pylori*;
NSAID = nonsteroidal anti-inflammatory drug; PPI = proton pump inhibitor.

Clinical features of meniscus & knee ligament injuries	
Meniscus	• Acute or subacute symptoms • **Small effusion** • **Locking** sensation with extension • Inability to fully extend • Joint line tenderness • Positive McMurray test
Cruciate ligaments	• Acute "pop" with rapid-onset, **large effusion**/hemarthrosis • **Anterior cruciate**: Anterior drawer & Lachman tests • Posterior cruciate (uncommon): Posterior drawer test
Collateral ligaments	• Effusion uncommon • **Medial collateral**: Instability with lateral movement, valgus laxity • **Lateral collateral**: Instability with medial movement, varus laxity

Positive nontreponemal testing after syphilis treatment		
Reason	**Clinical & laboratory findings**	**Management**
Treatment failure	Persistent symptoms, failure of nontreponemal titer to fall 4-fold	Lumbar puncture to evaluate for neurosyphilis, retreatment
Reinfection	New sexual partner, new symptoms of primary or secondary syphilis	Retreatment depending on stage
Serofast state	Asymptomatic, titers fall by >4-fold	Repeat VDRL testing & check HIV status. No treatment indicated.

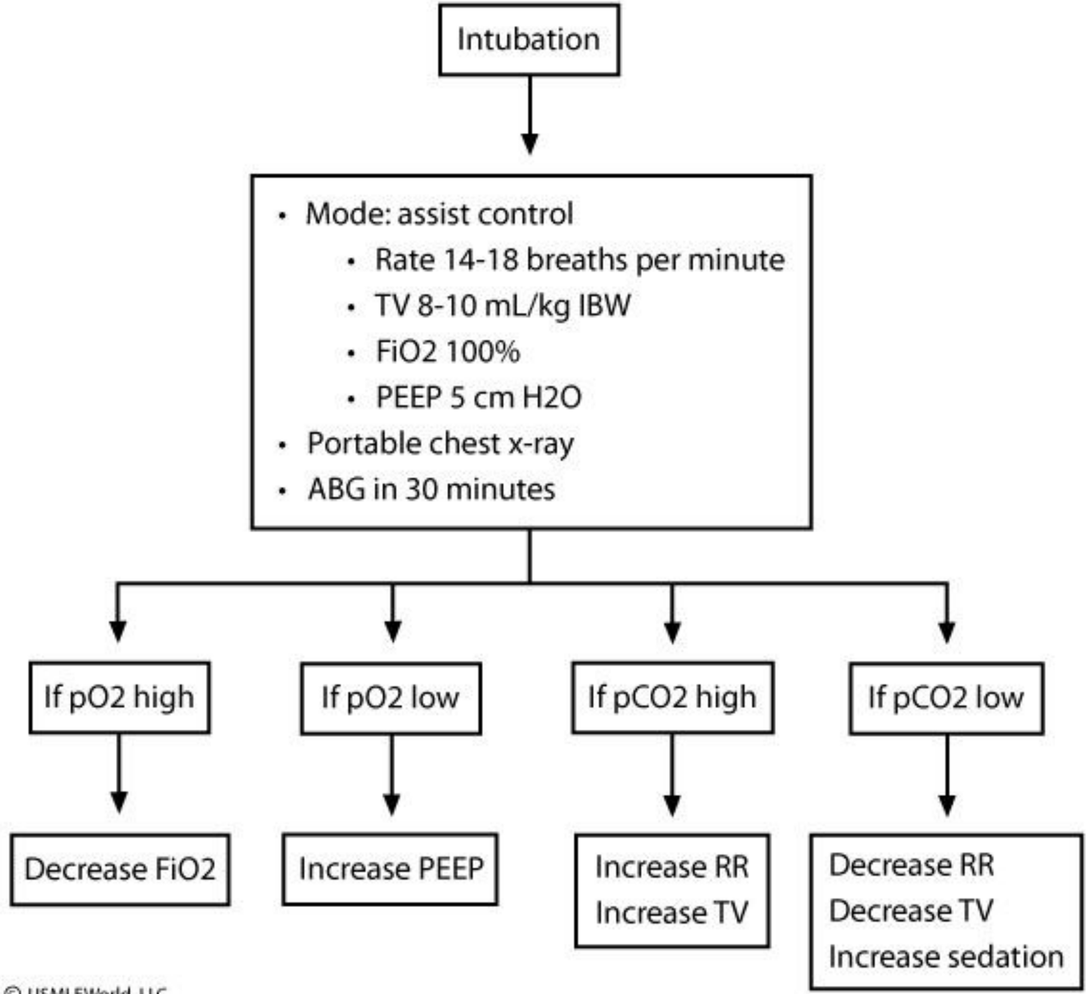
Clinical features of fibromyalgia	
Clinical presentation	• **Widespread musculoskeletal pain** • **Fatigue** • Cognitive difficulties (eg, impaired attention) • Nonspecific GI symptoms • Depression/anxiety symptoms • **Examination**: Tender points (eg, mid trapezius, costochondral junction, greater trochanter)
Diagnosis	• Symptoms ≥3 months • Normal inflammatory markers • WPI ≥7 & SS score ≥5 **OR** • WPI 3-6 & SS ≥9
Treatment	• **Lifestyle**: Aerobic exercise, sleep hygiene • **Medication**: TCA, SNRI

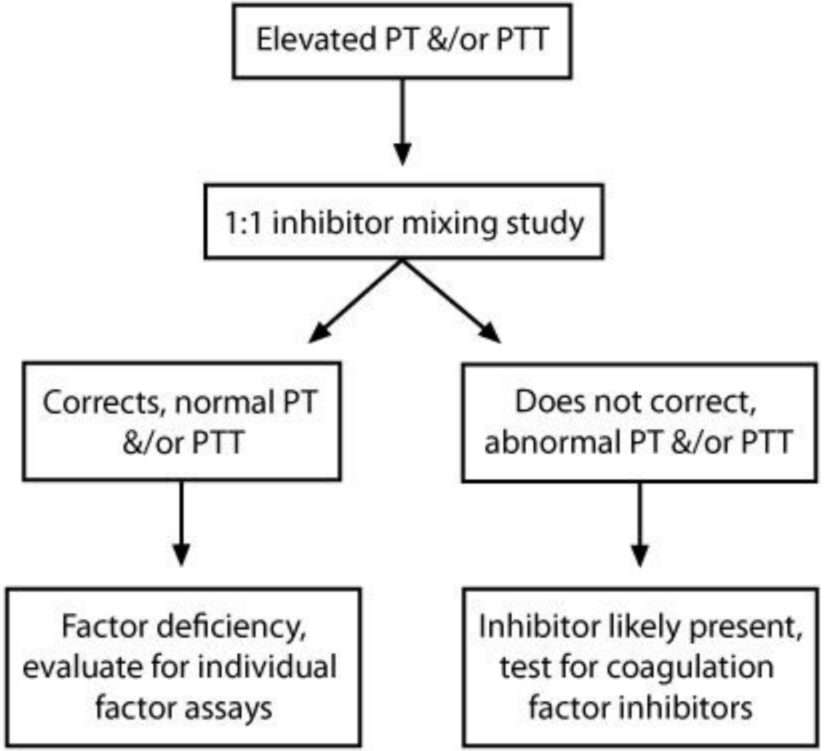
GI = gastrointestinal; **SS** = symptoms severity; **WPI** = widespread pain index;
TCA = tricyclic antidepressant; **SNRI** = serotonin-norepinephrine reuptake inhibitor.

Clinical features of peripartum cardiomyopathy
• Onset of heart failure during last month of pregnancy or within 5 months following delivery • LV systolic dysfunction with LV ejection fraction <45% • Absence of other causes of heart failure • Absence of heart disease prior to final month of pregnancy

LV = left ventricular.

Asthma severity	Symptom frequency	Nighttime awakenings	Medication management (preferred)
Mild intermittent	≤ 2 days week	≤ twice per month	• Albuterol as needed
Mild persistent	> 2 days per week but not daily	> twice per month	• Low-dose inhaled corticosteroid (ICS)
Moderate persistent	Daily symptoms	> once per week	• Low-dose ICS/salmeterol or medium-dose ICS ± salmeterol
Severe persistent	Throughout the day	≥ 4 times per week	• High-dose ICS/salmeterol & oral corticosteroid if needed


Intubation
• Mode: assist control
• Rate 14-18 breaths per minute
• TV 8-10 mL/kg IBW
• FiO2 100%
• PEEP 5 cm H2O
• Portable chest x-ray
• ABG in 30 minutes
If pO2 high
If pO2 low
If pCO2 high
If pCO2 low
Decrease FiO2
Increase PEEP
Increase RR
Increase TV
Decrease RR
Decrease TV
Increase sedation



Elevated PT &/or PTT
1:1 inhibitor mixing study
Corrects, normal PT &/or PTT
Does not correct, abnormal PT &/or PTT
Factor deficiency, evaluate for individual factor assays
Inhibitor likely present, test for coagulation factor inhibitors

Complex regional pain syndrome (CRPS)

Inciting events (usually Type I CRPS)

- Soft tissue injury
- Fractures
- Joint surgery (e.g., knee arthroscopy)
- MI or CVA
- No apparent cause in many patients

Clinical features

Stage 1 (Early disease)

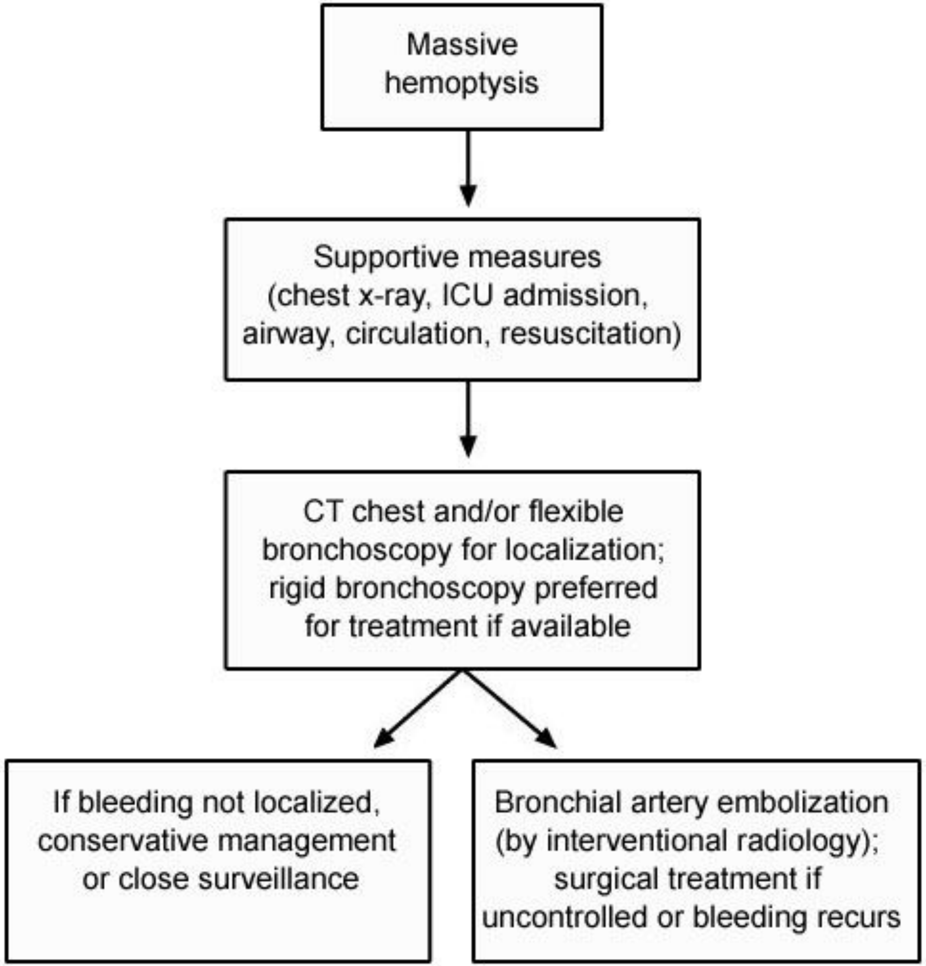
- Limb pain (burning, throbbing, aching, sensitive to cold & touch)
- Local edema in extremity
- Changes in color and temperature across > 1 nerve root distribution
- Patchy bone demineralization on x-ray

Stage 2 (Subacute disease for 3-6 months)

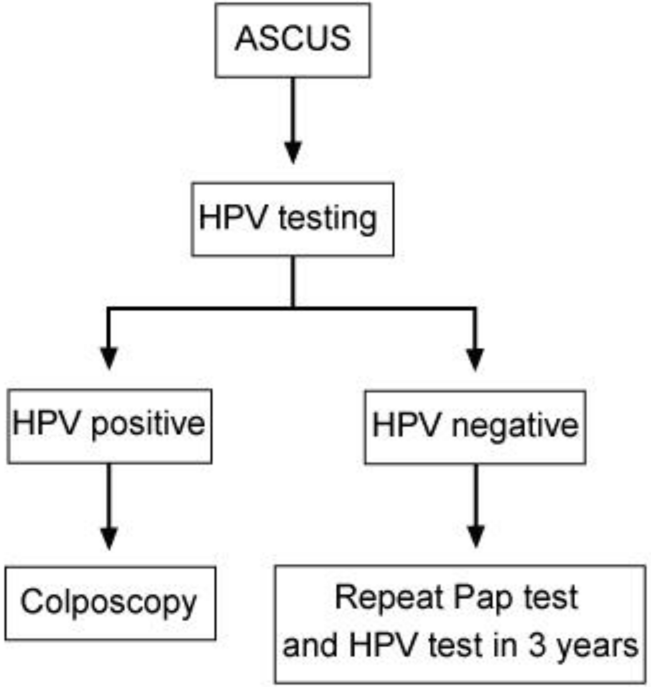
- Progressively worsening edema, skin thickening, muscle wasting

Stage 3 (Chronic disease)

- Severely limited movement (e.g., shoulder and hand)
- Digit contractures, brittle nails, skin changes
- Severe bone demineralization on x-rays


Massive hemoptysis
Supportive measures (chest x-ray, ICU admission, airway, circulation, resuscitation)
CT chest and/or flexible bronchoscopy for localization; rigid bronchoscopy preferred for treatment if available
If bleeding not localized, conservative management or close surveillance
Bronchial artery embolization (by interventional radiology); surgical treatment if uncontrolled or bleeding recurs

Management of ASCUS in women age ≥25



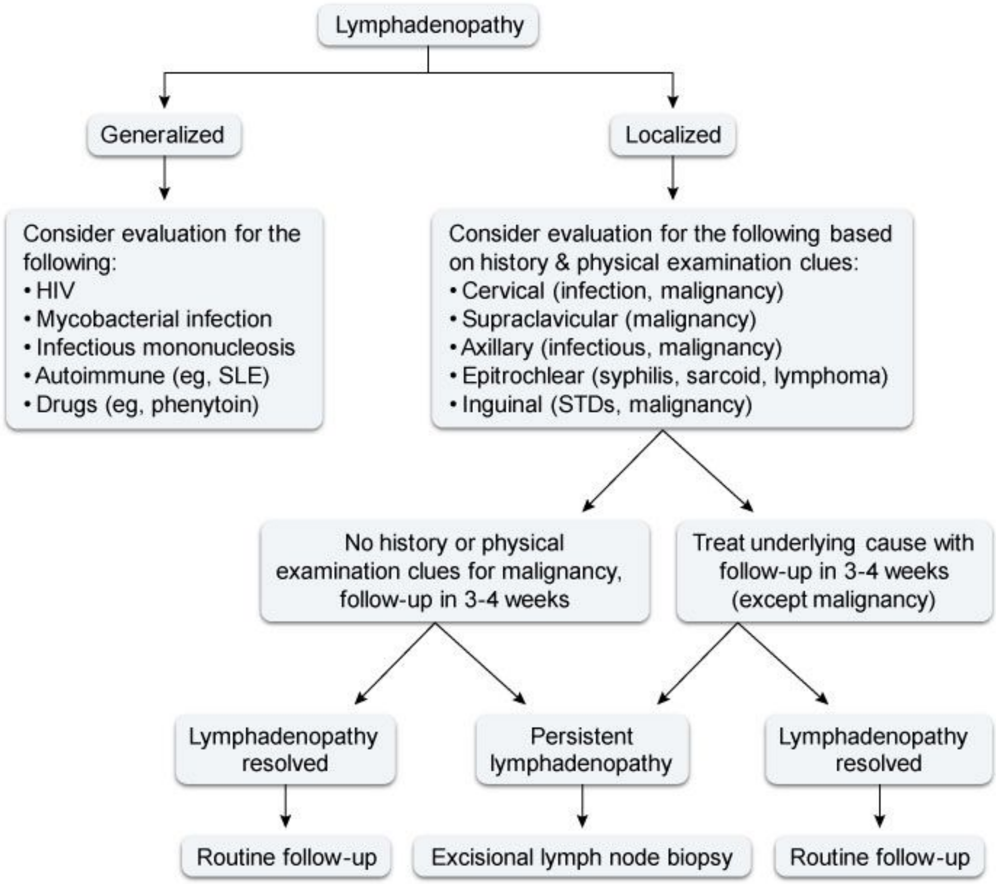
ASCUS = atypical squamous cells of undetermined significance;
HPV = human papillomavirus.

Evaluation of syncope with electrocardiographic monitoring	
Type of event monitor	**Indications**
In-hospital (telemetric)	• Seriously ill patients who may have life-threatening cardiac arrhythmias
Ambulatory (Holter) electrocardiography (continuous or intermittent recorders)	• Patients with frequent episodes of syncope (≥1/week) • Can store 24-48 hours of data
External loop recorder	• Syncope-free interval ≤1 month
Implantable loop recorder	• Recurrent, infrequent (≤1/month) syncope due to unidentifiable cause • Can provide monitoring for months to years

Features of constrictive pericarditis	
Etiology	• Idiopathic or viral pericarditis • Cardiac surgery or radiation therapy • Tuberculous pericarditis (in endemic areas)
Clinical presentation	• Fatigue & dyspnea on exertion • Peripheral edema & ascites • ↑ Jugular venous pressure • Pericardial knock may be heard • Pulsus paradoxus • Kussmaul's sign
Diagnostic findings	• ECG: Nonspecific, atrial fibrillation, low-voltage QRS complex • Chest imaging: Pericardial thickening, calcification • Echocardiogram: Increased pericardial thickness, abnormal septal motion, biatrial enlargement • Right-heart catheterization: Prominent x & y descents, equalization of left & right ventricular end-diastolic pressures

ECG = electrocardiogram.

Evaluation of lymphadenopathy

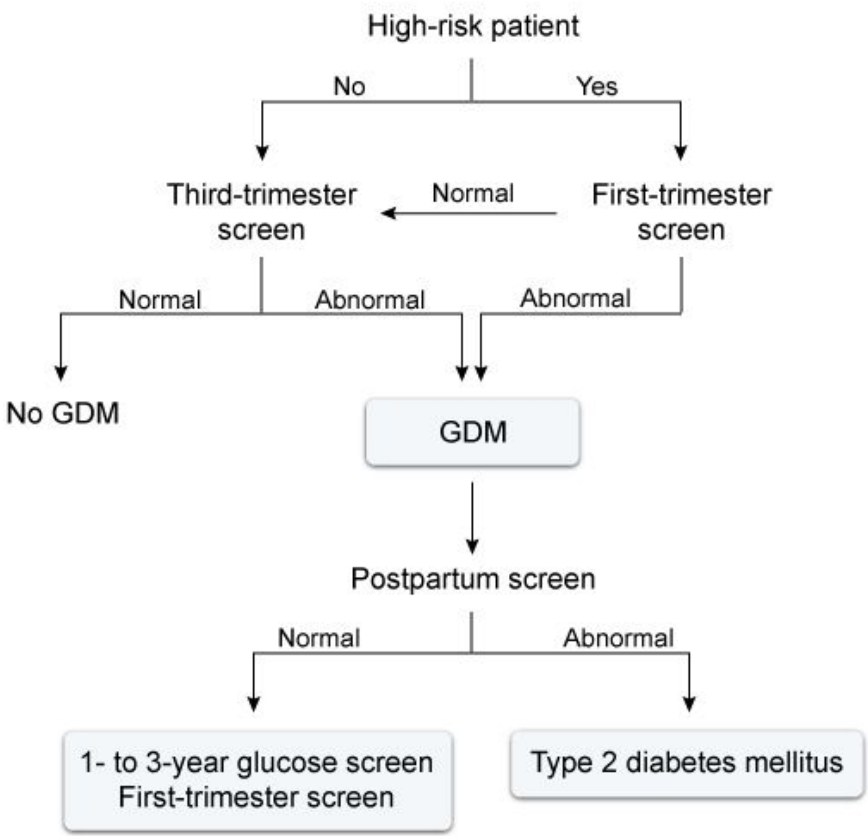


SLE = systemic lupus erythematosus; **STDs =** sexually transmitted diseases.

Differential diagnosis of thrombocytopenia			
Clinical features	**HUS**	**TTP**	**ITP**
Prodromal illness of diarrhea, abdominal pain & vomiting	Yes	No	Preceding viral infection
Fever	Less common	Yes	No
Neurologic changes	Less common	More common	No
Microangiopathic hemolytic anemia	Yes	Yes	No
Thrombocytopenia	Yes	Yes	Yes
Acute kidney injury	More common	Less common	No
↑ LDH, ↑ bilirubin & ↓ haptoglobin	Yes	Yes	No
ADAMTS13 activity <10%	No	Yes	No

HUS = hemolytic-uremic syndrome; **ITP** = immune thrombocytopenic purpura;
LDH = lactate dehydrogenase; **TTP** = thrombotic thrombocytopenic purpura.

Prenatal diabetes screening

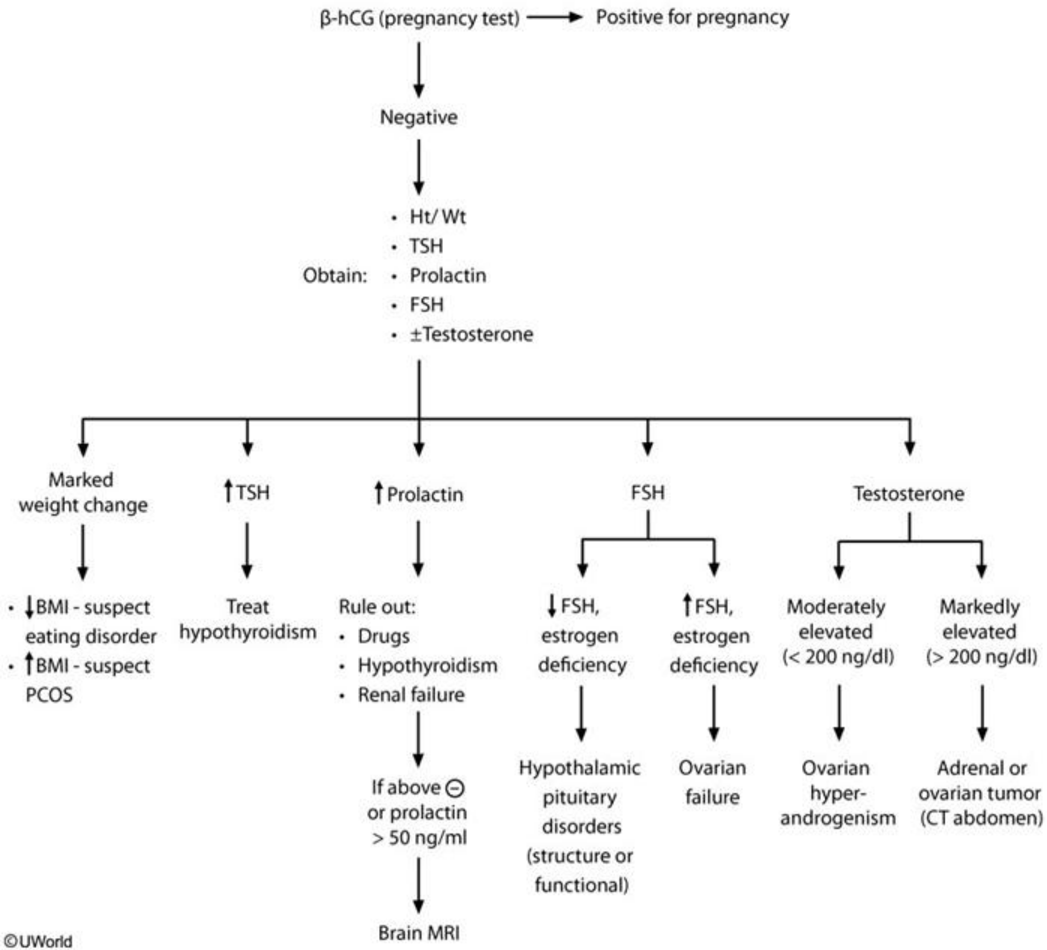


GDM = gestational diabetes mellitus.

Clinical features of central retinal artery occlusion	
Etiologies	• Carotid artery atherosclerosis (most common) • Cardiogenic embolism • Small artery disease due to diabetes &/or hypertension • Carotid artery dissection • Hematologic disease (eg, sickle cell, hypercoagulability) • Vasculitis (eg, giant cell arteritis)
Clinical presentation	• Painless acute vision loss in 1 eye • Complete or relative afferent pupillary defect • Retinal whitening/cherry red spot in macula on funduscopy
Treatment	• Urgent ophthalmology consult • Lower intraocular pressure (eg, ocular massage) • Possible intra-arterial thrombolytics • Long-term atherosclerosis risk factor modification

Clinical presentation of CML	
Chronic stable phase (at time of diagnosis in 85% of patients)	• Can be asymptomatic • Nonspecific systemic symptoms (eg, fatigue, weight loss, malaise) • Abdominal pain (usually due to splenomegaly) • Anemia, leukocytosis, thrombocytosis • <10% blasts in bone marrow and periphery
Accelerated phase	• Progressively worsening leukocytosis • 10-20% blasts in bone marrow or periphery
Blast crisis	• >30% blasts in bone marrow or >20% blasts in periphery • Resembles acute leukemia • Extramedullary blast infiltrates (eg, sarcoma) possible

Evaluation of secondary amenorrhea



Clinical features of renal cell cancer

Signs/symptoms

- Can be asymptomatic
- Hematuria (only with invasion of collecting system)
- Abdominal mass, abdominal pain, weight loss
- Scrotal varicocele

Paraneoplastic symptoms

- Anemia (microcytic or normocytic)
- Erythrocytosis (increased erythropoietin production)
- Non-metastatic liver function abnormalities
- Fever and weight loss
- Hypercalcemia (due to bone metastases, ↑PTHrP, or ↑prostaglandins)
- AA amyloidosis

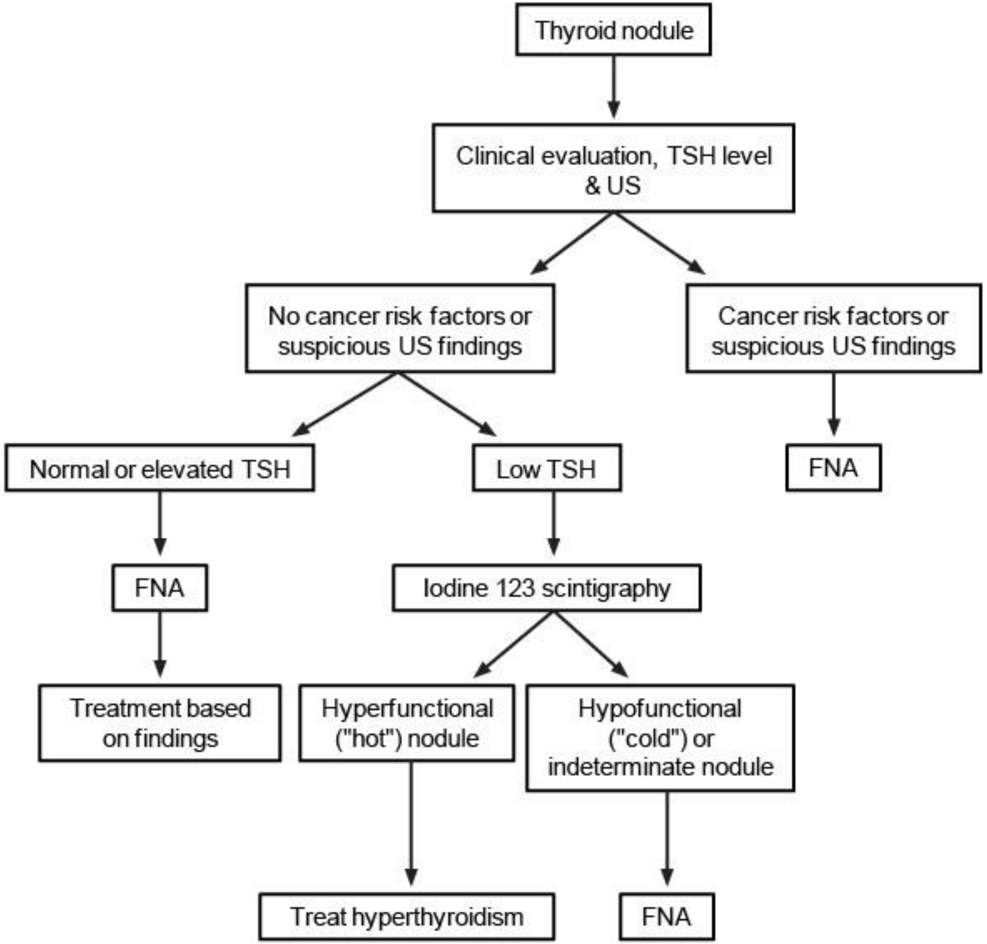
Clinical features of *Neisseria meningitidis*

Signs/symptoms

- Fever & severe myalgias mimicking flu
- Meningitis (triad of fever, neck stiffness & altered mental status seen less often than in *S pneumoniae* meningitis)
- Purpura fulminans (petechial rash progressing to ecchymoses, bullae & vesicles & ultimately gangrenous necrosis)
- Adrenal insufficiency (Waterhouse-Friderichsen syndrome)
- Myocarditis
- DIC and shock

Treatment, vaccination & chemoprophylaxis

- Initially, third-generation cephalosporin (eg, cefotaxime or ceftriaxone) **plus** vancomycin (for resistant pneumococcal coverage)
- Glucocorticoids not helpful
- Protein C concentrate for patients with protein C deficiency & purpura fulminans
- Vaccinate asplenic patients 14 days prior to or post splenectomy & revaccinate every 5 years
- Prophylactic antibiotics for close respiratory contacts: rifampin, ciprofloxacin, or ceftriaxone



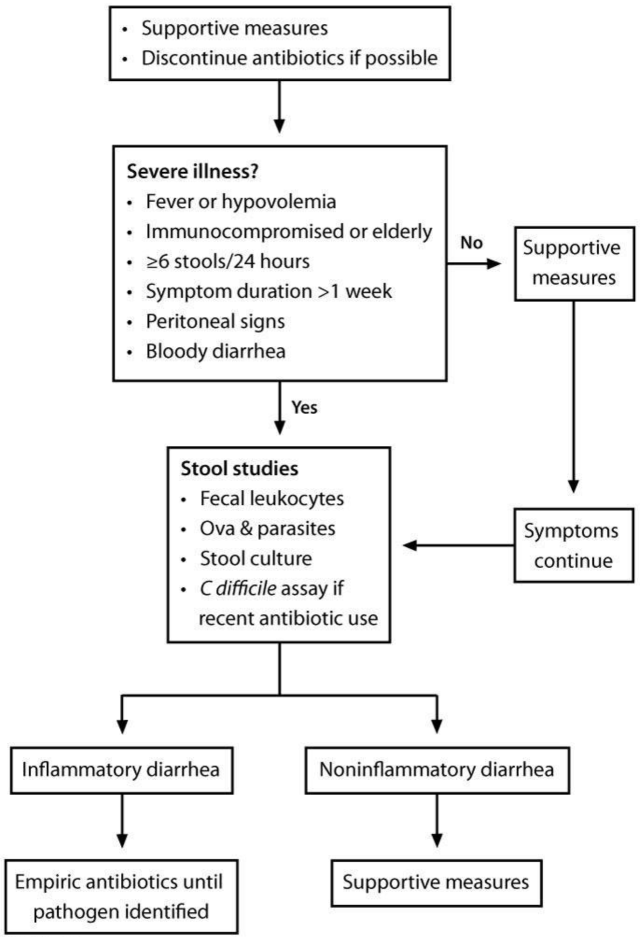
FNA = fine needle aspiration; **US** = ultrasound.

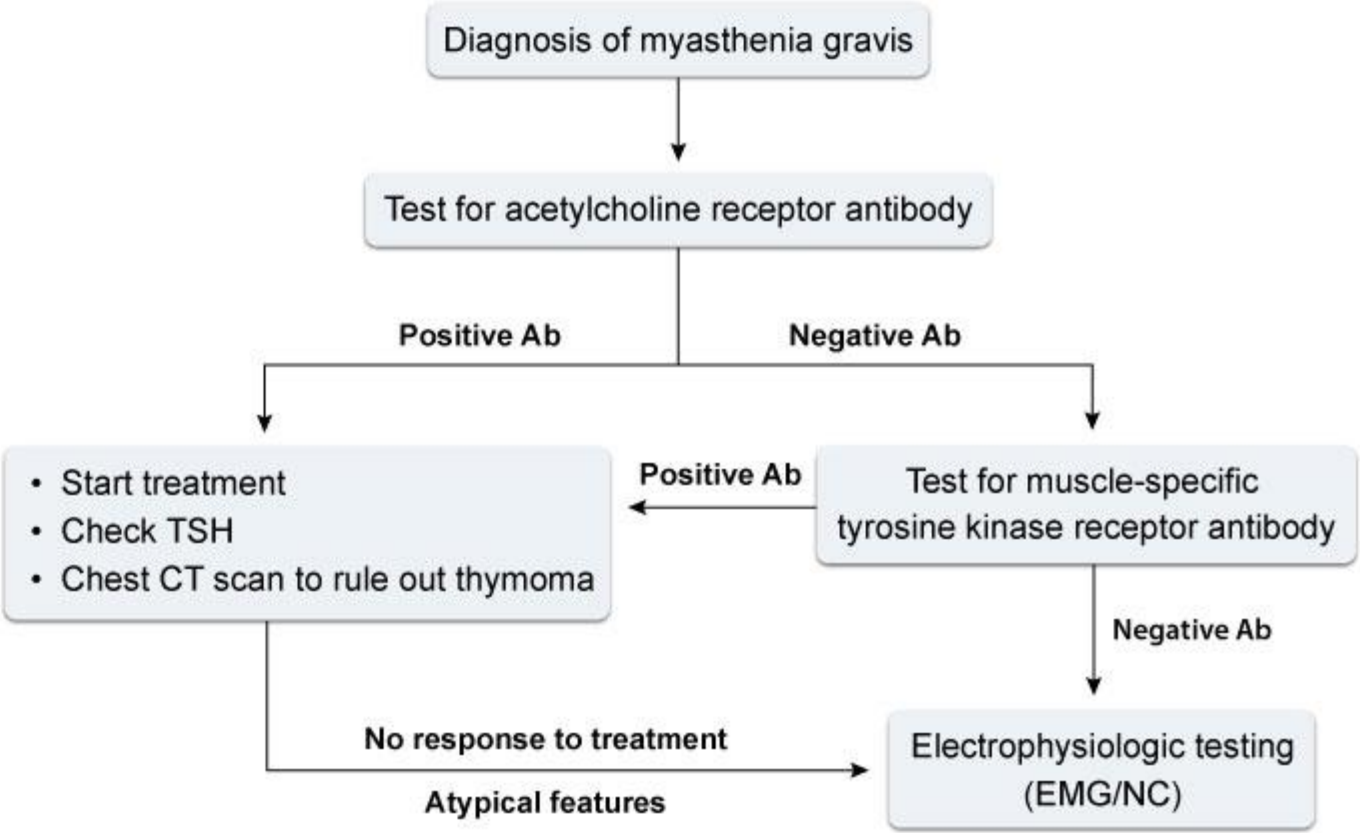
Features of carcinoid syndrome	
Clinical manifestations	• Skin: flushing, telangiectasias, cyanosis • Gastrointestinal: diarrhea, cramping • Cardiac: valvular lesions (right > left side) • Pulmonary: bronchospasm • Miscellaneous: Niacin deficiency (dermatitis, diarrhea & dementia)
Diagnosis	• Elevated 24-hour urinary excretion of 5-HIAA • CT/MRI of abdomen & pelvis to localize tumor • OctreoScan to detect metastases • Echocardiogram (if symptoms of carcinoid heart disease are present)
Treatment	• Octreotide for symptomatic patients **&** prior to surgery/anesthesia • Surgery for liver metastases

Mixed connective tissue disease
Diagnosis • Serologic: Anti-U1-RNP antibodies **PLUS** at least 3 of the following clinical criteria: ◦ Swollen hands or fingers ◦ Myositis or myalgia ◦ Synovitis ◦ Raynaud's phenomenon

Condition	Antibodies to antigen
Drug-induced lupus	DNA-histone complex (95%)
SLE	ds DNA (97%), Sm
Systemic sclerosis	• RNA polymerase II and III • Scl-70/anti-topoisomerase I
MCTD	RNP (100%)
Sjögren's syndrome	SS-A/Ro, SS-B/La
Limited scleroderma	Centromere
Polymyositis, dermatomyositis	Aminoacyl-tRNA synthetases (eg, Jo-1)
Microscopic polyangiitis	Myeloperoxidase
Granulomatosis with polyangiitis (Wegener's)	Proteinase 3 (PR3)

Management of acute diarrhea



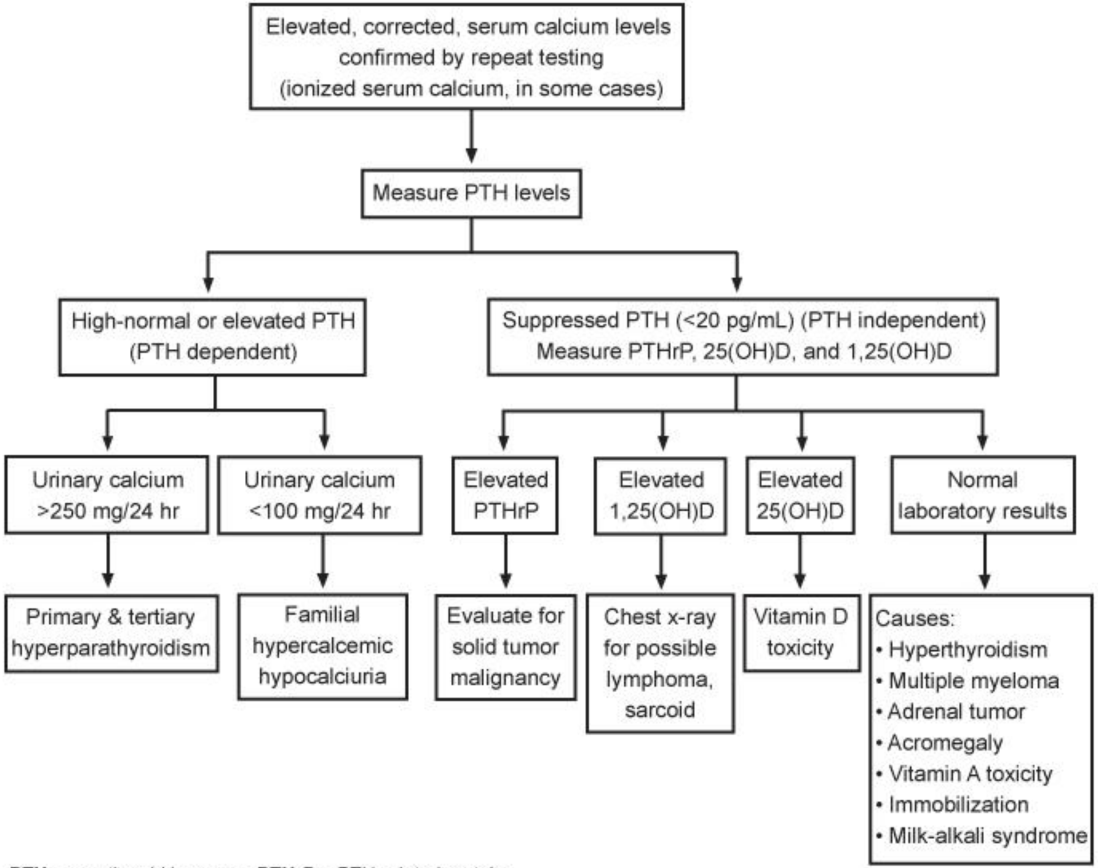


Ab = antibody; **EMG** = electromyogram; **NC** = nerve conduction.

Thalassemias			
	Disorder (genotype)	**Hb electrophoresis**	**Anemia severity**
Alpha thalassemia	Silent carrier (α α / α -)	Normal	Asymptomatic
	Trait (α α / - - **OR** α - / α -)	Normal	Mild symptoms
	Hb H disease (α - / - -)	5%-30% Hb H (adults)	Chronic hemolysis
	Major (fetal hydrops) (- - / - -)	• Hb Barts, Hb Portland & Hb H present • Absent Hb A, Hb F & Hb A2	Fatal in utero
Beta thalassemia	Trait (β / β^{o})	Increased Hb A2	Mild
	Intermediate (β^{+}/ β^{+}, others)	Increased Hb F	Moderate
	Major (β^{o}/ β^{o})	Absent Hb A, only Hb A2 & Hb F present	Severe

	Acute mesenteric ischemia	Chronic mesenteric ischemia
Presentation	• Rapid onset of periumbilical pain (often severe) • Hematochezia is a late complication	• Dull postprandial epigastric pain, food aversion & weight loss (present in 80% of patients)
Risk factors	• Advanced age • Atherosclerosis, atrial fibrillation, congestive heart failure • Hypercoagulable disorders	• Cigarette smoking • Atherosclerosis
Laboratory	• Leukocytosis; elevated serum **lactate**, amylase & phosphate levels; **metabolic acidosis**	• None
Diagnosis	• CT (preferred) or MR angiography • Mesenteric angiography, if diagnosis is unclear after initial imaging	• CT or MR angiography • Duplex ultrasonography • Conventional angiography
Treatment	• Resuscitative measures • Broad-spectrum antibiotics • Nasogastric tube for decompression • Surgery for bowel infarction or perforation	• Surgical reconstruction • Percutaneous transluminal angioplasty +/- stent

Diagnostic approach to hypercalcemia



PTH = parathyroid hormone; **PTHrP** = PTH-related protein.

Features of acute pericarditis	
Etiology	• Viral (most common), bacterial • Iatrogenic (surgery, radiation & drug-mediated) • Connective tissue disease • Post-myocardial infarction • Uremia • Malignancy
Clinical presentation	• Pleuritic chest pain (improved by sitting up & leaning forward) • Systemic symptoms (eg, fever, fatigue) • Pericardial friction rub
Diagnostic studies	• Leukocytosis, elevated ESR & CRP • **ECG**: Diffuse ST elevation & PR depression (except in uremic pericarditis) • **Echocardiogram**: Possible pericardial effusion
Treatment	• NSAIDs & colchicine (viral or idiopathic pericarditis)

ESR = erythrocyte sedimentation rate; **CRP** = C-reactive protein; **NSAID** = nonsteroidal anti-inflammatory drug.

Common RBC abnormalities seen on peripheral smear		
Type	**Appearance**	**Common associated conditions**
Bite cell	• "Bites" secondary to phagocytes extracting denatured hemoglobin (Heinz bodies) precipitates	• α-thalassemia • G6PD deficiency
Schistocyte (helmet cell)	• RBC fragments	• DIC, TTP/HUS, traumatic hemolysis
Spherocyte	• Spheroidal RBC, absent central pallor	• Hereditary spherocytosis • Autoimmune hemolysis
Target cell	• Red blood cell with "bull's eye" extra hemoglobin drop in center	• Thalassemia • HbC disease • Postsplenectomy states • Liver disease (especially obstructive)
Teardrop cell	• Teardrop shaped RBC	• Extramedullary hematopoiesis (e.g., myelofibrosis, major thalassemia)
Basophilic stippling	• Blue granules of various sizes in RBC cytoplasm • Represent ribosomal precipitates	• Thalassemias and iron deficiency • Alcohol abuse • Lead and heavy metal poisoning
Howell-Jolly bodies	• Retained RBC nuclear remnants normally removed by spleen • Usually single, round, dark, purple-red, and peripheral in location	• Asplenia or splenic hypofunction (e.g., sickle cell disease)

Prevention of catheter-related infections	
Catheter location & duration	• Higher risk: lower extremity > upper extremity • Higher risk: femoral > internal jugular > subclavian • Replace peripheral catheter after 72-96 hours • Remove within 24-48 hours if not inserted with sterile technique • Routine replacement not indicated for central vein catheters • PICC line can be used for months • Remove catheter when no longer essential
Catheter material	• Antimicrobial-impregnated catheters and fewer ports may reduce central catheter-related infections
Catheter & site care	• Daily chlorhexidine bathing in ICU patients ↓ central catheter-related infections • Routine hand washing before and after catheter site palpation • Sterile technique and chlorhexidine skin disinfection before catheter insertion • Replace catheter for suspected infection (eg, purulence, fever, and hemodynamic instability) • Do not use guidewire technique to replace

Neurogenic & vascular claudication		
	Neurogenic claudication (pseudoclaudication)	**Vascular claudication**
Symptoms	• Posture-dependent pain • Lumbar extension worsens pain (eg, walking downhill) • Lumbar flexion relieves pain (eg, walking while bent forward) • Lower-extremity numbness & tingling • Lower-extremity weakness • Low back pain	• Exertionally dependent pain • Pain relieved with rest, but not with bending forward while walking • Lower-extremity cramping/tightness • No significant lower-extremity weakness • Possible buttock, thigh, calf, or foot pain
Examination	• Normal pulses • Frequently normal examination	• Decreased pulses • Cool extremities • Decreased hair growth • Pallor with leg elevation
Diagnosis	• MRI of the spine	• Ankle-brachial index

Indications for repeat endoscopy
• Inadequate ulcer biopsy • Patients from endemic areas (eg, Asia, Latin America) • No recent NSAID use • Family history of gastric cancer • Presence of *H pylori* and/or gastric atrophy • Age >50 years • Large ulcers >2 cm • Absent duodenal ulcers (require higher acid secretion, lower risk for gastric cancers)

Epidemiology and extrapulmonary clinical manifestations of endemic mycoses		
Syndrome	**Epidemiology**	**Clinical clues**
Blastomycosis	Mississippi/Ohio River Valley Midwestern United States	• Cutaneous plaques/ulcerations • Bone lesions with sinus tracts • Genitourinary involvement • CNS involvement (very rare)
Histoplasmosis	Mississippi/Ohio River Valley Central and South America	• Hilar/mediastinal lymphadenopathy • Hepatosplenomegaly • Pancytopenia • Adrenal insufficiency
Coccidioidomycosis	Southern Arizona/California Northern Mexico	• Skin lesions • Lymph node involvement • Meningitis • Osteoarticular infection

Treatment of SLE dermatitis

Mild disease/ first-line
- Sunscreens & sun-protective clothing
- **Topical glucocorticoids**

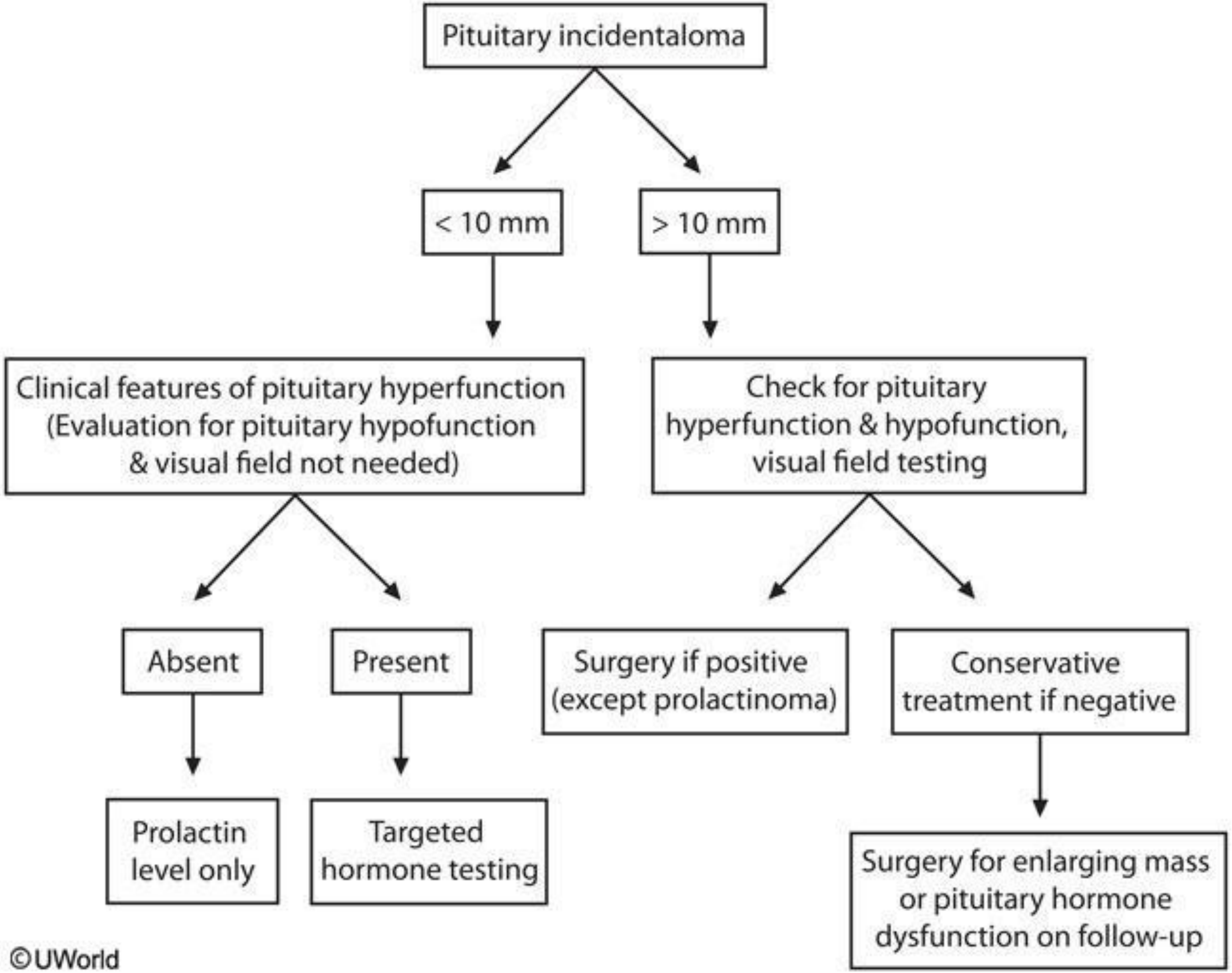
↓

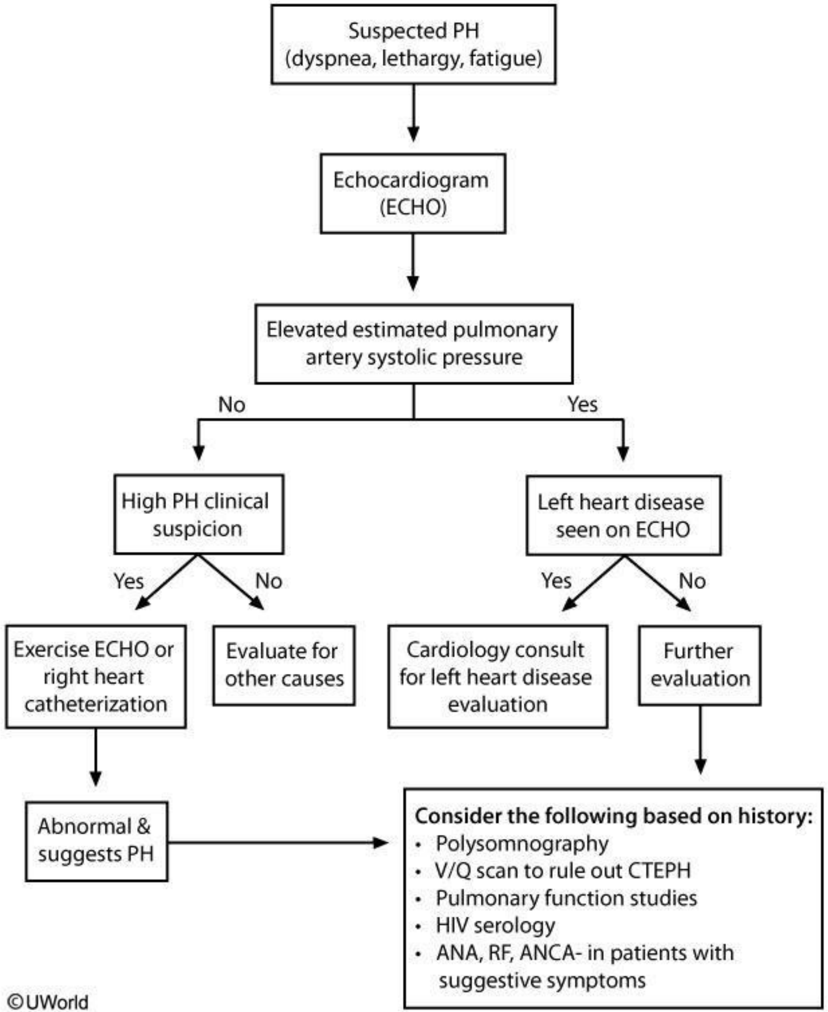
Moderate disease/ second-line
- **Antimalarial drugs**
  Hydroxychloroquine, chloroquine

  ↓
- Low-dose **oral glucocorticoids**
  (eg, prednisone10-15 mg/day)

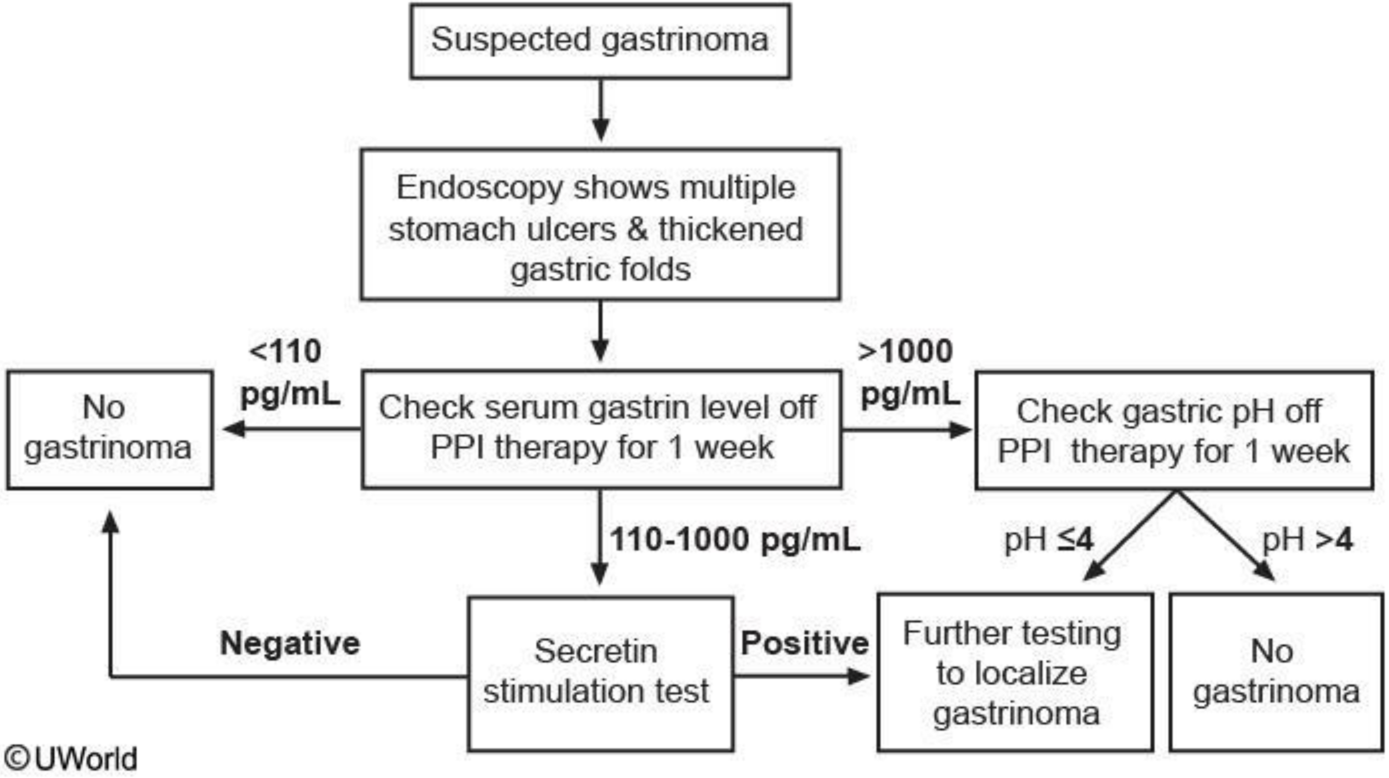
↓

Severe disease/ third-line
- **Immunosuppressive drugs** & retinoids
  - Methotrexate
  - Isotretinoin
  - Azathioprine
  - Cyclophosphamide
  - Cyclosporine

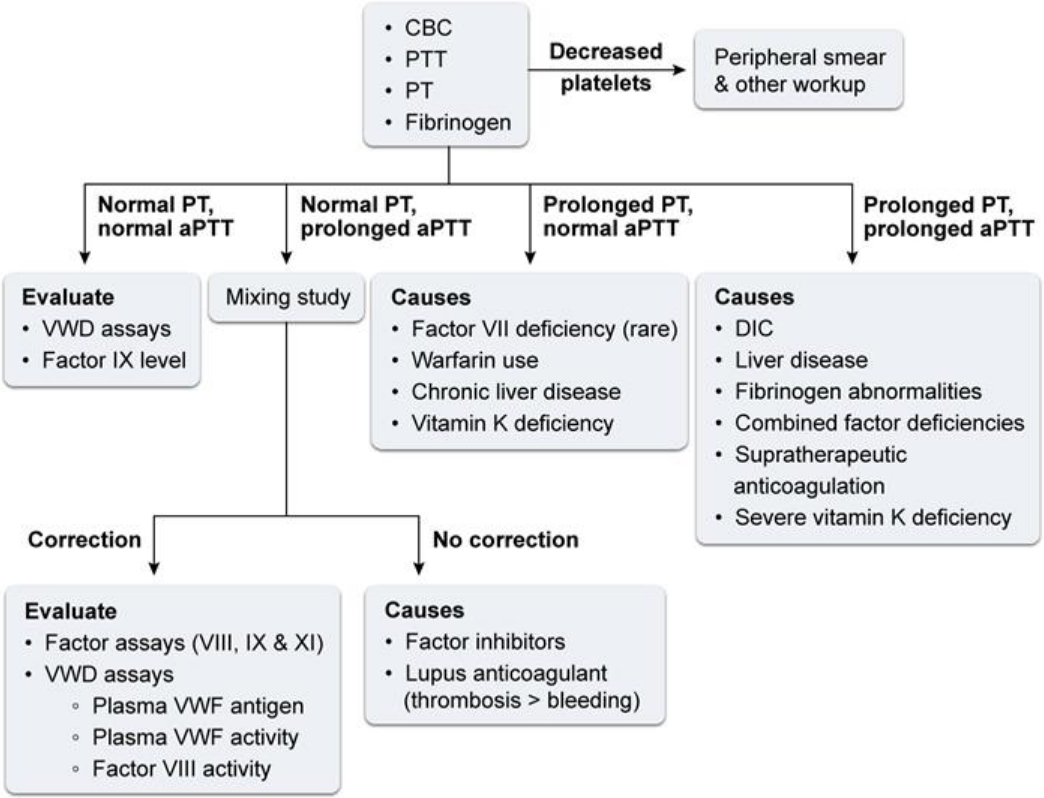

Pituitary incidentaloma
< 10 mm
> 10 mm
Clinical features of pituitary hyperfunction (Evaluation for pituitary hypofunction & visual field not needed)
Check for pituitary hyperfunction & hypofunction, visual field testing
Absent
Present
Surgery if positive (except prolactinoma)
Conservative treatment if negative
Prolactin level only
Targeted hormone testing
Surgery for enlarging mass or pituitary hormone dysfunction on follow-up


Suspected PH
(dyspnea, lethargy, fatigue)
Echocardiogram
(ECHO)
Elevated estimated pulmonary
artery systolic pressure
No
Yes
High PH clinical
suspicion
Left heart disease
seen on ECHO
Yes
No
Yes
No
Exercise ECHO or
right heart
catheterization
Evaluate for
other causes
Cardiology consult
for left heart disease
evaluation
Further
evaluation
Abnormal &
suggests PH
Consider the following based on history:
• Polysomnography
• V/Q scan to rule out CTEPH
• Pulmonary function studies
• HIV serology
• ANA, RF, ANCA- in patients with
suggestive symptoms

Clinical features of dengue fever	
Classic dengue fever	• Flu-like febrile illness with marked myalgias & joint pains ("break-bone fever") • **Retro-orbital pain** • Rash ("white islands in sea of red")
Dengue hemorrhagic fever	• Increased vascular permeability • Thrombocytopenia (<100,000/mm^3) • Spontaneous bleeding → **shock** • **Positive tourniquet test** (petechiae after sphygmomanometer cuff inflation for 5 minutes)
Management	• Supportive care

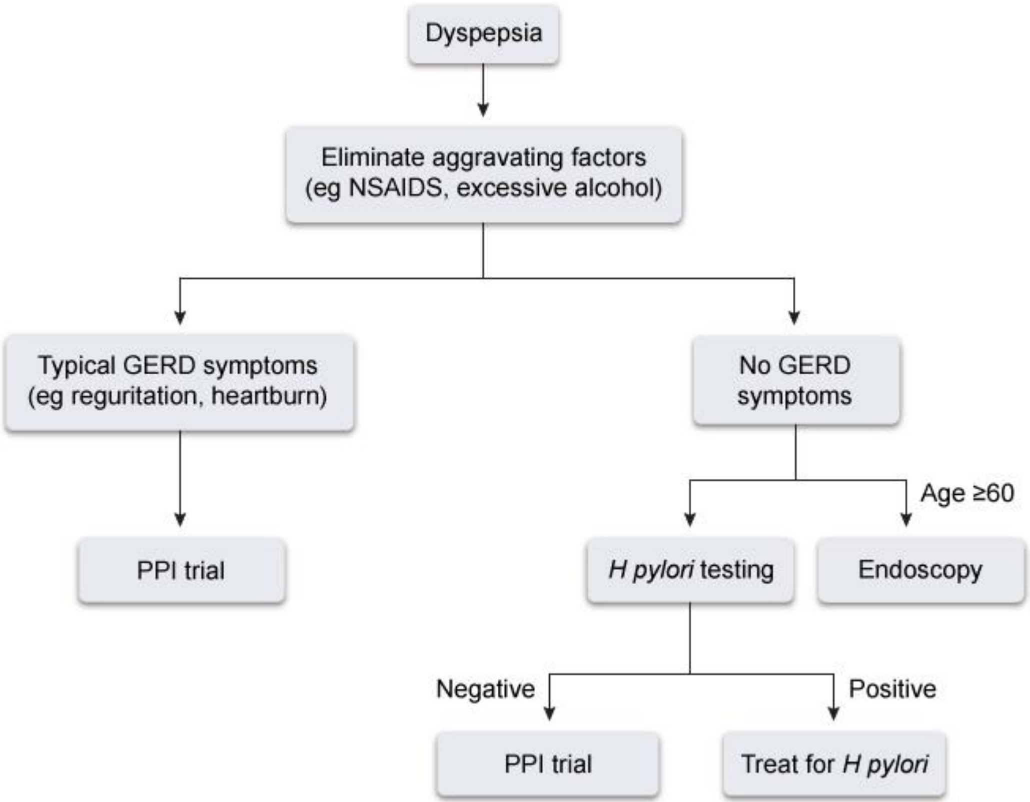

Suspected gastrinoma
Endoscopy shows multiple stomach ulcers & thickened gastric folds
Check serum gastrin level off PPI therapy for 1 week
<110 pg/mL
No gastrinoma
>1000 pg/mL
Check gastric pH off PPI therapy for 1 week
110-1000 pg/mL
pH ≤4
pH >4
Secretin stimulation test
Negative
Positive
Further testing to localize gastrinoma
No gastrinoma
©UWorld

Evaluation of bleeding disorders



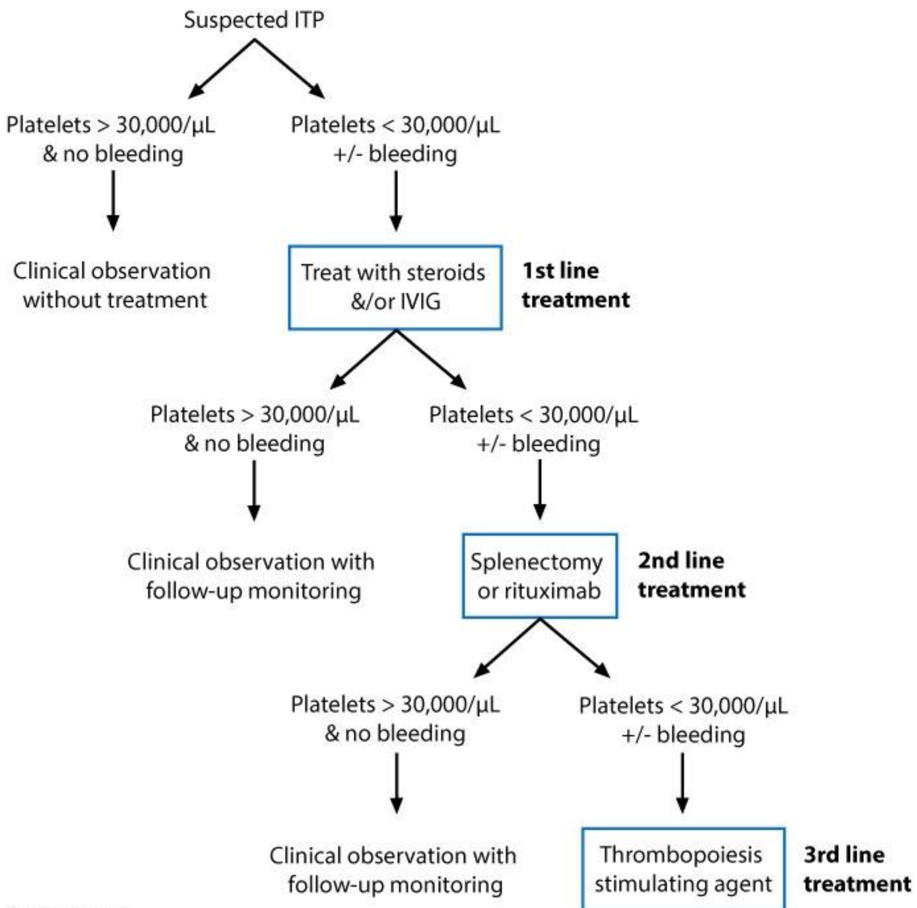
aPTT = activated PTT; **CBC** = complete blood count; **DIC** = disseminated intravascular coagulation; **VWD** = von Willebrand disease; **VWF** = von Willebrand factor.

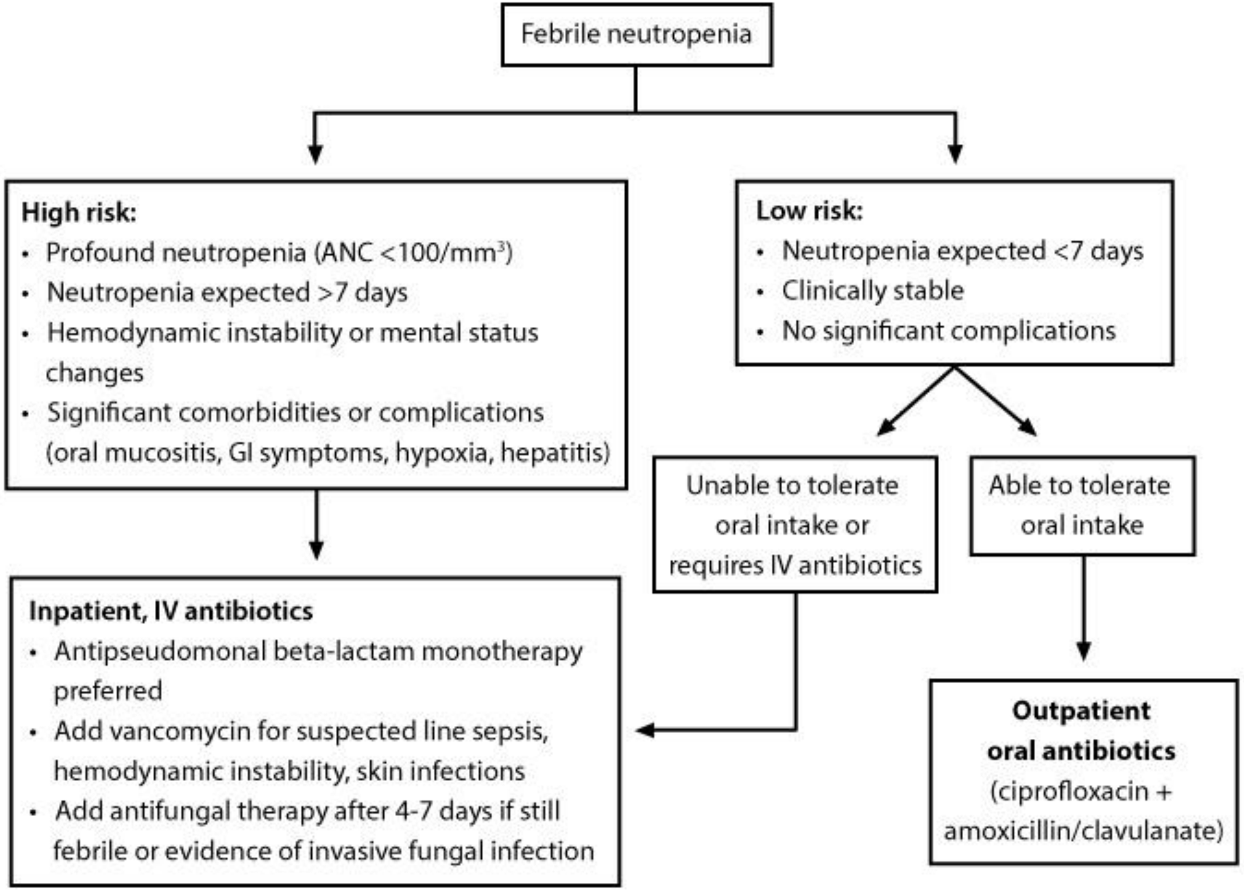
Initial management of dyspepsia



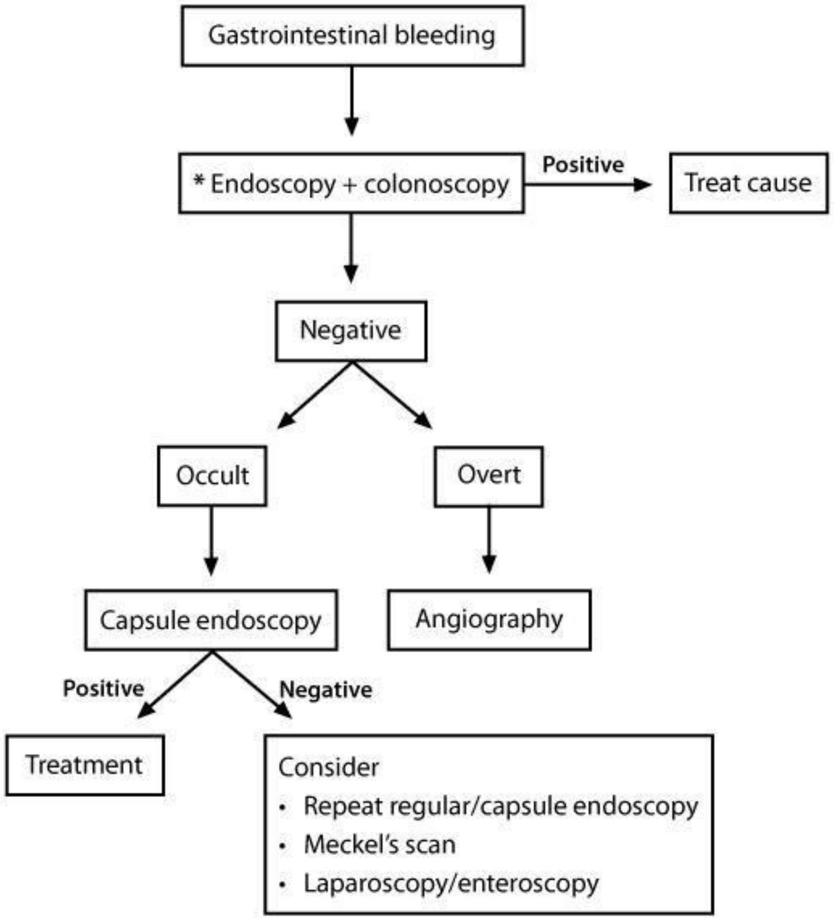
GERD = gastrointestinal reflux disease; *H pylori* = *Helicobacter pylori*;
NSAID = nonsteroidal anti-inflammatory drug; PPI = proton pump inhibitor.

ITP treatment in adults

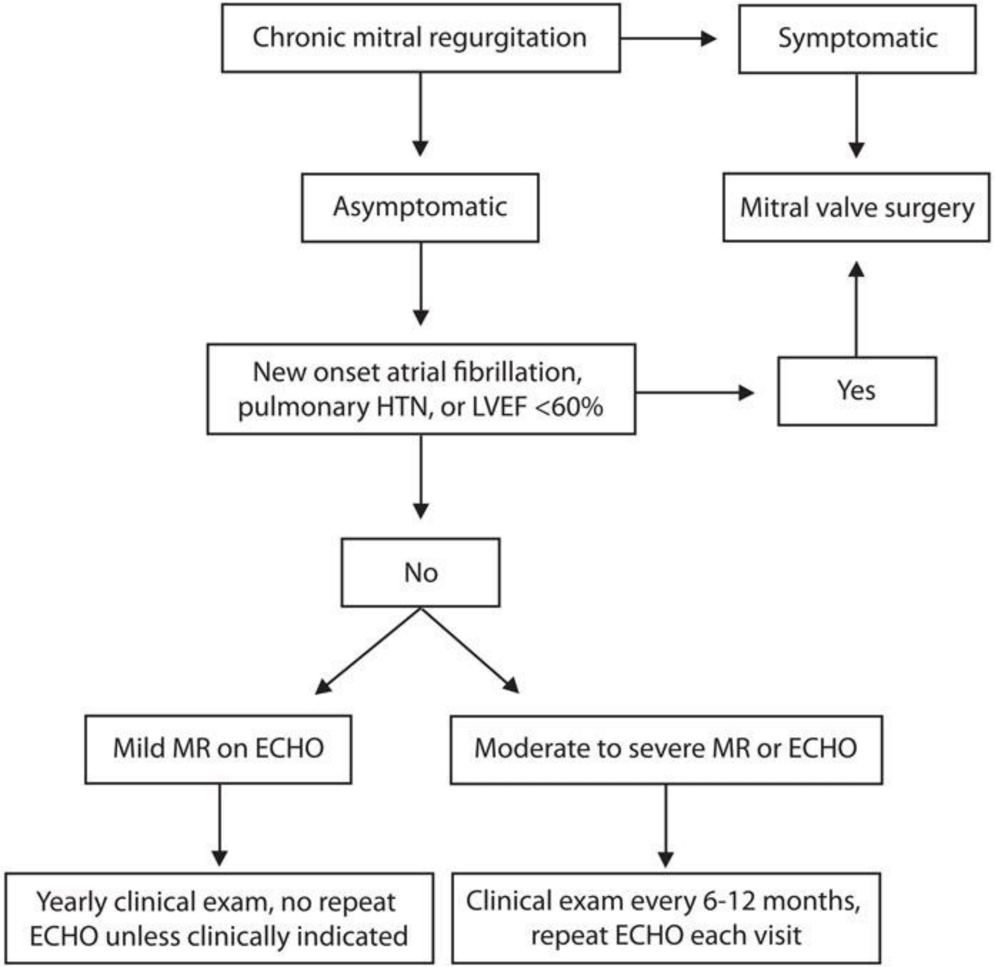




ANC = absolute neutrophil count; **GI** = gastrointestinal; **IV** = intravenous.



* For patients with iron deficiency anemia PLUS positive fecal occult blood test


Chronic mitral regurgitation
Symptomatic
Asymptomatic
Mitral valve surgery
New onset atrial fibrillation, pulmonary HTN, or LVEF <60%
Yes
No
Mild MR on ECHO
Moderate to severe MR or ECHO
Yearly clinical exam, no repeat ECHO unless clinically indicated
Clinical exam every 6-12 months, repeat ECHO each visit

Clinical features of Lemierre's syndrome

Signs/symptoms

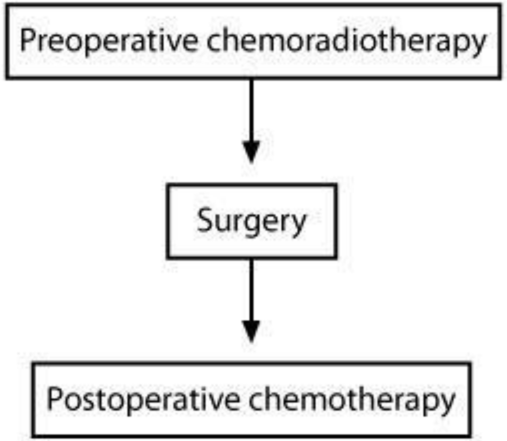
- Recent sore throat
- Persistent fever and systemic symptoms while taking appropriate antibiotic therapy for sore throat
- Neck pain and swelling
- Septic embolism

Laboratory/radiographic findings

- WBC > 15,000 cells/μL
- Chest x-ray and/or CT may show evidence of pulmonary emboli
- Neck CT shows internal jugular thrombophlebitis

Follow-up surveillance of papillary thyroid cancer	
Year 1	• Serum TSH, free T4, thyroglobulin at 6 months after initial therapy • Neck ultrasound 6-12 months after initial therapy • Serum TSH, free T4, thyroglobulin, neck ultrasound at 12 months • Diagnostic radioiodine scan for abnormalities
Years 2-20	• Serum TSH, free T4, thyroglobulin annually • Neck ultrasound every 1-3 years or less frequently in low-risk patients • Diagnostic radioiodine scan for abnormalities
>Year 20	• Serum TSH, free T4, thyroglobulin annually • Neck ultrasound every 3-5 years or less frequently in low-risk patients • Diagnostic radioiodine scan for abnormalities

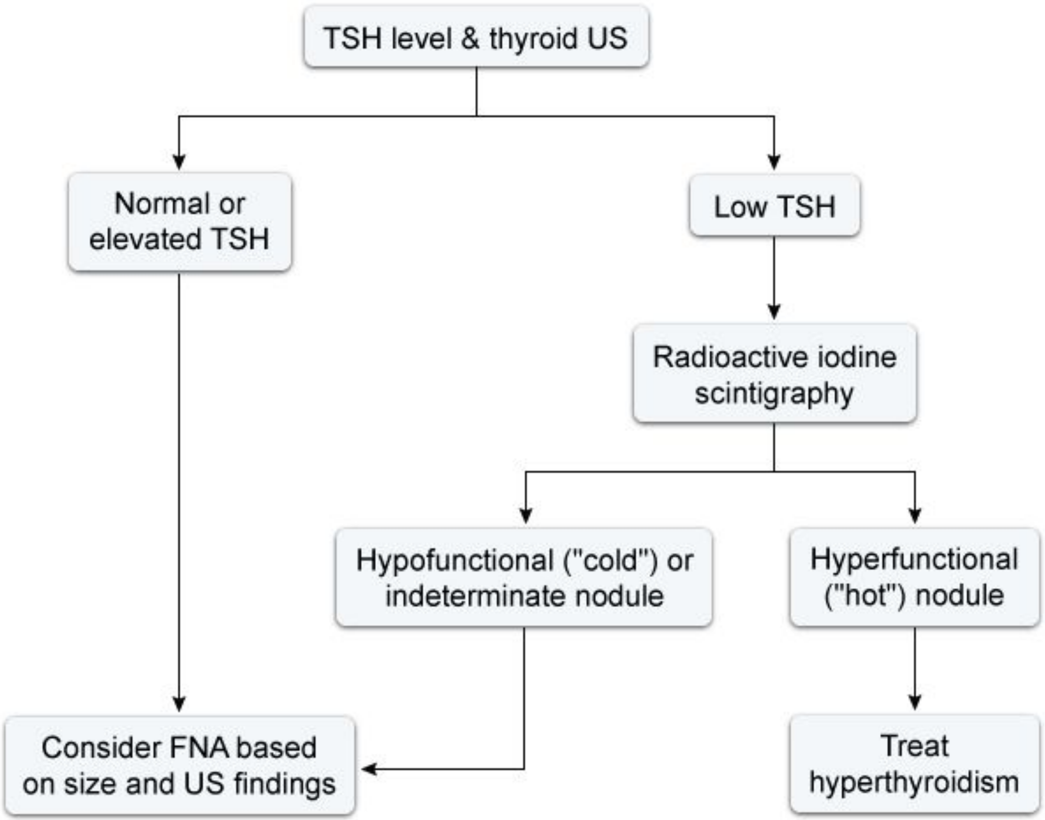
Treatment of locally advanced rectal cancer



Type of hearing loss	Rinne test	Weber test	Possible causes
Conductive	Abnormal in affected ear (bone >air)	Localizes to affected ear	Cerumen impaction, cholesteatoma, otosclerosis, external/middle ear tumors, tympanic membrane rupture, severe otitis media
Sensorineural	Normal in both ears (air >bone)	Localizes to unaffected ear	Meniere disease, acoustic neuroma, presbycusis, ototoxic drugs (eg, aminoglycosides)

Clinical features of multiple sclerosis	
Features suggesting multiple sclerosis	• Onset at age 15-50 • Optic neuritis • Lhermitte's sign • Internuclear ophthalmoplegia • Fatigue • Uhthoff's phenomenon (heat sensitivity) • Sensory symptoms (numbness & paresthesia) • Motor symptoms (paraparesis & spasticity) • Bowel/bladder dysfunction
Disease pattern	• Relapsing-remitting (majority) • Primary progressive • Secondary progressive • Progressive relapsing
Diagnosis	• T2 MRI lesions disseminated in time & space (periventricular, juxtacortical, infratentorial, or spinal cord) • Oligoclonal IgG bands on cerebrospinal fluid analysis

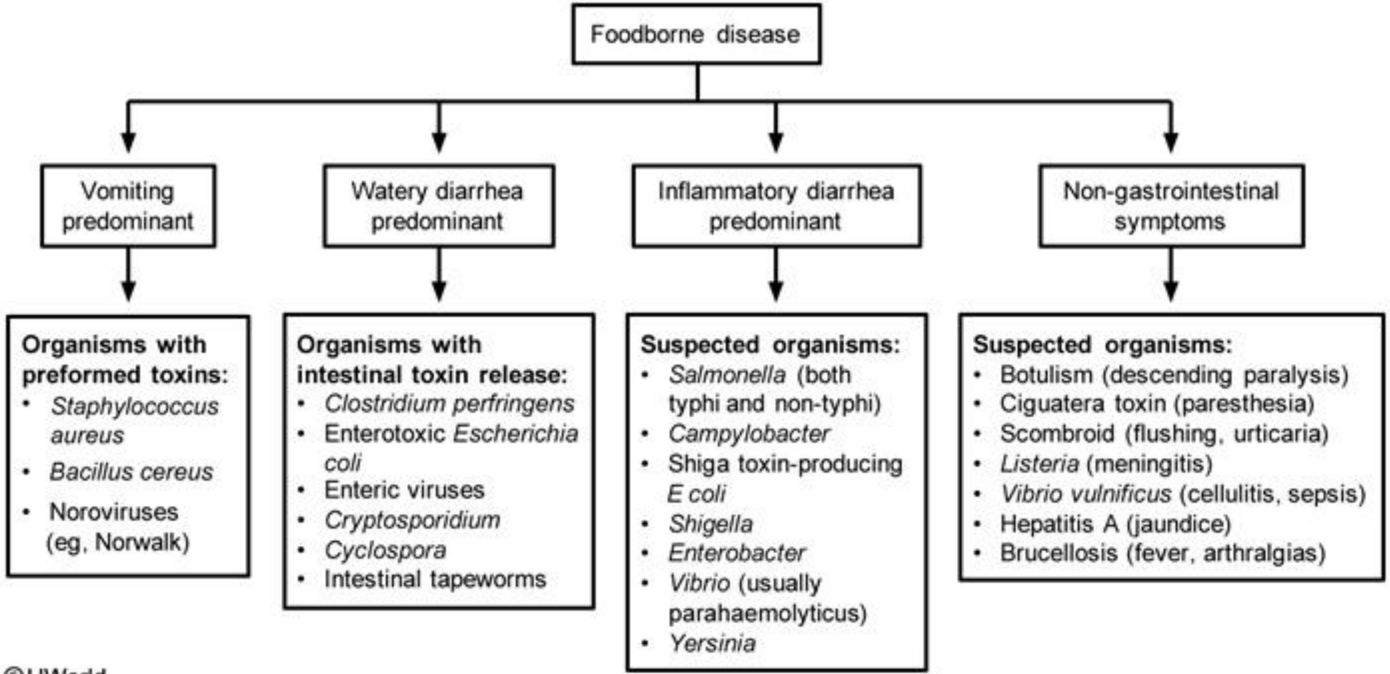
Evaluation of thyroid nodules



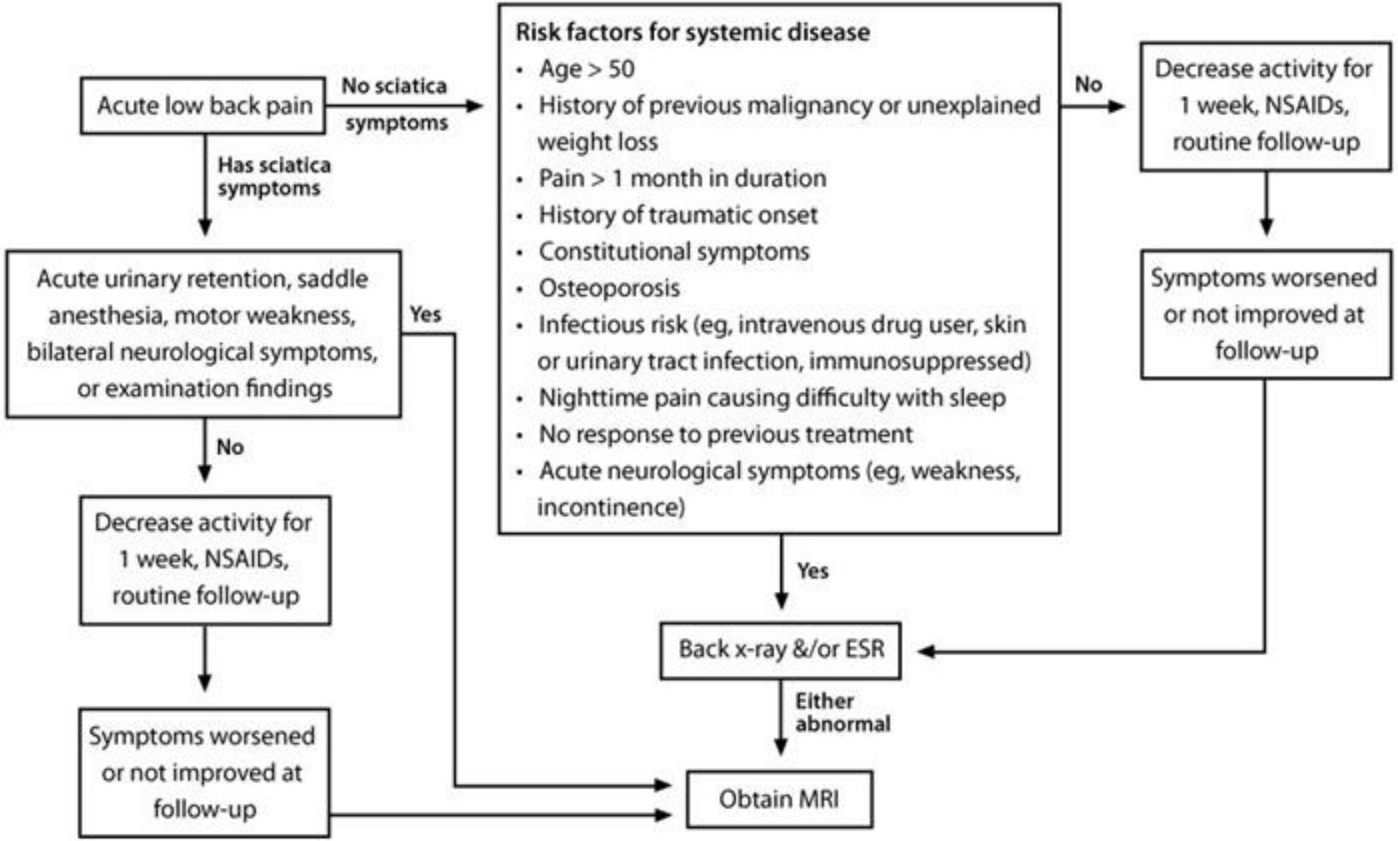
FNA = fine-needle aspiration; **US** = ultrasound.

Parameter	Painless thyroiditis	Graves' disease
Etiology	Preformed thyroid hormone release	Excess thyroid hormone formation
Symptom onset	1-2 months, usually no exophthalmos	Gradual onset, exophthalmos usually present
Goiter	Usually mild to none	Usually present
Thyroglobulin level	High	High
RAIU scan	Low iodine uptake	Markedly increased uptake

Cause of acute scrotal pain	Location of pain	Cremasteric reflex	Clinical features
Testicular torsion	Testis	Absent	• Acute onset of severe pain • Profound testicular swelling • High riding testis • Bell clapper deformity • Ultrasound preferred test
Epididymitis	Epididymis	Present	• Fever, urinary frequency/urgency (acute) • Induration, swelling, and exquisite tenderness of the involved epididymis • Pyuria and/or bacteriuria on urinalysis • Urine culture recommended
Fournier's gangrene	Diffuse scrotum	Present	• Severe pain from anterior abdomen into scrotum and penis • Skin edema, crepitus, blisters/bullae • CT/MRI preferred but urgent surgical evaluation required


Foodborne disease
Vomiting predominant
Watery diarrhea predominant
Inflammatory diarrhea predominant
Non-gastrointestinal symptoms
Organisms with preformed toxins:
• Staphylococcus aureus
• Bacillus cereus
• Noroviruses (eg, Norwalk)
Organisms with intestinal toxin release:
• Clostridium perfringens
• Enterotoxic Escherichia coli
• Enteric viruses
• Cryptosporidium
• Cyclospora
• Intestinal tapeworms
Suspected organisms:
• Salmonella (both typhi and non-typhi)
• Campylobacter
• Shiga toxin-producing E coli
• Shigella
• Enterobacter
• Vibrio (usually parahaemolyticus)
• Yersinia
Suspected organisms:
• Botulism (descending paralysis)
• Ciguatera toxin (paresthesia)
• Scombroid (flushing, urticaria)
• Listeria (meningitis)
• Vibrio vulnificus (cellulitis, sepsis)
• Hepatitis A (jaundice)
• Brucellosis (fever, arthralgias)

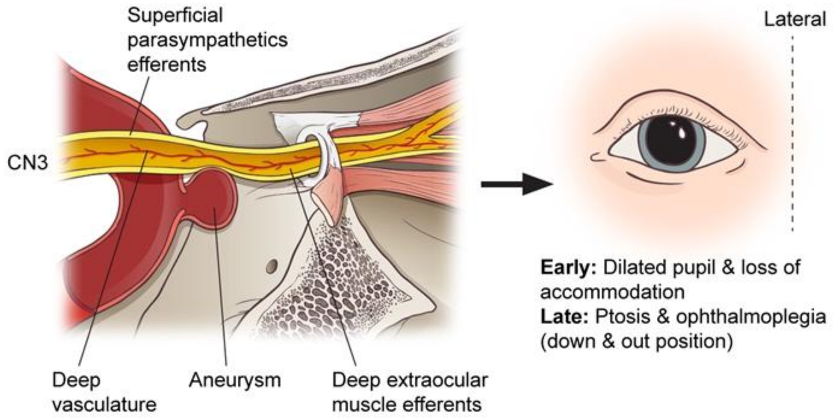
Management of lower back pain



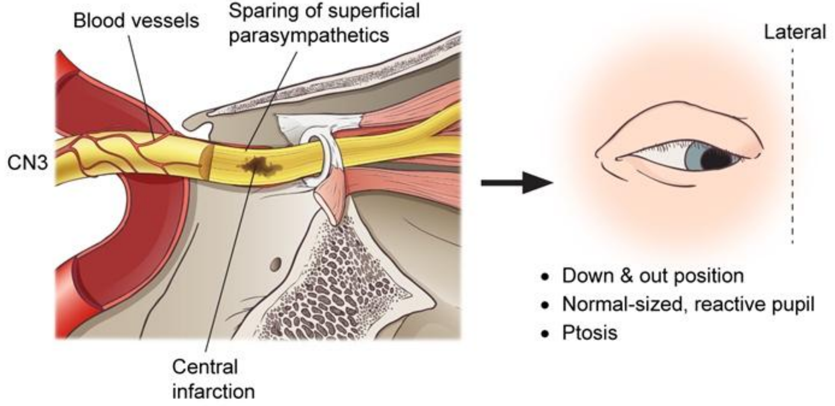
Clinical features of hypothermia	
Classification	**Mild:** 32-35 C (90-95 F) • Tachycardia, tachypnea • Ataxia, dysarthria, **increased shivering** **Moderate:** 28-32 C (82-90 F) • Bradycardia, lethargy, hypoventilation, **decreased shivering**, atrial arrhythmias **Severe:** <28 C (82 F) • Coma, cardiovascular collapse, ventricular arrhythmias
Treatment	**General** • Warmed (42 C [107 F]) crystalloid for hypotension • Endotracheal intubation in comatose patients **Rewarming techniques** • **Mild hypothermia**: Passive external warming (remove wet clothing, cover with blankets) • **Moderate hypothermia**: Active external warming (warm blankets, heating pads, warm baths) • **Severe hypothermia**: Active internal rewarming (warmed pleural or peritoneal irrigation, warmed humidified oxygen)

Micronutrient deficiencies after gastric bypass surgery	
Deficiency	**Complications/presentation**
Vitamin B12	• Macrocytic anemia, cognitive defects, peripheral neuropathy
Fat-soluble vitamins (A, D, E & K)	• **Vitamin A**: Night/complete blindness, poor bone growth, skin rashes, immune deficiencies • **Vitamin D**: Bone disease, secondary hyperparathyroidism • **Vitamin K**: Coagulopathy
Trace minerals (iron, zinc, copper & selenium)	• **Iron:** Anemia • **Zinc**: Alopecia, night blindness, skin rashes • **Copper:** Fragile hair, skin depigmentation, muscle weakness, ataxia, neuropathy, osteoporosis
Calcium	• Osteopenia, metabolic bone disease, secondary hyperparathyroidism
Folate	• Macrocytic anemia
Thiamine	• Possible intractable vomiting and Wernicke's encephalopathy

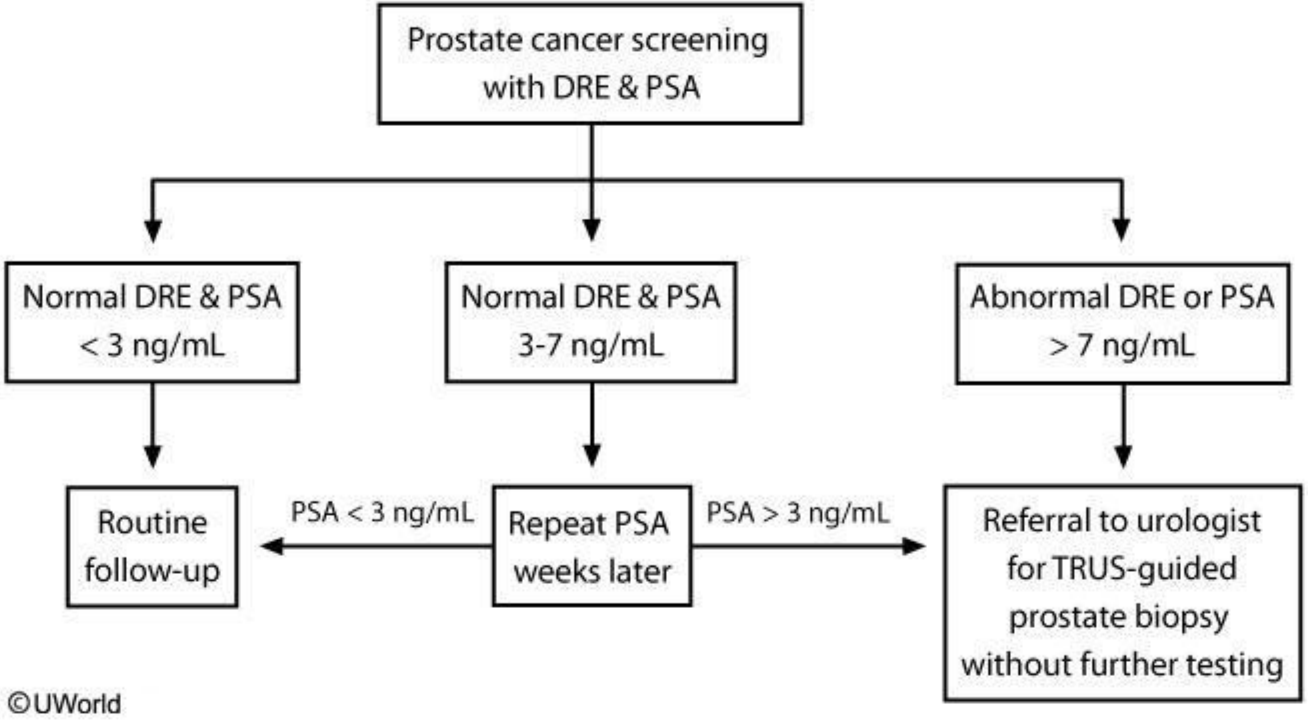
Aneurysmal compression of oculomotor nerve



Diabetic ophthalmoplegia



Management of elevated PSA

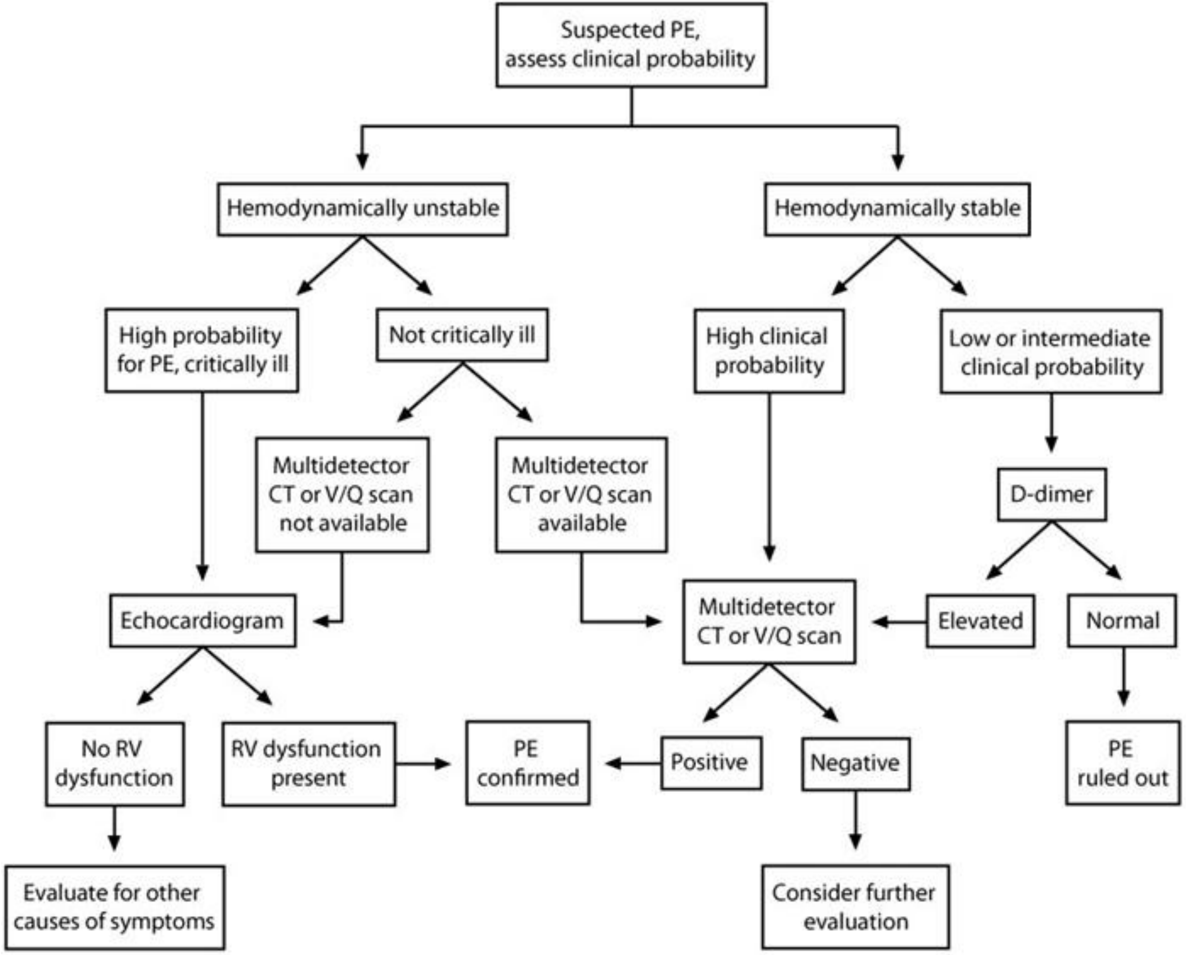


Secondary causes of premenopausal osteoporosis
• Hyperthyroidism • Hyperparathyroidism • Vitamin D and/or calcium deficiency • GI malabsorption (eg, celiac sprue, IBD) • Cushing's syndrome • Estrogen deficiency (eg, premature ovarian failure) • Rheumatoid arthritis • Medications (eg, steroids, chronic heparin, phenytoin) • Chronic kidney or liver disease • Hypercalciuria • Alcoholism

GI = gastrointestinal; **IBD** = inflammatory bowel disease.

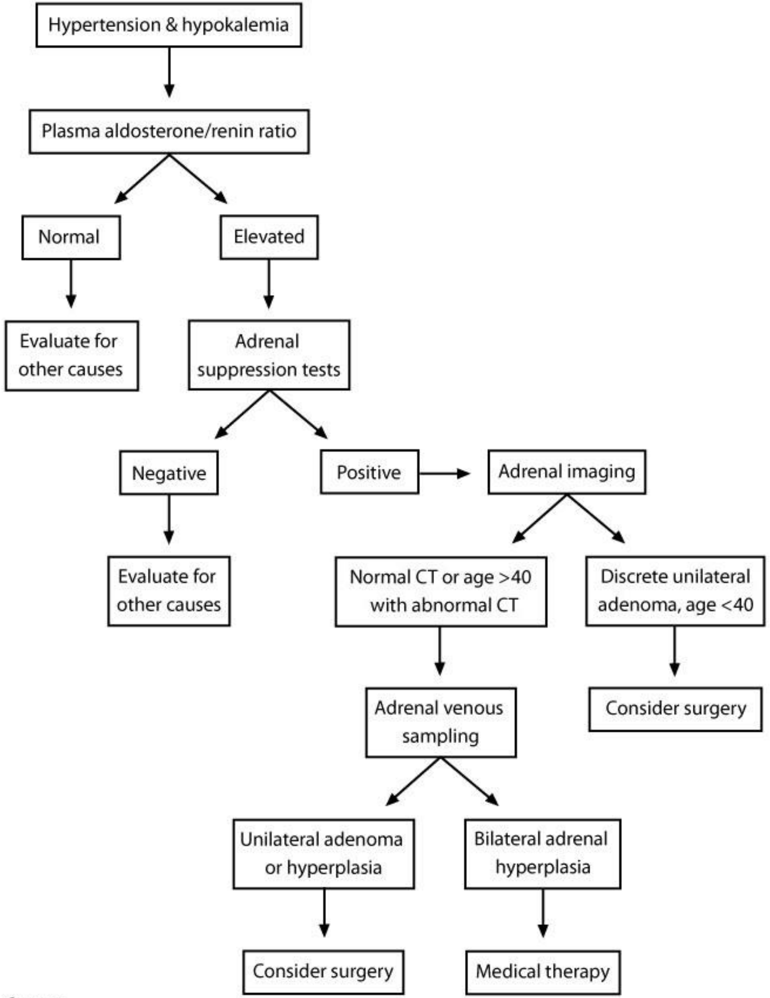
Screening criteria for latent autoimmune diabetes of adulthood (LADA)
• Age of onset >35 but <50 • Acute onset of symptoms • BMI <25 kg/m2 • Personal or family history of autoimmune disease Two or more of the above criteria is associated with 90% sensitivity and 70% specificity for LADA

Clinical features of irritable bowel syndrome	
Rome diagnostic criteria	**Recurrent abdominal pain/discomfort ≥3 days/month for the past 3 months & ≥2 of the following:** • Symptom improvement with bowel movement • Change in frequency of stool • Change in form of stool
Warning signs/ symptoms	**Signs/symptoms suggesting etiologies other than IBS** • Rectal bleeding • Nocturnal (awakens from or prevents sleep) or worsening abdominal pain • Weight loss • Abnormal laboratory findings (eg, anemia & electrolyte disorders)
Classification	**IBS with constipation** • Firm/lumpy stools ≥25%, loose/fluid stools <25% **IBS with diarrhea** • Firm/lumpy stools <5%, loose/fluid stools ≥25% **Mixed IBS** • Firm/lumpy stools ≥25%, loose/fluid stools ≥25% **Undifferentiated IBS** • Insufficient abnormality of stool to categorize
Evaluation	**IBS with diarrhea** • Stool cultures • Celiac disease screening • 24-hour stool collection • Colonoscopy or flexible sigmoidoscopy & biopsy **IBS with constipation** • Radiography • Flexible sigmoidoscopy & colonoscopy



Suspected PE, assess clinical probability
Hemodynamically unstable
Hemodynamically stable
High probability for PE, critically ill
Not critically ill
High clinical probability
Low or intermediate clinical probability
Multidetector CT or V/Q scan not available
Multidetector CT or V/Q scan available
D-dimer
Echocardiogram
Multidetector CT or V/Q scan
Elevated
Normal
No RV dysfunction
RV dysfunction present
PE confirmed
Positive
Negative
PE ruled out
Evaluate for other causes of symptoms
Consider further evaluation

Indications for specialized RBC treatments	
Irradiated	• Bone marrow transplant (BMT) recipients • Acquired or congenital cellular immunodeficiency. • Blood components donated by first or second degree relatives
Leukoreduced	• Chronically transfused patients • CMV seronegative at-risk patients (e.g., AIDS, transplant patients) • Potential transplant recipients • Previous febrile nonhemolytic transfusion reaction
Washed	• IgA deficiency • Complement-dependent autoimmune hemolytic anemia • Continued allergic reactions (e.g., hives) with red cell transfusion despite antihistamine treatment



Hypertension & hypokalemia
Plasma aldosterone/renin ratio
Normal
Elevated
Evaluate for other causes
Adrenal suppression tests
Negative
Positive
Adrenal imaging
Evaluate for other causes
Normal CT or age >40 with abnormal CT
Discrete unilateral adenoma, age <40
Adrenal venous sampling
Consider surgery
Unilateral adenoma or hyperplasia
Bilateral adrenal hyperplasia
Consider surgery
Medical therapy

Clinical features of HSV encephalitis	
Symptoms/signs	• Rapid onset (<1 week) of fever & headache • Focal neurologic deficits (eg, cranial nerve palsies, hemiparesis, ataxia) • Mental status changes & seizures
Imaging/laboratory results	• CSF may be HSV PCR positive & show lymphocytic pleocytosis • CT or MRI may be normal or show temporal lobe abnormalities • EEG with high amplitude slow waves in >80% of patients

CSF = cerebrospinal fluid; **EEG** = electroencephalogram; **HSV** = herpes simplex virus;
PCR = polymerase chain reaction

Ottawa ankle rules

X-ray of the *ankle* is required if:

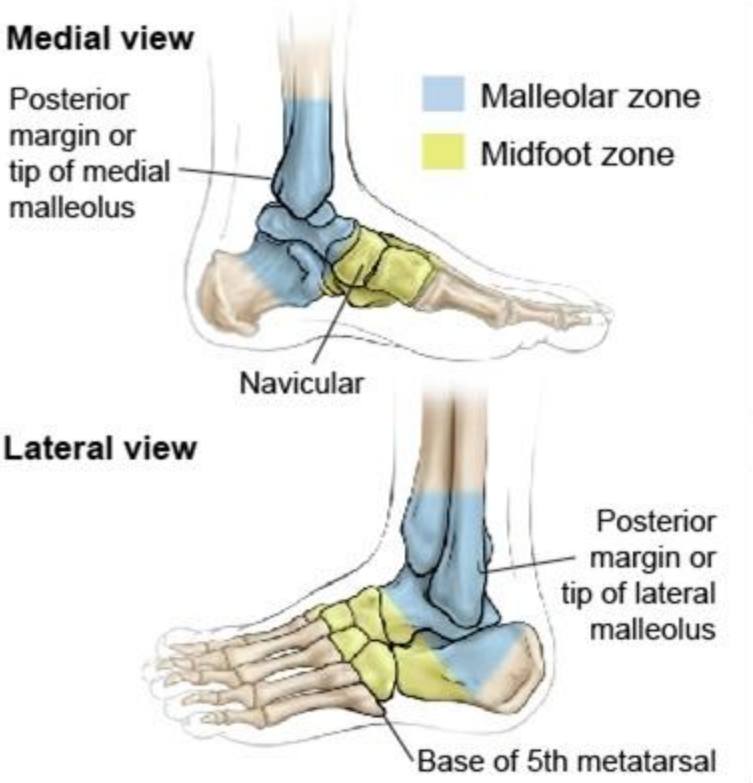
Pain at the *malleolar* zone **and**

- Tender at posterior margin/tip of medial malleolus **OR**
- Tender at posterior margin/tip of lateral malleolus **OR**
- Unable to bear weight 4 steps (2 on each foot)

X-ray of the *foot* is required if:

Pain at the *midfoot* zone **and**

- Tender at the navicular **OR**
- Tender at the base of the 5th metatarsal **OR**
- Unable to bear weight 4 steps (2 on each foot)



Features of brachial plexopathy in cancer patients	
Suggestive of cancer involvement	• Severe pain in brachial plexus • Pain occurs early in symptom course • Affects lower brachial plexus • +/- Horner's syndrome
Suggestive of radiation injury	• Less painful • Pain occurs later in symptom course • Affects upper brachial plexus • More paresthesias and weakness • Fasciculations and myokymia on EMG

Evaluation of male hypogonadism

Specific symptoms/signs of hypogonadism
(sexual dysfunction/infertility, loss of body hair, gynecomastia, hot flashes, low bone density, testicular atrophy)

↓

Low serum testosterone confirmed on repeat testing
- Consider free testosterone/SHBG

↓

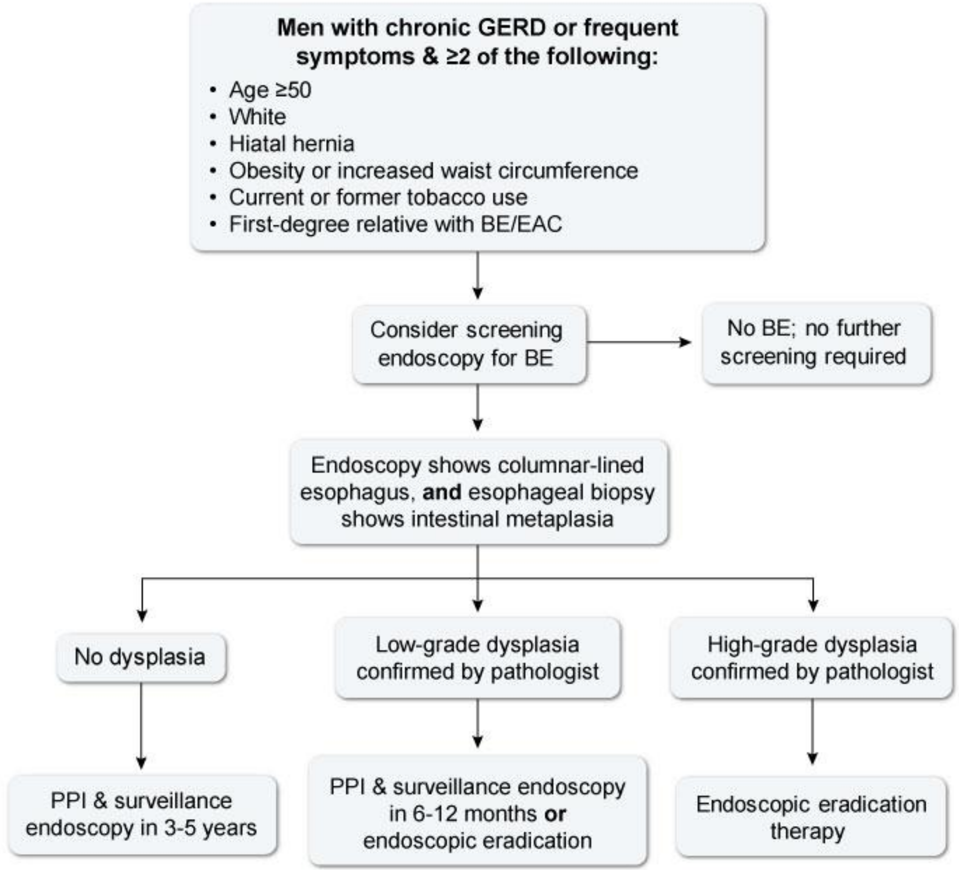
Check LH & FSH

Elevated → **Primary hypogonadism**
- Consider karyotype analysis

Low/normal → **Secondary hypogonadism**
- Consider additional testing (prolactin, iron studies, other pituitary hormones, MRI of pituitary)

SHBG = sex hormone-binding globulin.

Evaluation & management of Barrett esophagus



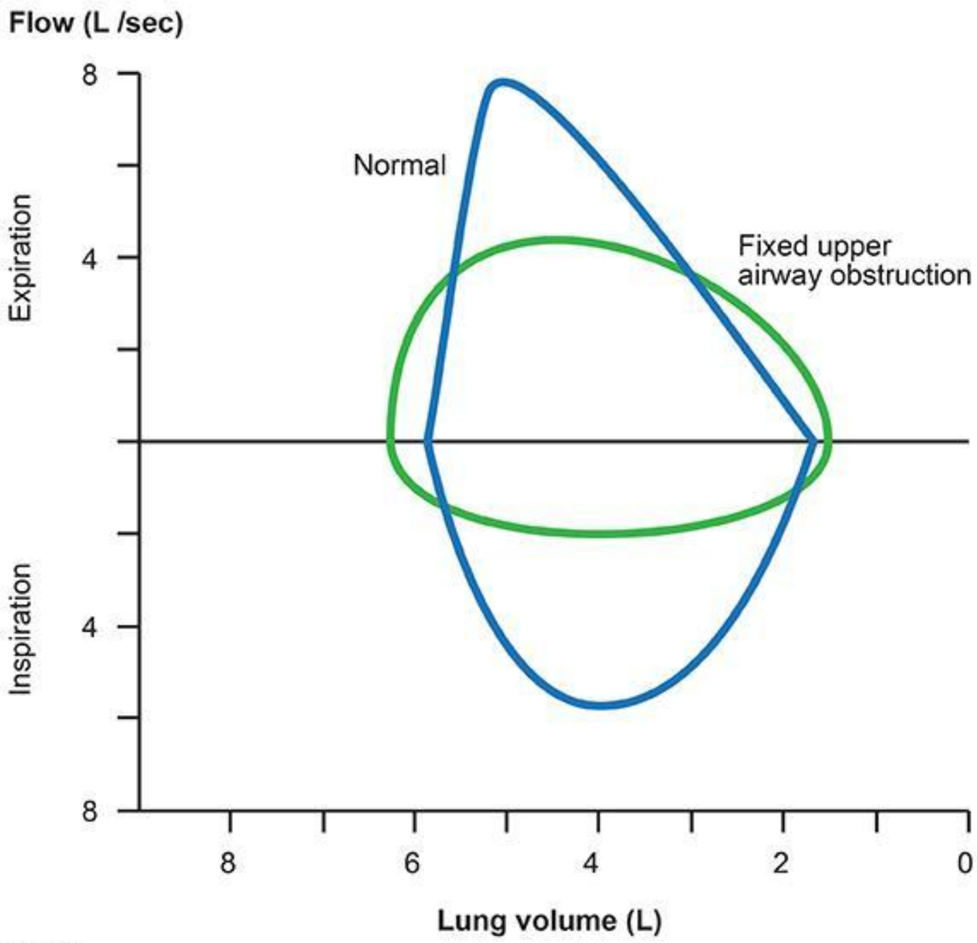
BE = Barrett esophagus; **EAC** = esophageal adenocarcinoma;
GERD = gastroesophageal reflux disease; **PPI** = proton pump inhibitor.

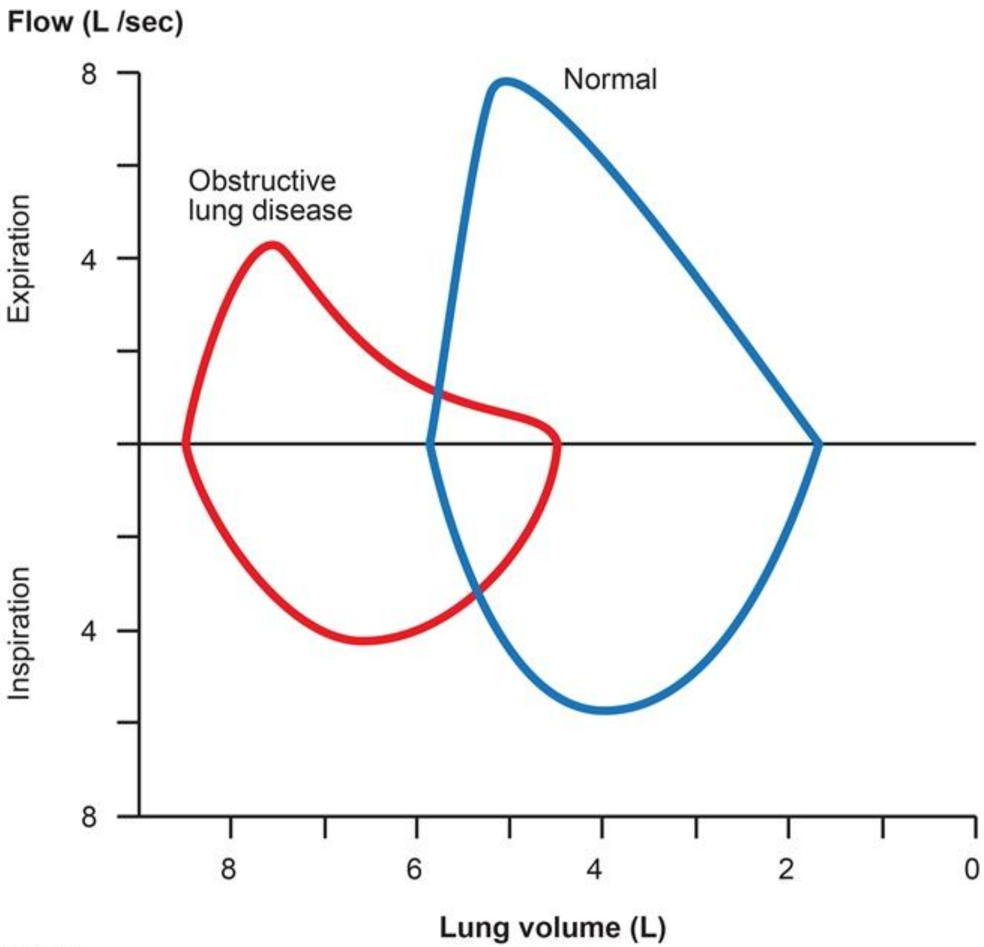
Classification of multiple endocrine neoplasia (MEN)	
Type I	• Primary hyperparathyroidism (>90%) • Enteropancreatic tumors (60%-70%) • Pituitary tumors (10%-20%)
Type 2A	• MTC (>90%) • Pheochromocytoma (40%-50%) • Parathyroid hyperplasia (10%-20%)

Type 2B	• MTC
	• Pheochromocytoma
	• Other
	  ◦ Mucosal & intestinal neuromas
	  ◦ Marfanoid habitus

Stage	Clinical features
Early	• Pain in the low back, flank, or abdomen (usually dull) • Weight loss, malaise, anorexia • Claudication due to arterial compression • Gross hematuria; possible testicular pain • IVC obstruction leading to edema or DVT
Late	• Progressive ureteral obstruction (usually mid and distal) • Oliguria, flank pain
Laboratory findings	• Elevated ESR and CRP • Elevated ANA (up to 60% of patients) • Anemia • Normal urine sediment (usually)
Imaging	• Ultrasound showing a poorly-marginated periaortic mass (usually hypoechoic) with hydronephrosis • CT scan with a periaortic retroperitoneal mass, possible IVC compression, and lymphadenopathy

Retroperitoneal fibrosis syndrome


Flow (L /sec)
8
4
4
8
Expiration
Inspiration
Normal
Fixed upper
airway obstruction
8
6
4
2
0
Lung volume (L)

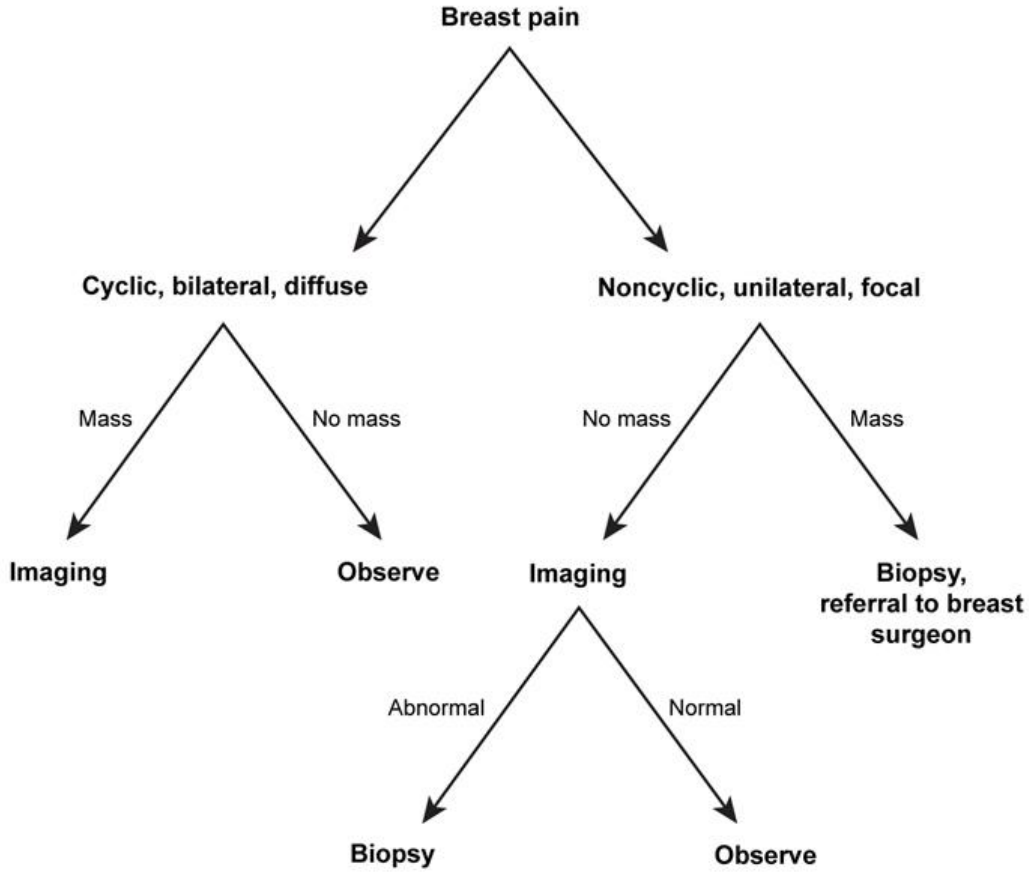


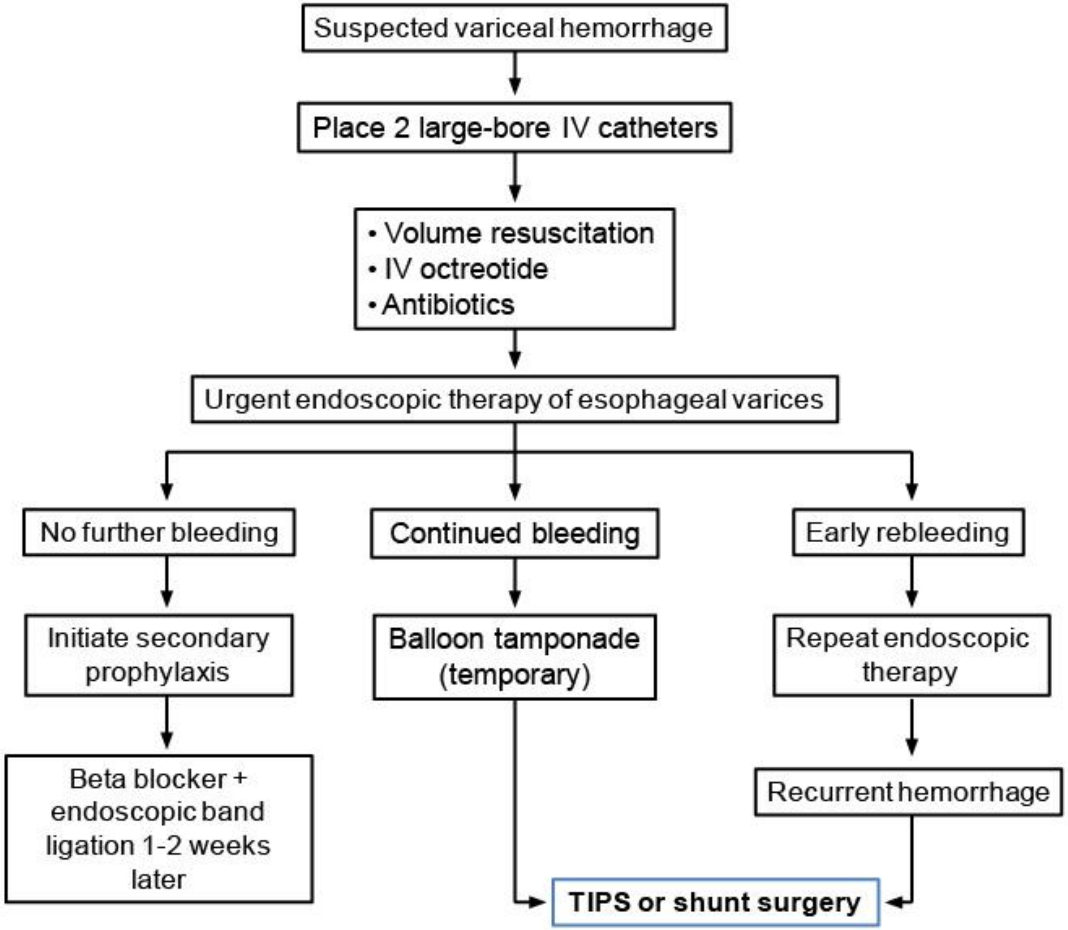
Flow (L /sec)
8
4
4
8
Expiration
Inspiration
Normal
Obstructive
lung disease
8
6
4
2
0
Lung volume (L)

Clinical features of nocardiosis	
Clinical presentation	**Pulmonary** • Fever, cough, dyspnea, hemoptysis, or chest pain • Commonly disseminates from lungs to CNS **Cutaneous** • Cellulitis, ulceration, abscess, lymphocutaneous lesions, or mycetoma **CNS** • Headache, focal neurologic deficits, seizures, or meningismus
Diagnosis	• Acid-fast, aerobic filamentous branching Gram-positive rod • Lung imaging: nodular infiltrates, consolidation, or abscesses • Head CT: multiple parenchymal lesions/abscesses
Treatment	• Trimethoprim-sulfamethoxazole (TMP-SMX) – First-line • Amikacin • Imipenem • Third-generation cephalosporins **Severe infections require combination of medications**

Features of HIV-negative MAC		
	Known underlying lung disease	**No known underlying lung disease**
Patient risk factors	• Usually middle-aged & elderly Caucasian men with COPD or alcohol abuse	• Usually non-smoking women > age 50
Symptoms	• Cough, weight loss, night sweats, and possible diarrhea	• Cough with purulent sputum, no fever or weight loss
Chest x-ray	• Upper lobe infiltrate/cavities similar to *M. tuberculosis* may be present	• Interstitial infiltrates or right middle lobe/lingular infiltrate • Nodules or bronchiectasis may be present

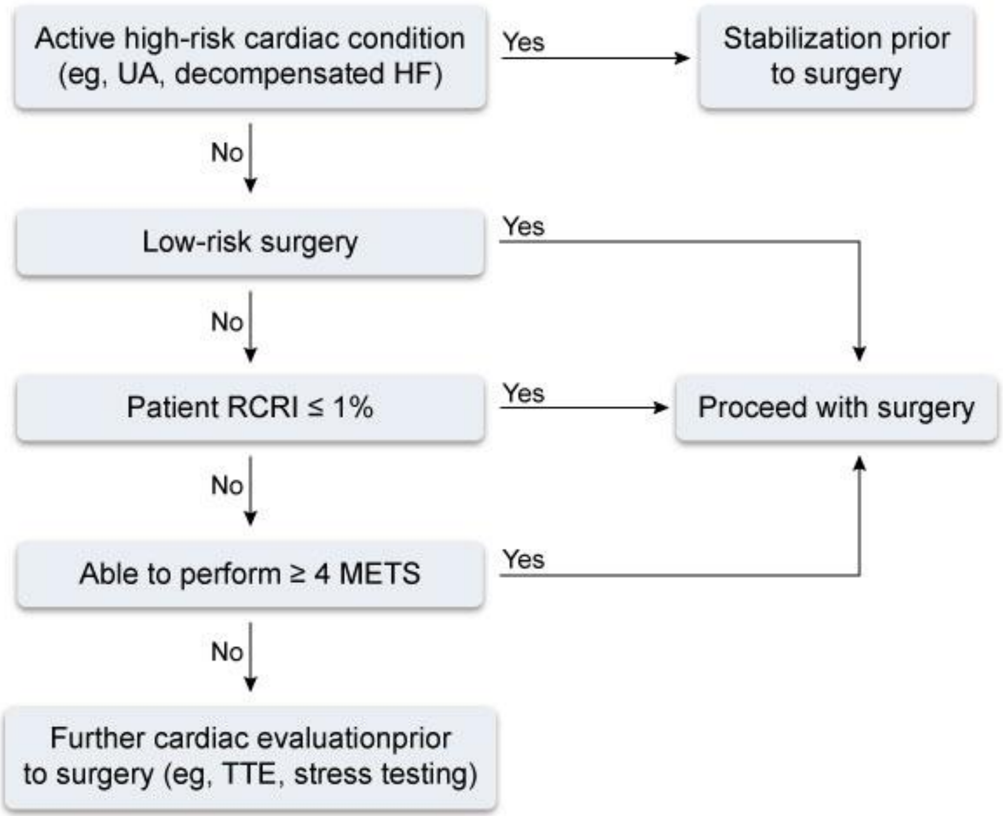
Management of breast pain





IV = intravenous; **TIPS** = transjugular intrahepatic portosystemic shunt.

Preoperative cardiac evaluation for noncardiac surgery



HF = heart failure; **RCRI** = revised cardiac risk index; **TTE** = transthoracic echocardiogram; **UA** = unstable angina.

Aspirin-exacerbated respiratory disease	

Clinical features	• Samter's triad
	  ○ Chronic rhinosinusitis with nasal polyposis
	  ○ Asthma
	  ○ Aspirin (or NSAID) sensitivity
	• Ingestion can result in acute asthma exacerbation, profuse rhinorrhea, conjunctival injection, and erythematous flushing of the face
	• Symptoms begin within 30 minutes to 3 hours after ingestion

Treatment	• Avoid NSAIDs, if possible • Add leukotriene receptor antagonist (eg, montelukast, zafirlukast) • Consider aspirin desensitization if aspirin use is required for other medical conditions

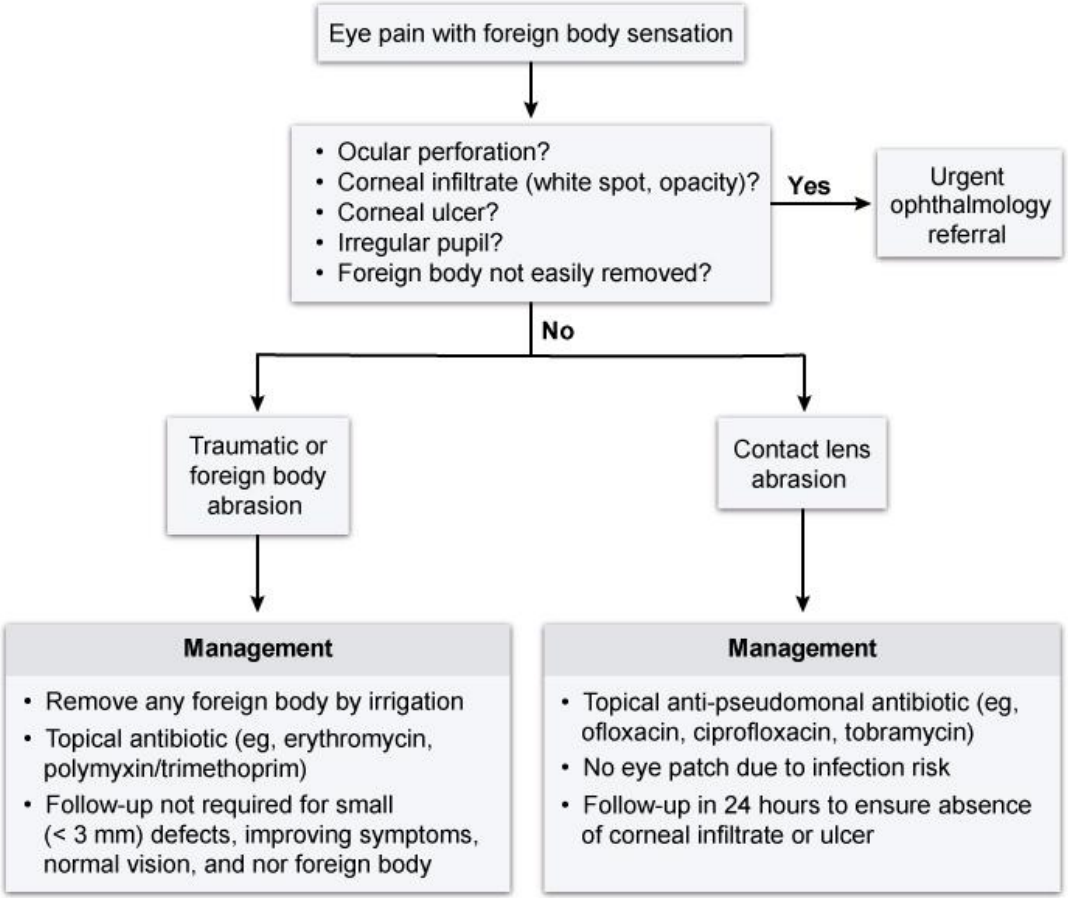
Treatment of syphilis		
Stage	**Preferred treatment**	**Alternate treatment (penicillin allergy)**
Early syphilis (primary, secondary, or latent <1 year)	Benzathine penicillin G 2.4 million units IM x 1 dose	Doxycycline x 2 weeks
Tertiary or late syphilis (gummas, cardiovascular complications, or latent >1 year/unknown duration)	Benzathine penicillin G 2.4 million units IM x 3 weekly doses	Doxycycline x 4 weeks
Neurosyphilis	Penicillin G 3-4 million units IV every 6 hours x 2 weeks	Ceftriaxone

IM = intramuscular; **IV** = intravenous.

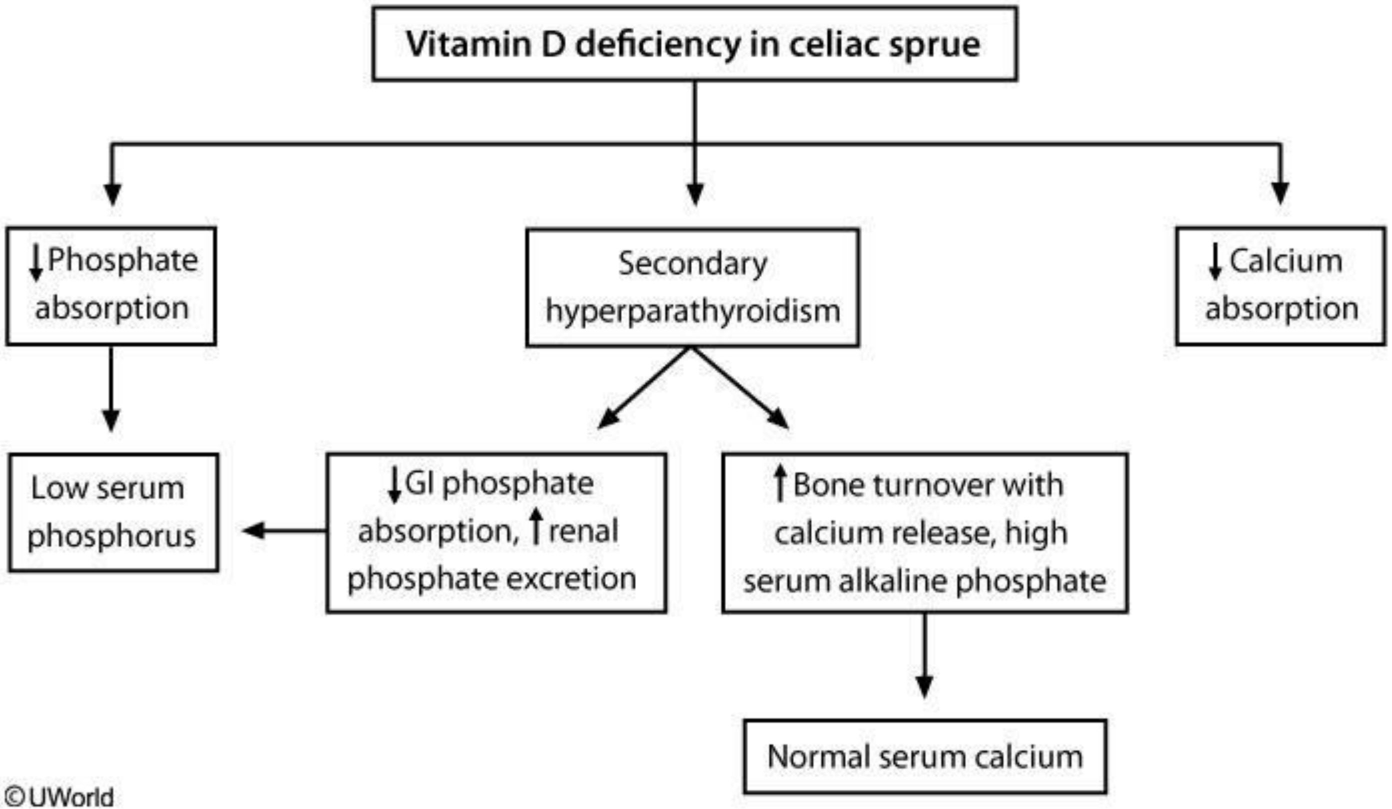
Clinical manifestations of amyloidosis (primary & secondary)	
Renal	• Heavy **proteinuria** with nephrotic syndrome • Peripheral edema
Cardiac	• **Restrictive cardiomyopathy** • Conduction defects, low voltage
CNS	• Peripheral &/or autonomic neuropathy • Stroke
GI	• Hepatomegaly • Dysmotility, malabsorption • GI bleeding due to vascular fragility
Pulmonary	• Pulmonary nodules, tracheobronchial infiltration, pleural effusions
Musculoskeletal	• Enlarged tongue, shoulder pad enlargement
Skin	• Thickening of skin, subcutaneous nodules/plaques • Ecchymoses, periorbital purpura
Heme	• Anemia, thrombocytopenia

CNS = central nervous system; **GI** = gastrointestinal.

Management of corneal abrasion

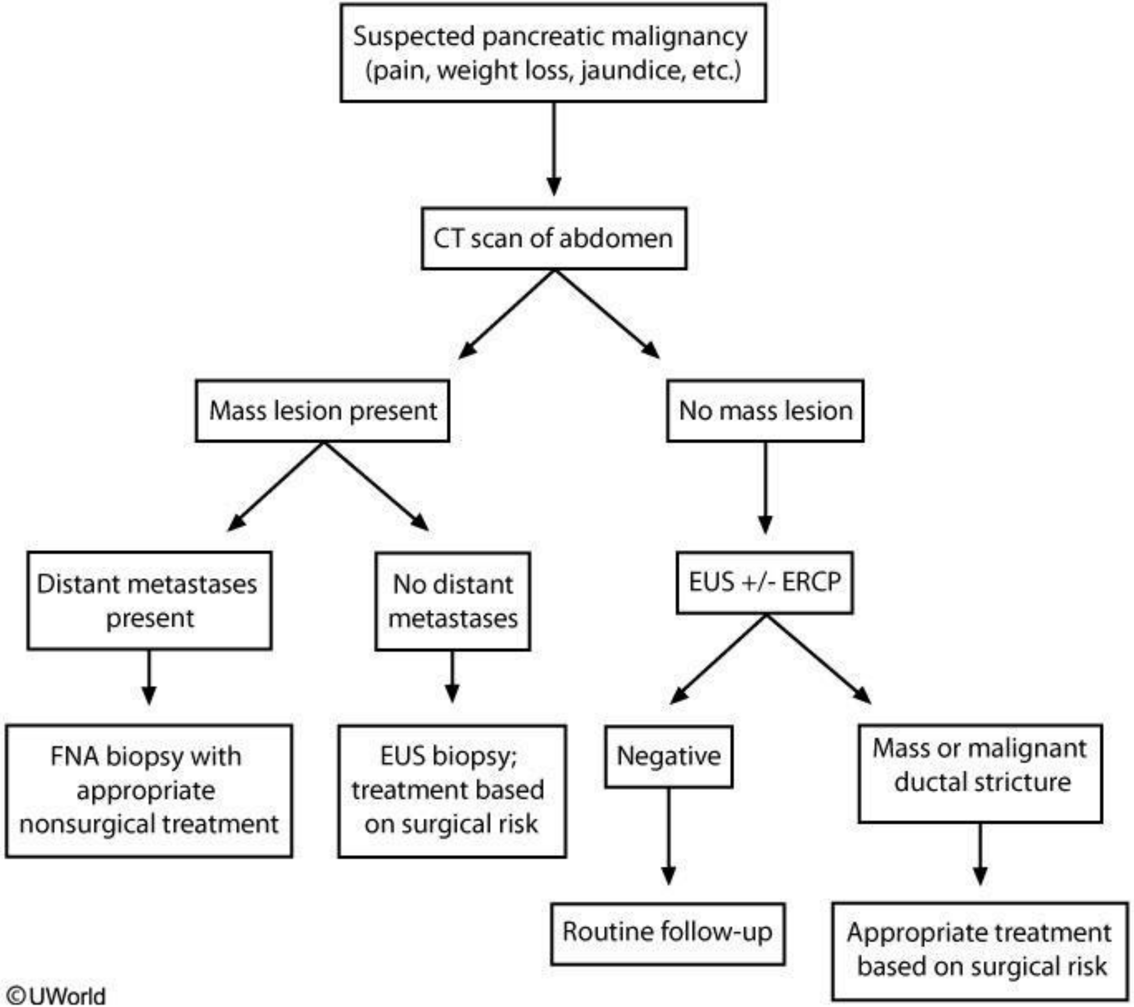


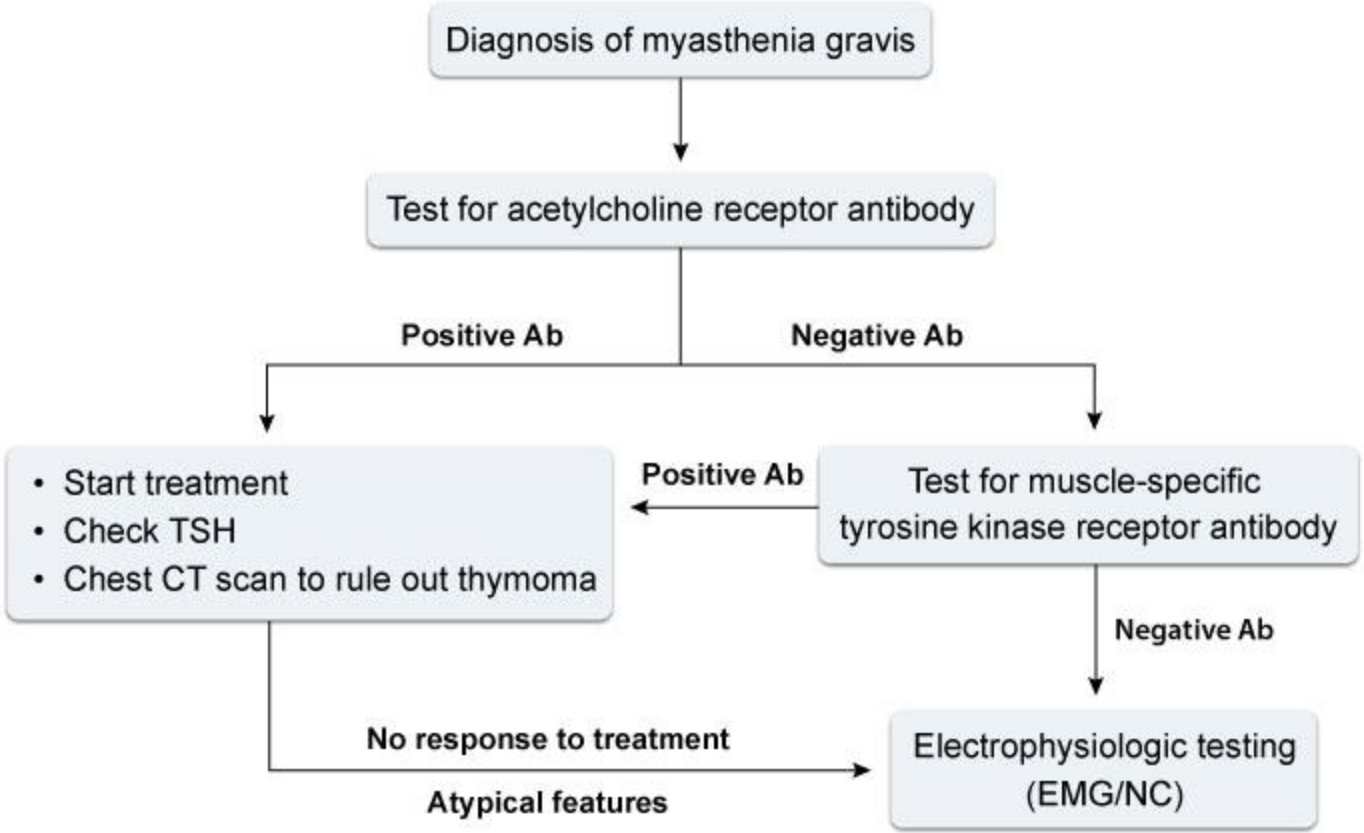
Sensitivity of anti-neutrophil cytoplasmic antibodies (ANCA) in active vasculitis			
	c-ANCA (anti-PR3)	**p-ANCA (anti-MPO)**	**c-ANCA or p-ANCA**
GPA	50-95%	5-23%	85-90%
Microscopic polyangiitis	8-26%	50-85%	65-95%
Churg-Strauss syndrome	30%	30%	50-60%



Vitamin D deficiency in celiac sprue
↓Phosphate absorption
Secondary hyperparathyroidism
↓Calcium absorption
Low serum phosphorus
↓GI phosphate absorption, ↑renal phosphate excretion
↑Bone turnover with calcium release, high serum alkaline phosphate
Normal serum calcium

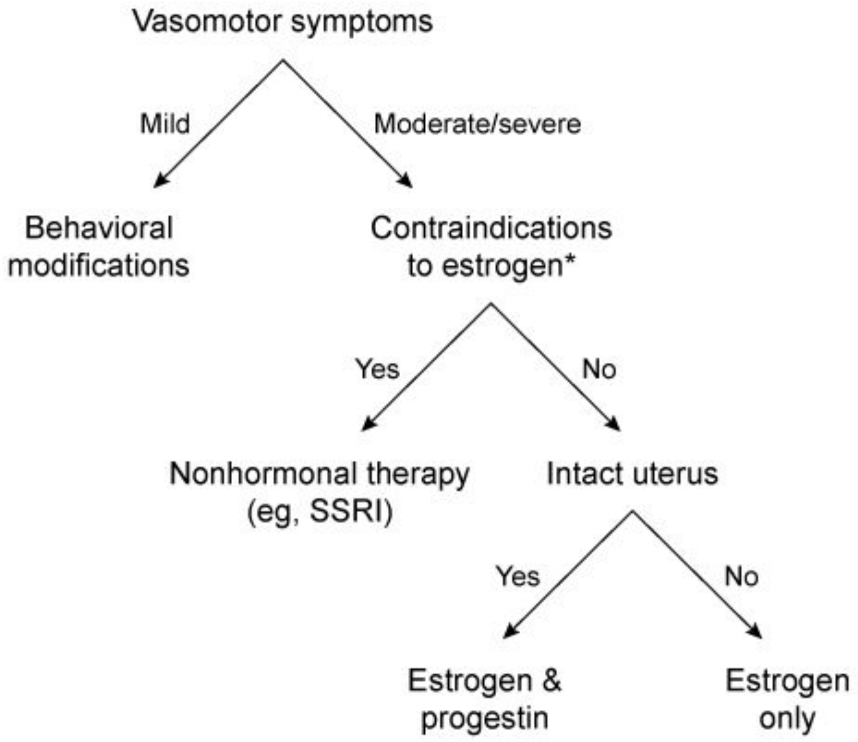
Clues for increasing index of suspicion for *Legionella* pneumonia	
Exposure to possibly contaminated water	• Recent travel (especially cruise or hotel stay) within the previous 2 weeks • Contaminated potable water in hospitals/nursing homes
Clinical clues	• Fever >39 C (102.2 F) • Bradycardia relative to high fever • Neurological symptoms (especially confusion) • Gastrointestinal symptoms (especially diarrhea) • Unresponsive to beta-lactam & aminoglycoside antibiotics
Laboratory clues	• Hyponatremia • Hepatic dysfunction • Hematuria & proteinuria • Sputum Gram stain showing many neutrophils, but few or no microorganisms


Suspected pancreatic malignancy (pain, weight loss, jaundice, etc.)
CT scan of abdomen
Mass lesion present
No mass lesion
Distant metastases present
No distant metastases
EUS +/- ERCP
FNA biopsy with appropriate nonsurgical treatment
EUS biopsy; treatment based on surgical risk
Negative
Mass or malignant ductal stricture
Routine follow-up
Appropriate treatment based on surgical risk



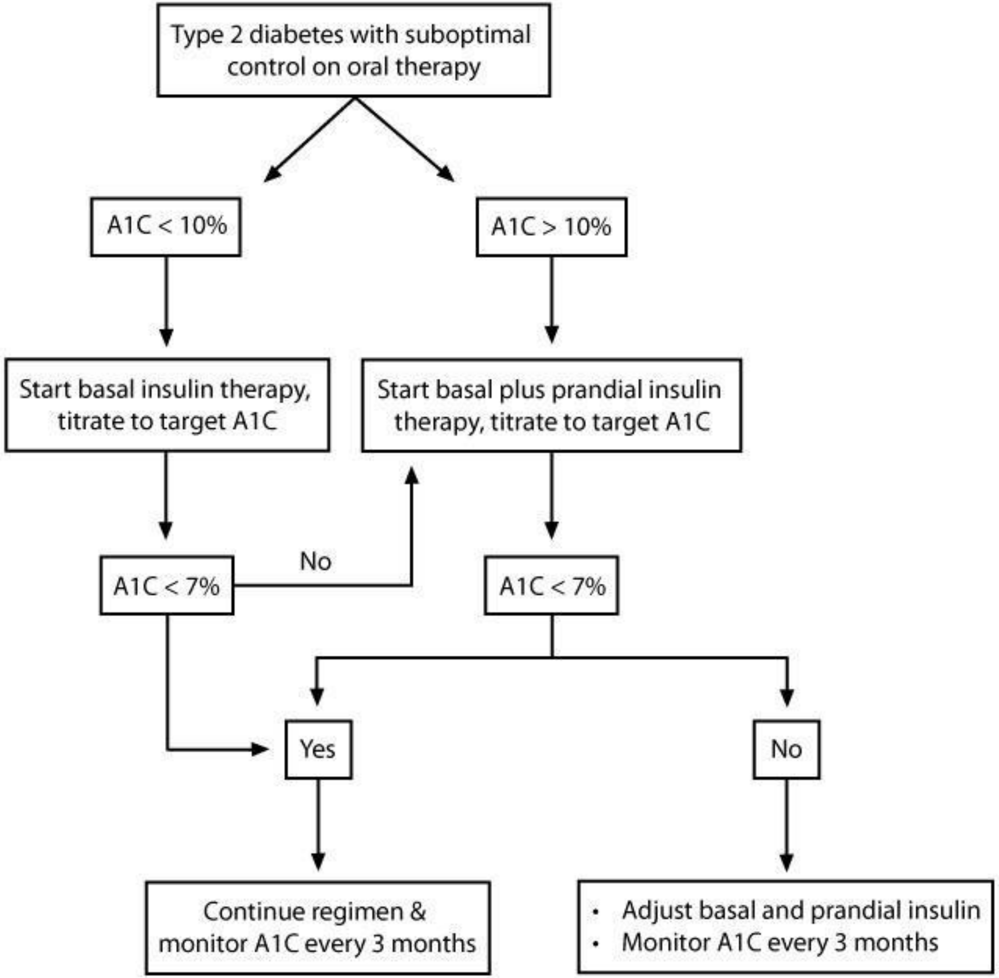
Ab = antibody; **EMG** = electromyogram; **NC** = nerve conduction.

Treatment of menopause



*Contraindications to estrogen: Breast cancer, coronary heart disease, endometrial cancer, liver disease, thromboembolism.

SSRI = selective serotonin reuptake inhibitor.


Type 2 diabetes with suboptimal control on oral therapy
A1C < 10%
A1C > 10%
Start basal insulin therapy, titrate to target A1C
Start basal plus prandial insulin therapy, titrate to target A1C
A1C < 7%
No
A1C < 7%
Yes
No
Continue regimen & monitor A1C every 3 months
• Adjust basal and prandial insulin
• Monitor A1C every 3 months

Clinical features of Paget's disease	
Signs & symptoms	• Majority of patients are asymptomatic • **Skull:** Deformity with enlargement, hearing loss, dizziness • **Spine & pelvis:** Bone pain, spinal stenosis, nerve compression • **Long bones:** Bowing deformities with ↑ fracture risk • **Bone tumors:** Osteosarcoma, giant cell tumors (usually benign)
Laboratory & imaging findings	• Elevated serum & bone-specific ALP • Bone markers may or may not be elevated with active disease (eg, PINP, CTx, NTx & urinary hydroxyproline). • Calcium & phosphorus are usually normal; may be elevated with fracture or immobilization. • Plain radiographs show osteolytic or mixed lytic/sclerotic lesions.
Diagnosis	• Combination of radiographic findings & elevated ALP • No additional bone turnover tests or imaging is required for diagnosis. • **Bone scan** is more sensitive than x-ray & is helpful to document the extent & locations of skeletal involvement.
Treatment	• **Bisphosphonates** are the preferred treatment.

ALP = alkaline phosphatase; **CTx** = C-telopeptide; **NTx** = N-telopeptide;
PINP = procollagen type I N propeptide.

Disseminated histoplasmosis	
Epidemiology	• Midwest & central United States (Ohio & Mississippi River Valleys) • Soil contaminated by **bird or bat droppings** • ↑ **dose exposure or immunocompromised** (eg, **AIDS**)
Symptoms	• Systemic (**fevers**, chills, malaise) • **Weight loss** & cachexia • Pulmonary (cough, dyspnea) • Mucocutaneous lesions (papules, nodules) • Reticuloendothelial (HSM, LAD)

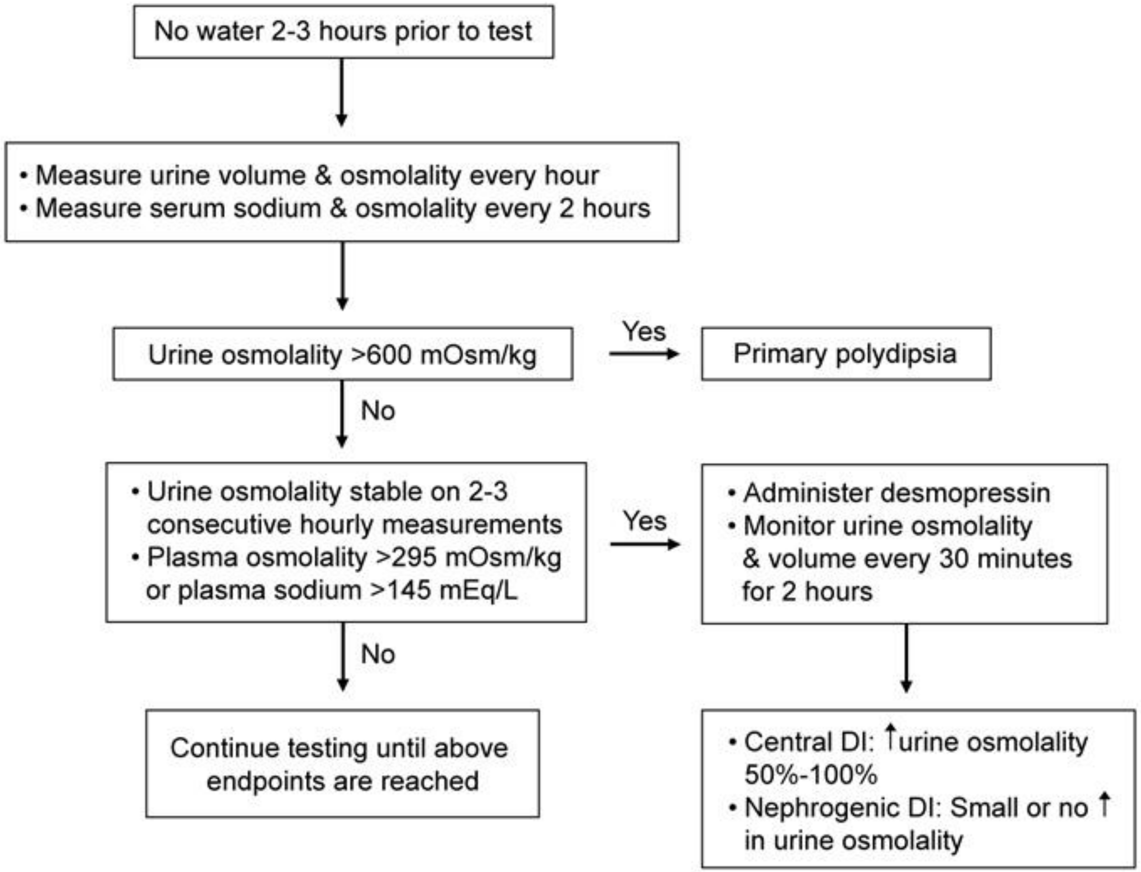
Diagnosis	• Laboratory findings:
	  ○ **Pancytopenia**
	  ○ Transaminitis
	  ○ ↑ LDH & ferritin
	• Chest x-ray (reticulonodular or interstitial infiltrate)
	• Urine/serum ***Histoplasma* antigen** (↑ sensitivity/rapid)
	• Serology
	• Culture (takes 4-6 weeks)

Treatment	• Amphotericin B (moderate-severe disease) • Itraconazole (mild disease/maintenance)

HSM = hepatosplenomegaly; **LAD** = lymphadenopathy; **LDH** = lactate dehydrogenase.

Acute HIV infection	
Epidemiology	• Typically presents 2-4 weeks after exposure
Clinical features	• **Mononucleosis-like syndrome** (eg, fever, lymphadenopathy, sore throat, arthralgias) • Generalized macular rash • Gastrointestinal symptoms
Diagnosis	• Viral load is markedly **elevated** (>100,000 copies/mL) • HIV antibody testing may be **negative** (not yet seroconverted) • CD4 count may be normal
Management	• Combination antiretroviral therapy • Partner notification, consider secondary prophylaxis

Water deprivation test

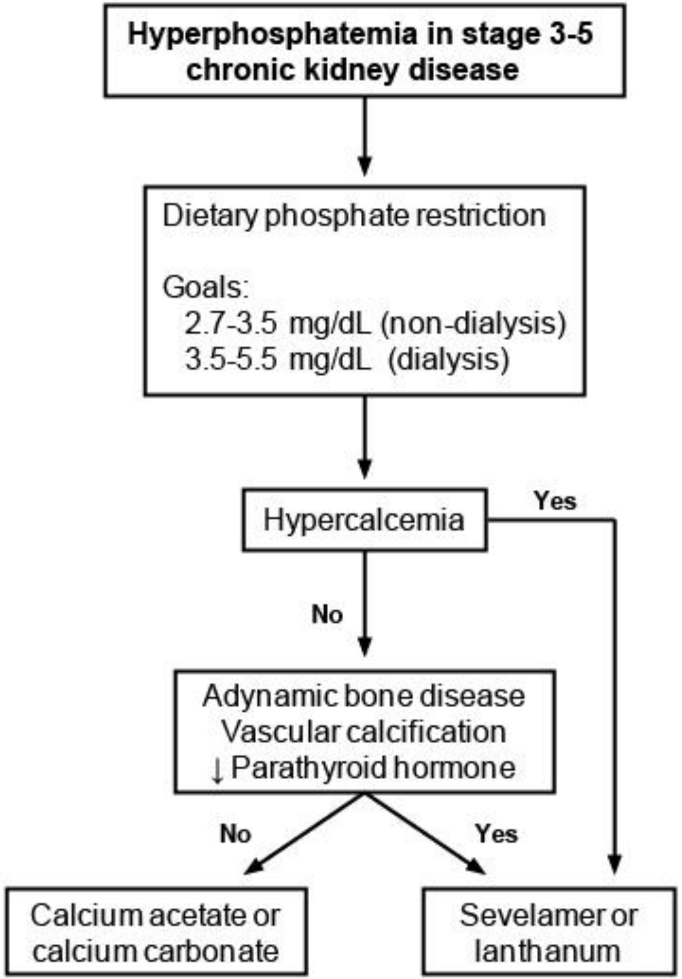


Clinical features of lithium toxicity	
Drugs that ↑ lithium levels	• Diuretics • NSAIDs • SSRIs • ACE inhibitors and ARBs • Non-dihydropyridine calcium channel blockers (eg, verapamil) • Antiepileptics (eg, carbamazepine, phenytoin)
Clinical presentation	• Neurologic: confusion, agitation, vertigo, ataxia and/or neuromuscular excitability (eg, irregular coarse tremors, fasciculations, myoclonic jerks) • Cardiac: bradycardia and prolonged QTc interval • Nephrogenic diabetes insipidus • Severe toxicity (lithium level 2.5-3.5 mEq/L): seizures, encephalopathy, coma

Recurrent aphthous stomatitis	
Recurrent aphthous ulcer minor	• **80% of cases** • Few, painful, small ulcers (<1 cm) • Shallow lesions heal **without scarring** in 7-10 days
Recurrent aphthous ulcer major	• **Large**, deep, painful ulcers (>1 cm) • Affects lips & soft palate • Lesions heal with **scarring** in 6 weeks
Herpetiform recurrent aphthous ulcer	• **Female predominance** • Up to 100 very small ulcers that can coalesce • Extremely painful • May have virtually continuous recurrence

Clinical features of CVID	
Clinical manifestations	• Recurrent infections (eg, rhinosinusitis, pneumonia, conjunctivitis) • Chronic lung disease (eg, bronchiectasis) • Autoimmune disorders (eg, vitiligo, rheumatoid arthritis, thyroid disorders) • Gastrointestinal disorders (eg, irritable bowel disease, malabsorption) • Granulomatous disease in lymphoid or solid organs • Increased risk of malignancy (eg, lymphoma, gastric cancers)
Diagnosis (requires all of these criteria)	• Extremely persistent low IgG levels • Low IgA and/or IgM levels • No response to vaccinations • Exclusion of other immunodeficiency disorders
Treatment and management	• Immunoglobulin replacement therapy • Age-appropriate cancer screening • Monitoring for lymphoma

Autoimmune hemolytic anemia		
	Warm agglutinin AIHA	**Cold agglutinin AIHA**
Etiology	• Drugs (eg, penicillin) • Viral infections • Autoimmune (eg, SLE) • Immunodeficiency states • Lymphoproliferative (eg, CLL)	• Infections (eg, *Mycoplasma pneumoniae* infection & infectious mononucleosis) • Lymphoproliferative diseases
Clinical presentation	• Asymptomatic to life-threatening anemia • Direct Coombs' positive with anti-**IgG**, anti-C3, or both	• Symptoms of anemia • Livedo reticularis & acral cyanosis with cold exposure that disappear with warming • Direct Coombs' positive with anti-C3 or anti-**IgM**, but usually not IgG
Treatment	• Corticosteroids • Splenectomy for refractory disease	• Avoidance of cold temperatures • Rituximab +/- fludarabine
Complications	• Venous thromboembolism • Lymphoproliferative disorders	• Ischemia & peripheral gangrene • Lymphoproliferative disorders

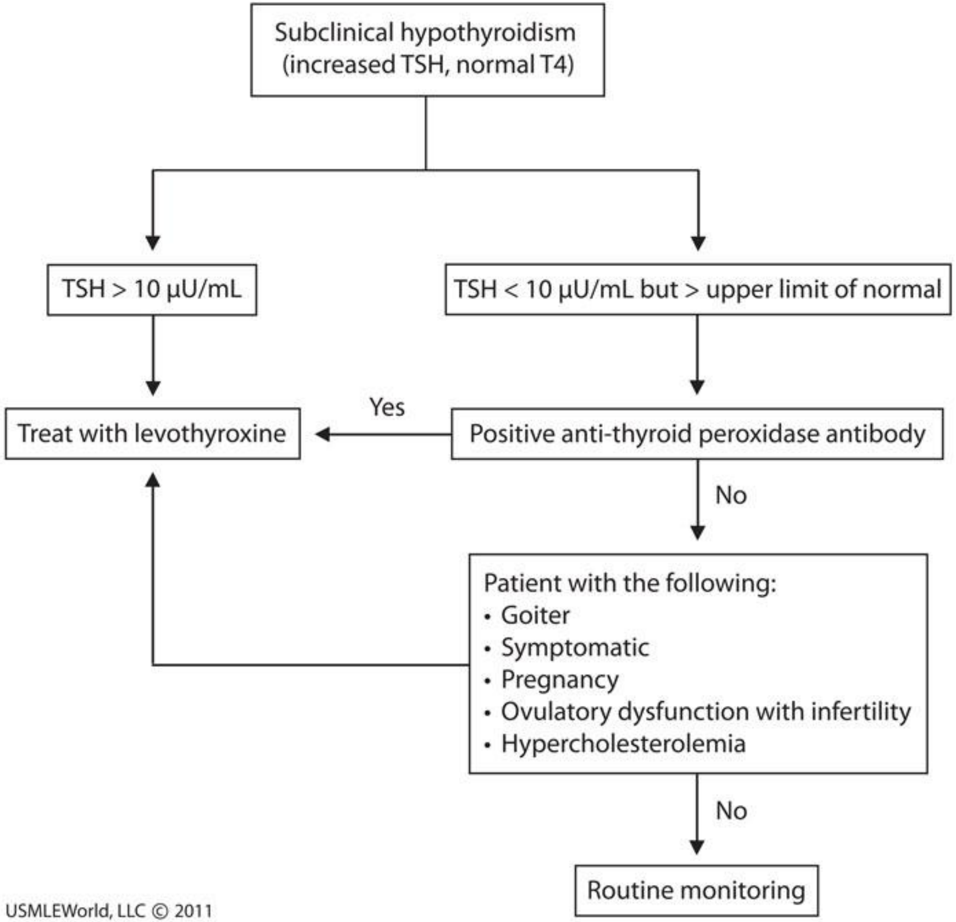


Hyperphosphatemia in stage 3-5 chronic kidney disease
Dietary phosphate restriction
Goals:
2.7-3.5 mg/dL (non-dialysis)
3.5-5.5 mg/dL (dialysis)
Hypercalcemia
Yes
No
Adynamic bone disease
Vascular calcification
↓ Parathyroid hormone
No
Yes
Calcium acetate or calcium carbonate
Sevelamer or lanthanum

Hereditary hemorrhagic telangiectasia	
Site & incidence	**Important clinical features**
Nasal telangiectasias (>90%)	Frequent epistaxis
Gastrointestinal telangiectasias (10%-40%)	Iron-deficiency anemia, acute gastrointestinal bleeding
Pulmonary AVMs (10%-30%)	Cyanosis, dyspnea, paradoxical embolism, **cerebral abscess**
Cerebral AVMs (10%-15%)	Epilepsy, ischemic or hemorrhagic stroke
Hepatic AVMs (up to 74%)	Usually asymptomatic but liver failure possible in some cases

Diagnostic criteria for RADS
• Chronic asthma-like illness after single high-level exposure to pulmonary irritant • Onset of symptoms within the first 24 hours • Absence of pre-existing lung disorder • Airflow obstruction on pulmonary function testing • Positive methacholine challenge test after exposure

Subclinical hypothyroidism

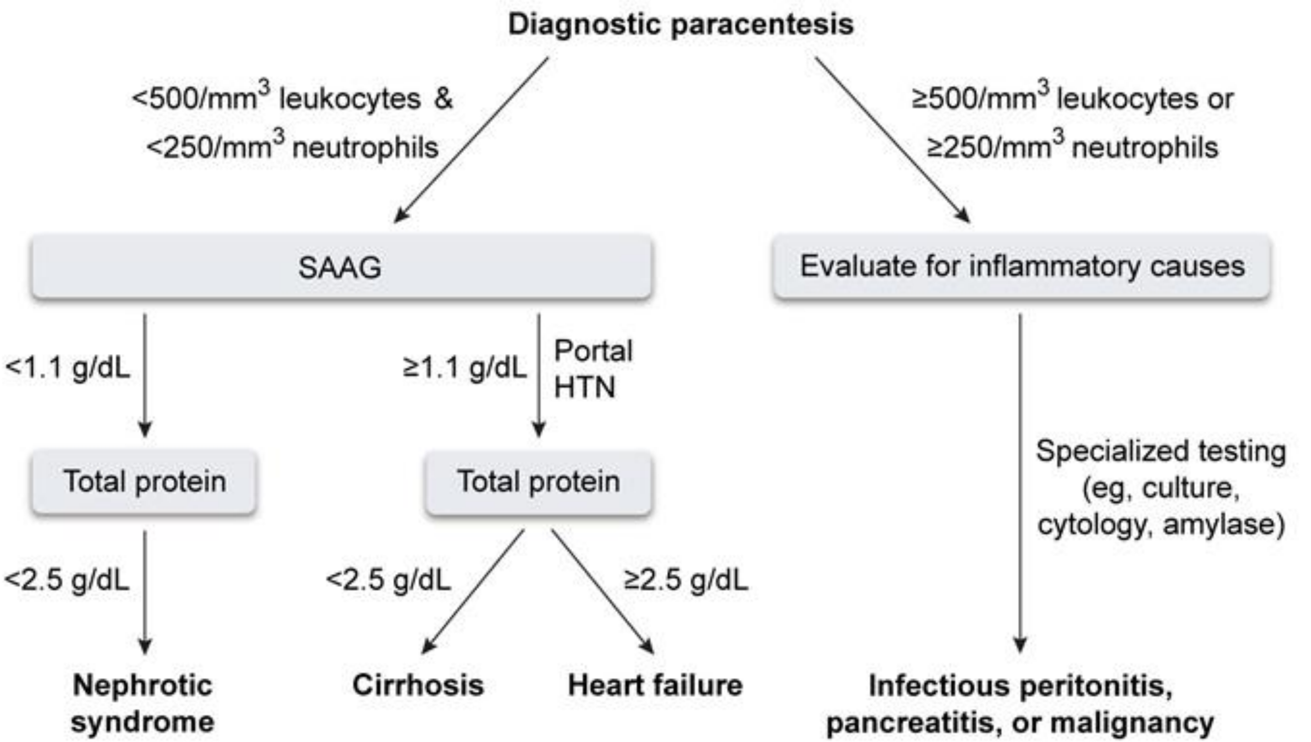


Features of adult minimal change disease	
Clinical	• Age >45 • Sudden onset of symptoms • Hypertension (43%)
Laboratory	• Nephrotic-range proteinuria (>3 g/day) • Serum albumin <2.2 g/dL • Serum cholesterol >420 mg/dL • Acute kidney injury (18%)

Clinical features of cervical actinomycosis	
Risk factors	• Dental caries, extractions, or infected tooth • Gingivitis and/or gingival trauma • Diabetes or immunosuppression • Malnutrition • Local tissue damage from malignancy or irradiation
Clinical presentation	• Chronic, slowly progressive, non-tender indurated mass • Extends through tissue planes to form abscess, fistula, & draining sinus tract • Mandible is the most commonly involved site

Adult-onset Still's disease	
Clinical features	
Most common	• High **fever** • **Arthritis** • Transient **rash**
Variable	• Lymphadenopathy • Sore throat • Hepatomegaly, splenomegaly • Pleuritis, pericarditis
Laboratory findings	
Most common	• **Leukocytosis** • Elevated inflammatory markers (eg, erythrocyte sedimentation rate, C-reactive protein) • Markedly **elevated ferritin** • Negative antinuclear antibodies, rheumatoid factor
Variable	• Normocytic anemia • Thrombocytosis • Elevated hepatic markers

Evaluation of ascites

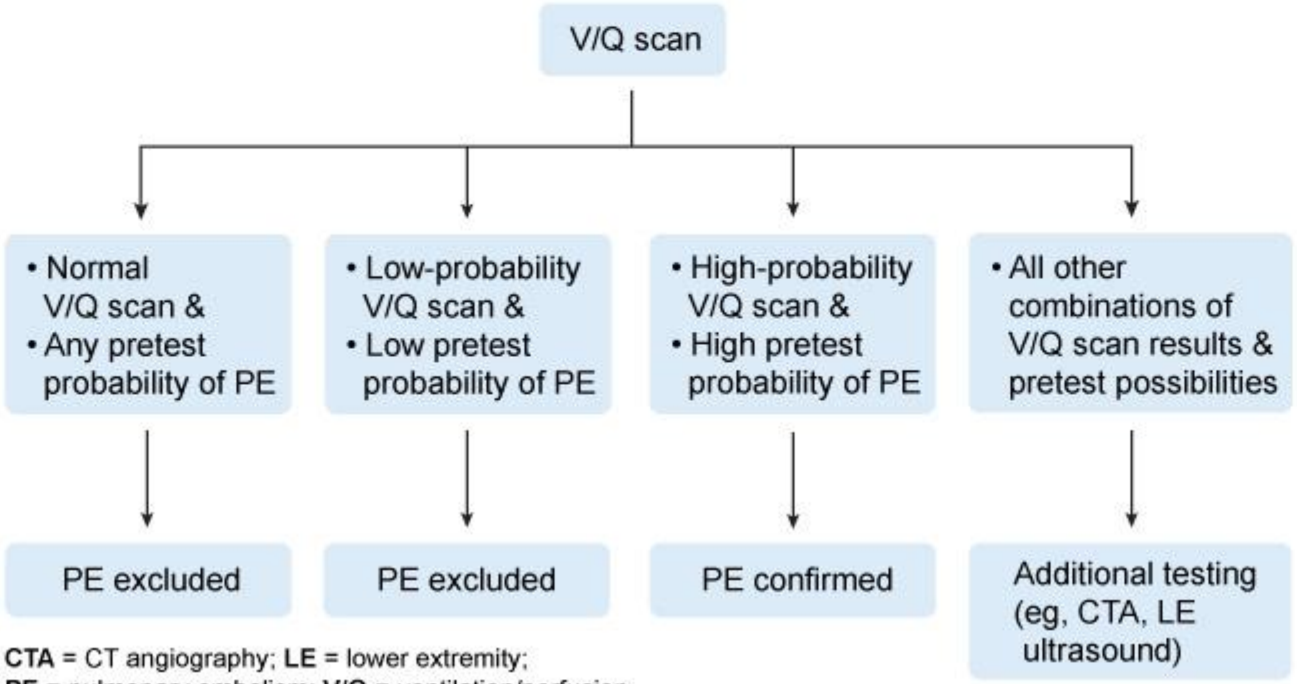


HTN = hypertension; **SAAG** = serum-to-ascites albumin gradient.

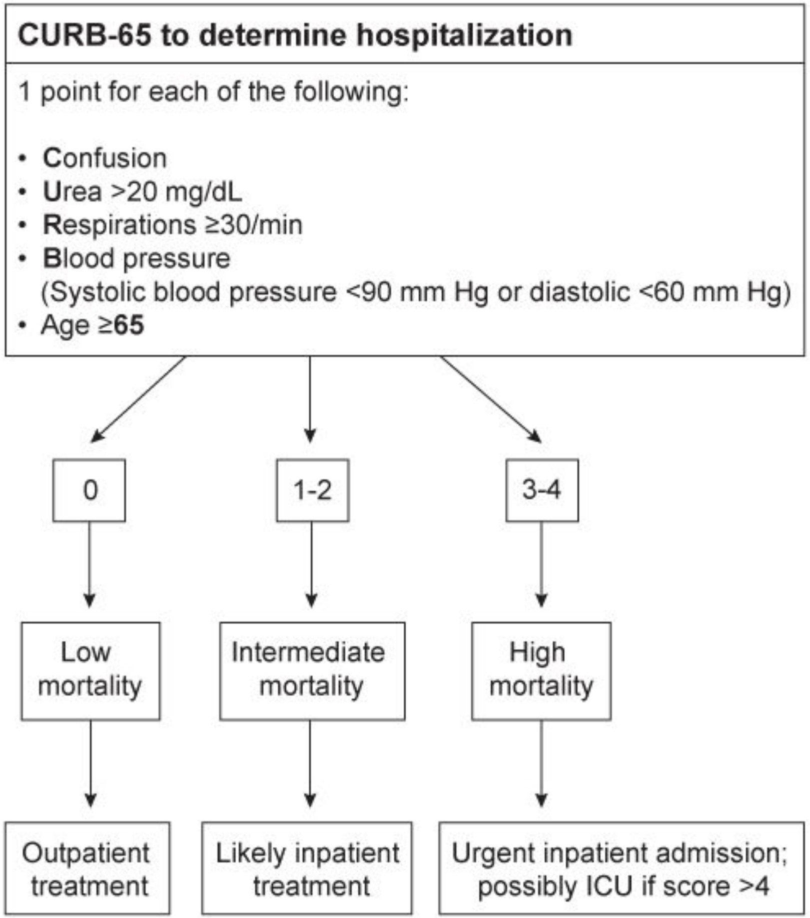
Management of reversible causes of transient urinary incontinence	
Severe constipation or stool impaction	• Increase mobility, fluid & fiber intake • Stool softeners, laxatives & bulking agents • Manual disimpaction if warranted
Urinary tract infection	• Antibiotic therapy
Atrophic vaginitis or urethritis	• Topical estrogen (may worsen urinary incontinence in some)
Metabolic conditions (hyperglycemia, hypercalcemia)	• Treatment of underlying cause
Medications	• Identification & discontinuation
Alcohol	• Provide counseling regarding cessation
Impaired ability to reach the toilet	• Use of bedside toilet • Consider physical therapy & assistive devices • Increase caregiver support

Porphyria cutanea tarda	
Clinical presentation	• Blisters, bullae, scarring, hypopigmentation/hyperpigmentation on sun-exposed skin (eg, back of hands, forearms, face) • Scarring and calcification similar to scleroderma
Associated conditions	• **Hepatitis C** • HIV • Excessive alcohol consumption • Estrogen use • Smoking
Diagnostic testing	• Mildly elevated liver enzymes and iron overload • Elevated plasma or urine **porphyrin** levels

Interpretation of V/Q scan results based on pretest probability for PE



CTA = CT angiography; **LE** = lower extremity;
PE = pulmonary embolism; **V/Q** = ventilation/perfusion.

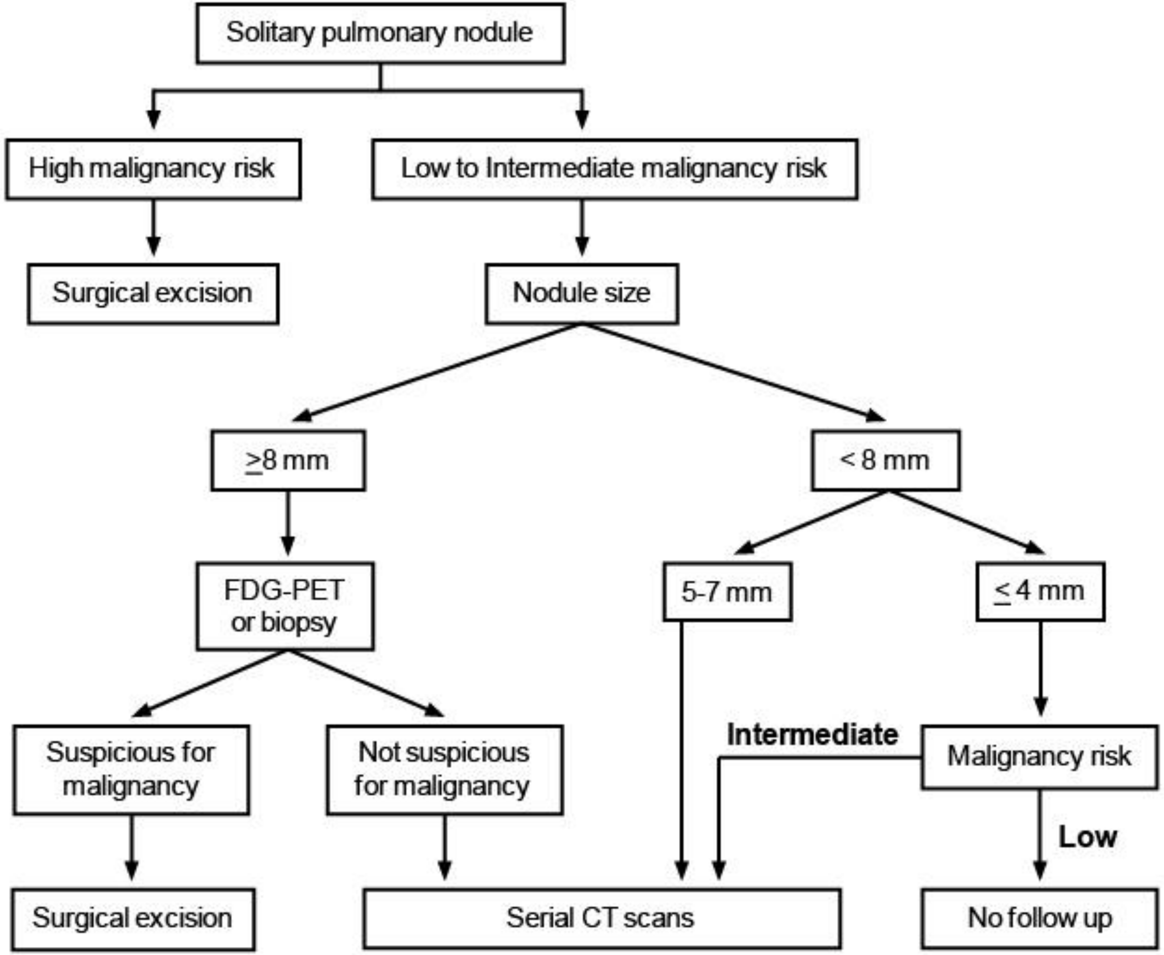


ICU = intensive care unit.

	Acute eosinophilic pneumonia	Chronic eosinophilic pneumonia
Clinical presentation	• Idiopathic, more common in new-onset/resumption of smoking • High fever (< 3 weeks duration) • Nonproductive cough, dyspnea • Bibasilar inspiratory crackles • Hypoxemic respiratory failure	• More common in women, nonsmokers, asthmatics, post-radiation therapy for breast cancer • Fever, cough, progressive dyspnea • Weight loss, night sweats • Rare respiratory failure
Diagnosis	• Initial neutrophilic predominant leukocytosis with subsequent eosinophilia • **X-ray/CT**: Bilateral diffuse ground glass & reticular opacities • **Bronchoalveolar lavage**: > 25% eosinophils • **Lung biopsy**: Diffuse alveolar damage, increased interstitial eosinophils	• Peripheral eosinophilia, ↑ESR & CRP • Iron deficiency anemia, ↑ platelets • **X-ray/CT**: bilateral peripheral/pleural-based infiltrates sparing perihilar & central lung regions • **Bronchoalveolar lavage**: > 25% eosinophils • **Lung biopsy**: interstitial & alveolar eosinophils, multinucleated giant cells

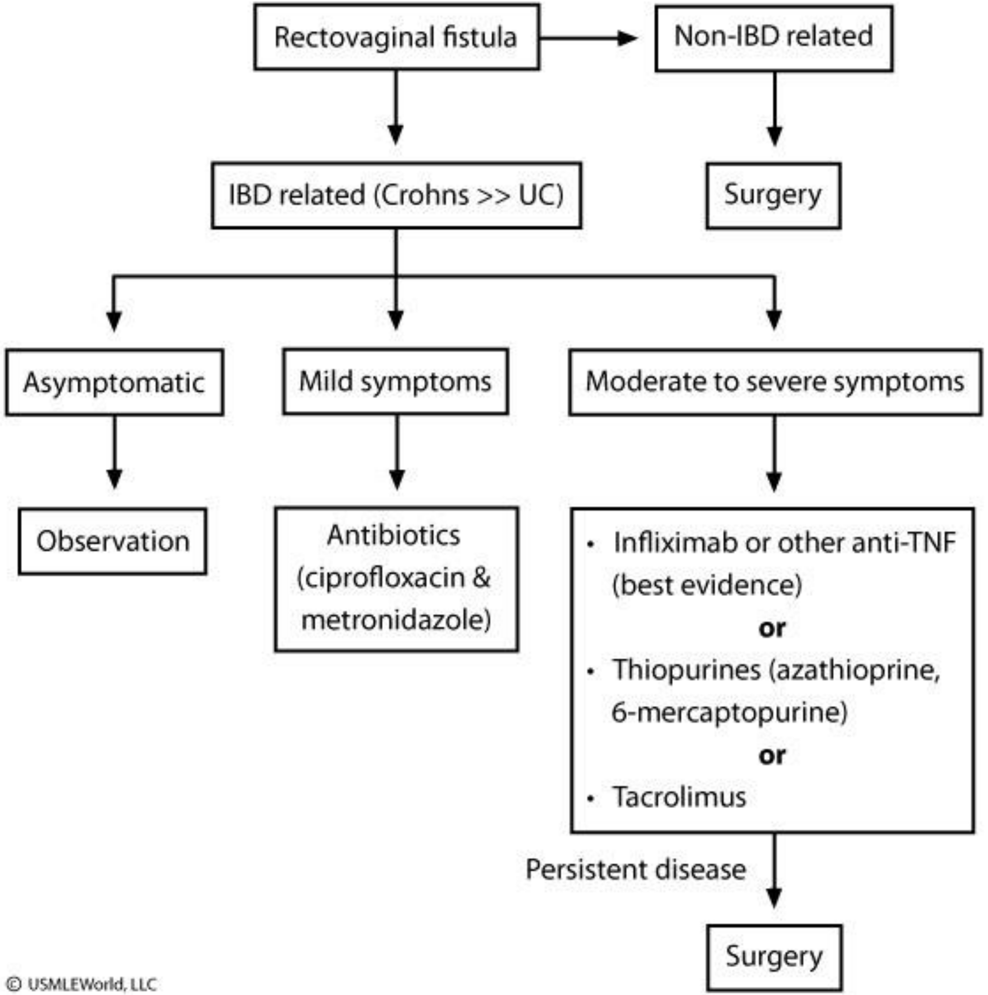
Intervals for follow-up colonoscopy after polypectomy	
Small rectal hyperplastic polyps	10 years
1 or 2 small (<1 cm) tubular adenomas	5 years
▪ 3-10 adenomas ▪ Any adenoma >1 cm ▪ Adenoma with high-grade dysplasia or **villous** features	3 years
More than 10 adenomas	<3 years Consider underlying familial syndrome
▪ Large (>2 cm) sessile polyp removed by piecemeal excision ▪ Polyp with adenocarcinoma (must have minimal invasion & ≥2 mm margin)	2-6 months 2-3 months

Assessment of malignancy risk for solitary pulmonary nodule			
Variable	**Low risk**	**Intermediate risk**	**High risk**
Nodule size (cm)	<0.8	0.8-2.0	≥2.0
Age (yr)	<40	40-60	>60
Smoking status	Never smoked	Current	Current
Smoking cessation (yr)	>15	5-15	<5
Nodule margin characteristics	Smooth	Scalloped	Corona radiata or spiculated

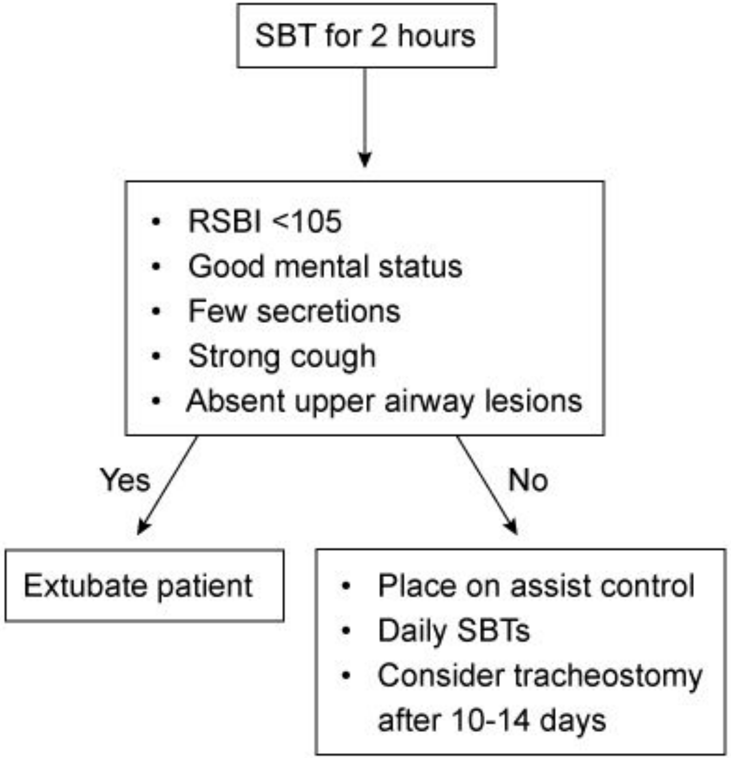


FDG-PET = 18-fluorodeoxyglucose positron emission tomography.

Clinical presentation of Wilson disease	
Hepatic	• Asymptomatic liver function abnormalities • Chronic hepatitis • Acute liver failure
Neuropsychiatric	• Parkinsonian-like tremor, rigidity, clumsiness of gait • Subtle personality changes to overt depression, paranoia, and catatonia
Other	• Hemolytic anemia • Fanconi syndrome • Recurrent nephrolithiasis • Premature arthropathy & chondrocalcinosis (common in knees)


Rectovaginal fistula
Non-IBD related
Surgery
IBD related (Crohns >> UC)
Asymptomatic
Observation
Mild symptoms
Antibiotics (ciprofloxacin & metronidazole)
Moderate to severe symptoms
• Infliximab or other anti-TNF (best evidence)
or
• Thiopurines (azathioprine, 6-mercaptopurine)
or
• Tacrolimus
Persistent disease
Surgery

Pulmonary weaning from vent



RSBI = rapid shallow breathing index; **SBT** = spontaneous breathing trial.

Clinical features of thyroid storm	
Precipitating factors	• Thyroid or non-thyroid surgery • Acute illness (eg, trauma, infection), childbirth • Acute iodine load (eg, iodine contrast)
Clinical presentation	• Fever as high as 40-41.1 C (104-106 F) • Tachycardia, hypertension, congestive heart failure, cardiac arrhythmias (eg, atrial fibrillation) • Agitation, delirium, seizure, coma • Goiter, lid lag, tremor, warm & moist skin • Nausea, vomiting, diarrhea, jaundice
Treatment	• Beta blocker (eg, propranolol) to ↓ adrenergic manifestations • PTU followed by iodine solution (SSKI) to ↓ hormone synthesis & release • Glucocorticoids (eg, hydrocortisone) to ↓ peripheral T4 to T3 conversion & improve vasomotor stability • Identify trigger & treat, supportive care

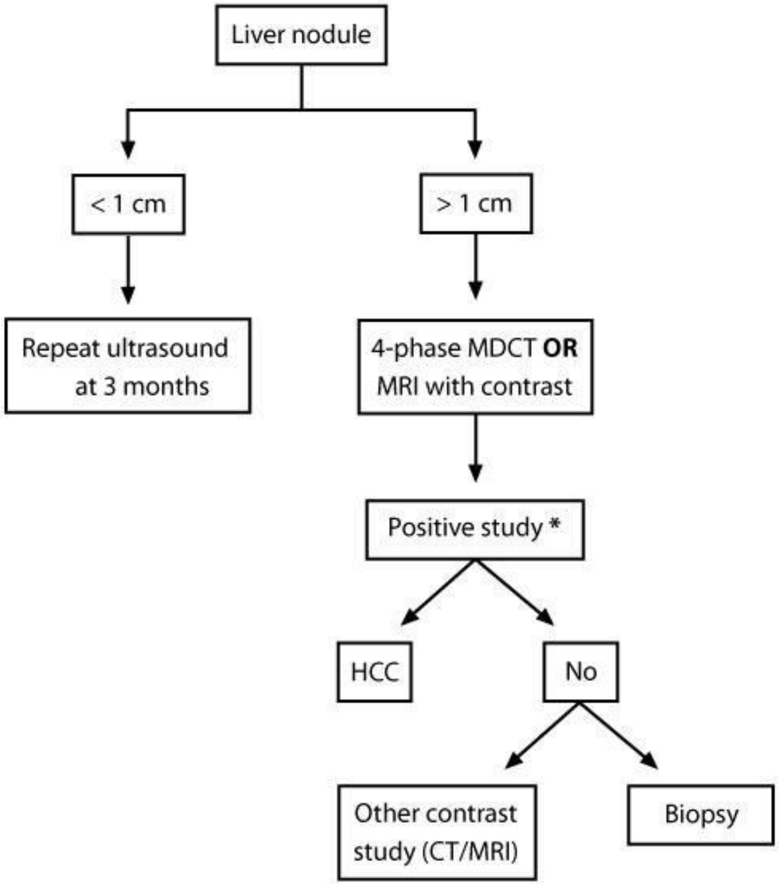
PTU = propylthiouracil; **SSKI** = potassium iodide.

Prevention of steroid-induced osteoporosis

- General measures
  - Use lowest possible steroid dose for shortest duration
  - Topical steroids preferred over oral/enteral steroids
  - Daily weight-bearing exercises
  - Stop tobacco & excessive alcohol use
  - Fall prevention
- Calcium & vitamin D
- Bisphosphonates (eg, alendronate, risedronate)
- Parathyroid hormone (eg, teriparatide) as second-line agent for severe osteoporosis

7.5mg/d >3 mo=tx for osteoporosis

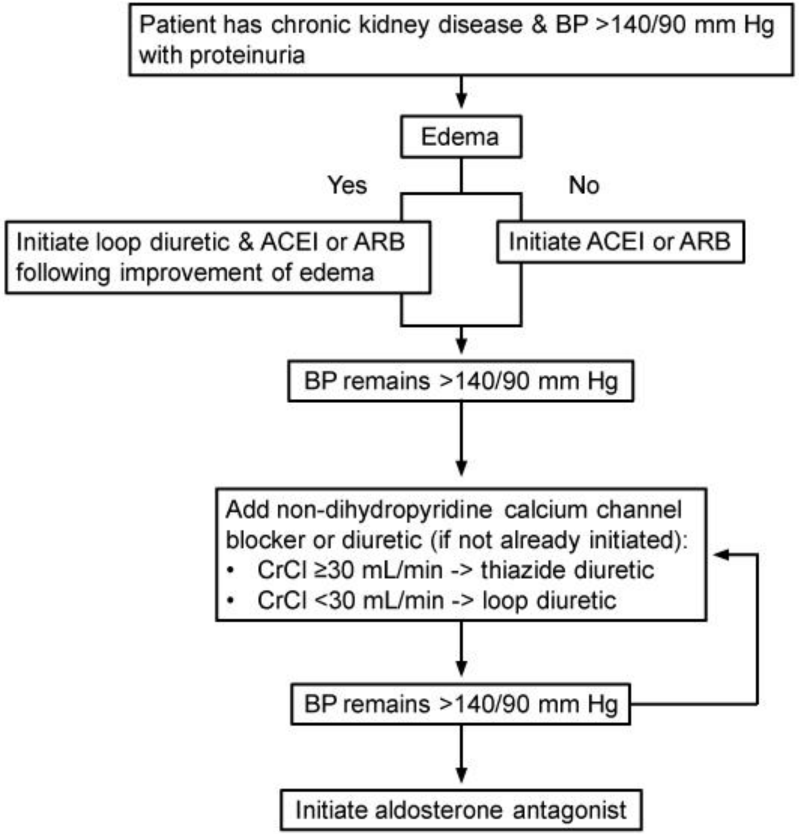
Major drug interactions of levothyroxine	
↓ levothyroxine absorption	• Bile acid binding agents (e.g., cholestyramine) • Iron, calcium, aluminum hydroxide • Proton pump inhibitors, sucralfate
↑ TBG concentration	• Estrogen (oral), tamoxifen, raloxifene • Heroin, methadone
↓ TBG concentration	• Androgens, glucocorticoids • Anabolic steroids • Slow-release nicotinic acid
↑ thyroid hormone metabolism	• Rifampin • Phenytoin • Carbamazepine



* Positive study: Arterial hypervascularity **AND** venous/delayed phase washout

Granulomatous disorders causing hypercalcemia	
Noninfectious	Infectious
• Sarcoidosis • Berylliosis • Crohn's disease • Eosinophilic granuloma • Granulomatosis with polyangiitis • Lymphomas	• Tuberculosis • Leprosy • Coccidioidomycosis • Histoplasmosis • *Pneumocystis carinii* pneumonia

Management of hypertension in proteinuric chronic kidney disease



ACEI = angiotensin-converting enzyme inhibitor; **ARB** = angiotensin II receptor blocker; **BP** = blood pressure; **CrCl** = creatinine clearance.

Treatment options for diabetic peripheral neuropathy
• Antidepressants (e.g., amitriptyline, duloxetine) • Anticonvulsants (e.g., pregabalin, valproic acid) • Topical capsaicin cream • Alpha-lipoic acid • Transcutaneous electrical nerve stimulation (TENS) • Lidocaine patch

Causes of acute, painful vision loss	
Acute angle-closure glaucoma	• **Headache, vomiting** • Red eye, corneal opacity • ↑ **Intraocular pressure** • **Treatment**: Timolol, apraclonidine & pilocarpine drops; intravenous acetazolamide
Optic neuritis	• Associated with **multiple sclerosis** • Pain with eye movement • Afferent pupillary defect • **Diagnosis**: Contrast-enhanced MRI • **Treatment**: Intravenous methylprednisolone
Keratitis	• Mucopurulent exudate • **Epithelial ulceration** on fluorescein staining • **Treatment**: Broad-spectrum antibiotic drops
Corneal ulcer	• **Foreign body sensation** • Traumatic, foreign body, infectious (eg, herpes simplex)

Osteoporosis risk factors in men

- Hypogonadism or androgen deprivation therapy
- Hyperthyroidism
- Hyperparathyroidism
- Medications (eg, glucocorticoids, anticonvulsants)
- Gastrointestinal (eg, subtotal or total gastrectomy, celiac disease, inflammatory bowel disease)
- Vitamin D deficiency
- Smoking or alcohol abuse
- History of fractures (with low impact) or falls

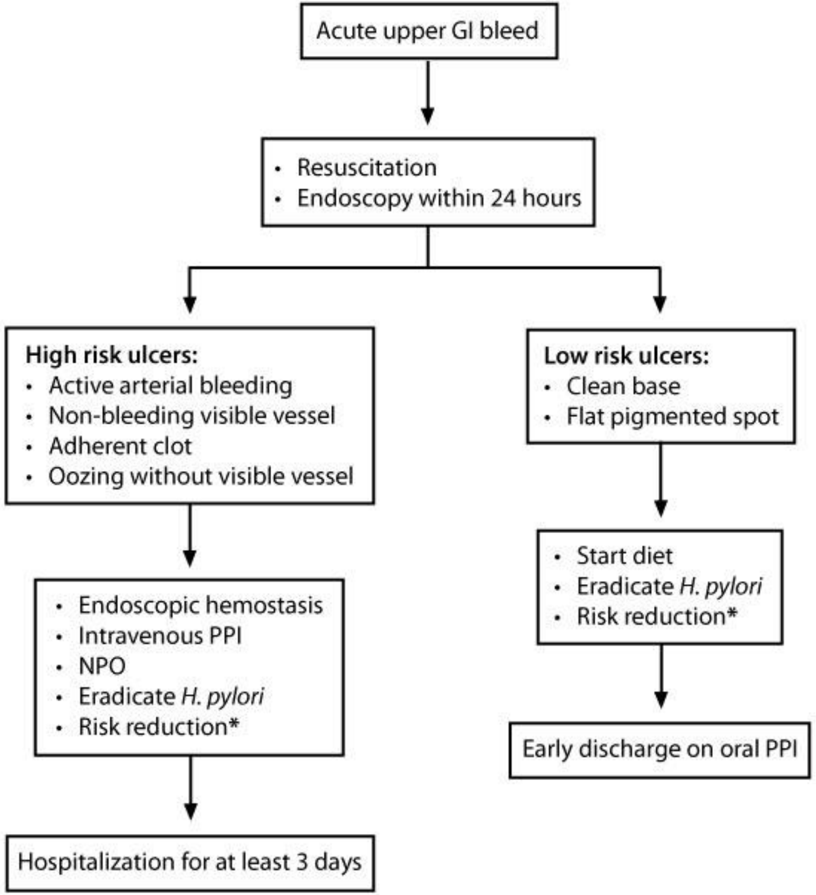
Criteria for the diagnosis of transfusion-associated lung injury (TRALI)

• New acute lung injury (ALI)
  ◦ Acute onset
  ◦ Hypoxemia (PaO_2/FiO2 <300 mm Hg, or O_2 saturation ≤90% on room air)
  ◦ Bilateral pulmonary infiltrates visualized on frontal chest x-ray
  ◦ No evidence of left atrial hypertension (i.e., no circulatory overload)
• No ALI immediately before transfusion
• Onset of ALI that occurs during or within 6 hours after the end of transfusion of ≥1 plasma-containing blood products
• No temporal relationship to an alternative risk factor for ALI

Characteristics of chronic primary headaches	
Headache subtype	**Clinical presentation**
Migraine	• Unilateral • Throbbing • Nausea/vomiting or photophobia
Tension-type	• Bilateral & nonthrobbing • Gradual onset • Mild to moderate without pericranial muscle tenderness
Medication-overuse	• Develops or worsens with daily medication use • Episodic migraine or tensionlike
Paroxysmal hemicrania	• Unilateral in trigeminal distribution • Ipsilateral autonomic symptoms • Duration 2-30 min • ≥5 attacks a day • Complete response to indomethacin
Cluster	• Unilateral in trigeminal distribution • Ipsilateral autonomic symptoms • Duration 15 min-3 hr • Circadian/predictable (“alarm clock headache”), can be provoked by alcohol or nicotine • More common in men

Recommended antimalaria chemoprophylaxis options for short-term travelers		
	Medication	**Adverse effects/comments**
Areas with **chloroquine-resistant *P falciparum*** (Sub-Saharan Africa, southern & Southeast Asia)	Atovaquone-proguanil	• Expensive • GI disturbance (eg, abdominal pain), ↑ liver function tests
	Doxycycline	• Inexpensive • GI disturbance, sun sensitivity, teratogenic
	Mefloquine	• Neuropsychiatric effects • **Agent of choice in pregnancy** • Weekly dosing
Areas with **chloroquine-susceptible *P falciparum*** (in addition to options above)	Chloroquine, hydroxychloroquine	• Need to be started 1-2 weeks in advance • Potential exacerbation of some skin conditions • Weekly dosing
Areas **without *P falciparum*** (parts of South America, Mexico, Korean peninsula)	Primaquine	• Potential teratogenicity • Hemolysis in patients with G6-PD deficiency • Weekly dosing

GI = gastrointestinal; G6-PD = glucose-6-phosphate dehydrogenase.



* Stop or limit NSAIDs, anticoagulants, & antiplatelet agents if possible

<table>
<tr><th colspan="2">Diagnostic criteria for MAC</th></tr>
<tr><td>Clinical</td><td>Pulmonary symptoms (eg, cough, fever, weight loss), chest x-ray showing nodular or cavitary opacities, or high-resolution CT scan with multifocal bronchiectasis and multiple small nodules,
AND
Appropriate exclusion of other diagnoses</td></tr>
<tr><td>Microbiology</td><td>At least 2 positive sputum culture results,
OR
One bronchial lavage (BAL) or wash with positive culture,
OR
Transbronchial or other lung biopsy with histopathological features of mycobacterial infection and positive culture for non-tuberculosis mycobacterial lung disease</td></tr>
</table>

Diagnostic criteria for multiple myeloma

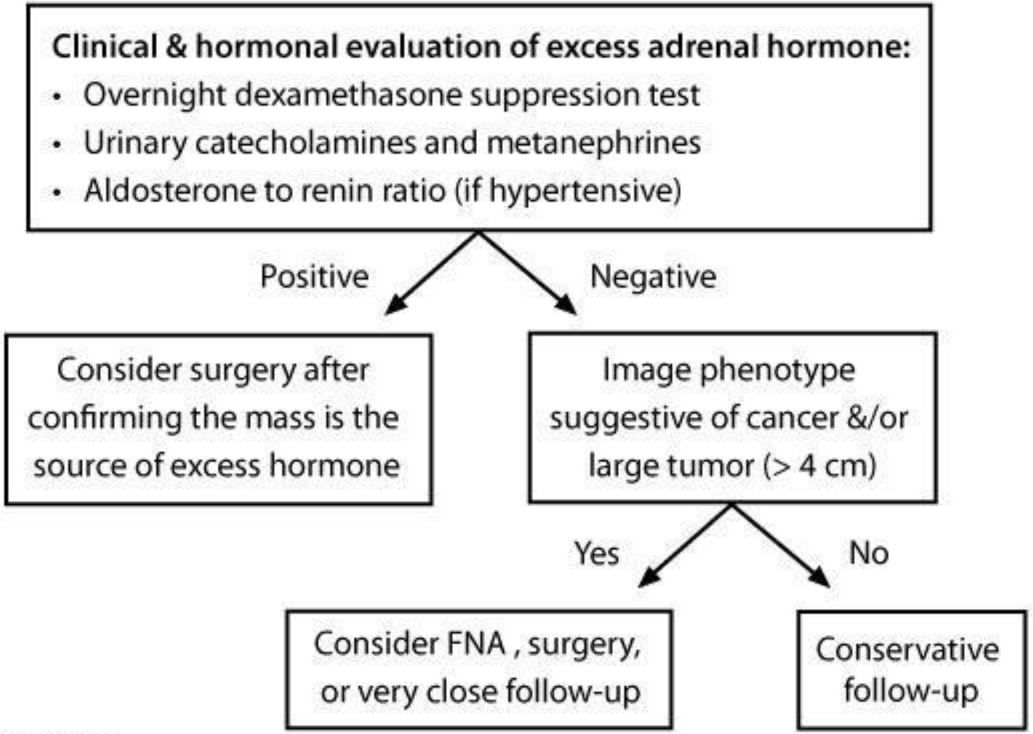
- Monoclonal protein in serum or urine
- ≥ 10% clonal plasma cells in bone marrow or soft tissue/bone plasmacytoma
- End-organ damage (CRAB)
  - **C**alcium elevation
  - **R**enal insufficiency
  - **A**nemia (usually normocytic)
  - **B**one pain (usually due to lytic lesions)

Ausculation of cardiac murmurs			
Maneuver	**What it does**	**Murmurs that get louder**	**Murmurs that get softer**
Valsalva	• **Early (Strain)** ↓ Venous return ↓ All murmurs except HCM & MVP • **Late (Release)** ↑ Venous return ↑ Right-sided murmurs	• **HCM** ↓ LV volume ↑ Gradient • **MVP** ↓ LV volume ↑ Leaflet prolapse	• **All others** ↓ Flow through stenotic or regurgitant valve during strain phase
Standing	• ↓ Venous return • Similar to the strain phase of Valsalva	• **HCM** ↓ LV volume ↑ Gradient • **MVP** ↓ LV volume ↑ Leaflet prolapse	• **All others** ↓ Flow through stenotic or regurgitant valve during strain phase
Squatting	• ↑ Venous return • ↑ Afterload by kinking of femoral arteries • ↑ Reverse flow	• AR • MR • VSD	• **HCM** ↑ Preload ↓ Gradient across outflow obstruction ↓ Obstruction • **MVP** ↑ LV size ↓ Leaflet prolapse
Handgrip	• ↑ Afterload • ↑ Blood pressure • ↑ Reverse flow across valve	• AR • MR • VSD	• **HCM** (↑ LV volume) • **AS** (↓ transvalvular pressure gradient)

AR = aortic regurgitation; **HCM** = hypertrophic cardiomyopathy; **LV** = left ventricular; **MR** = mitral regurgitation; **MVP** = mitral valve prolapse; **VSD** = ventricular septal defect; **AS** = aortic stenosis

Clinical features of dengue fever	
Classic dengue fever	• Flu-like febrile illness with marked myalgias & joint pains ("break-bone fever") • **Retro-orbital pain** • Rash ("white islands in sea of red")
Dengue hemorrhagic fever	• Increased vascular permeability • Thrombocytopenia ($<100,000/mm^3$) • Spontaneous bleeding → **shock** • **Positive tourniquet test** (petechiae after sphygmomanometer cuff inflation for 5 minutes)
Management	• Supportive care

Adrenal incidentaloma



Medication	↓ A1c	Points to remember
Metformin (biguanide)	1.0%-2.0%	• Initial therapeutic agent for most type 2 diabetics • Weight neutral, low risk of hypoglycemia • Lactic acidosis is a life-threatening complication
Sulfonylureas	1.0%-2.0%	• Generally added in patients with metformin failure • Weight gain & hypoglycemia are main side effects
Pioglitazone (TZDs)	1.0%-1.5%	• Used if unable to tolerate metformin or sulfonylureas • Side effects: weight gain, edema, CHF, bone fracture, bladder cancer • Low risk of hypoglycemia when used alone or with metformin • Can be used in renal insufficiency
DPP-IV inhibitors (eg, sitagliptin)	0.5%-0.8%	• Low risk of hypoglycemia • Weight neutral • Can be used in renal insufficiency
GLP-1 receptor agonist (eg, exenatide)	0.5%-1.0%	• Possible second agent for metformin failure, especially if weight loss is desired • Low hypoglycemia risk when used alone or with metformin

CHF = congestive heart failure; **DPP** = dipeptidyl peptidase-4; **GLP-1** = glucagonlike peptide-1; **IV** = intravenous; **TZDs** = thiazolidinediones.